

PEARSON EDEXCEL INTERNATIONAL GCSE (9-1)

INFORMATION AND COMMUNICATION TECHNOLOGY (ICT)

COMPLETE SUMMARY BOOK



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A Message to My Students

Dear Students,

Welcome to your journey through the world of Information and Communication Technology. The skills and concepts detailed within this book are designed not just to prepare you for your upcoming assessments, but to empower you as digital citizens in an increasingly interconnected world. Work hard, stay curious, and always challenge yourselves to explore deeper.

Wishing you all the absolute best in your studies and future endeavors!

— Teacher Jamal Mohamed Fahadh

Digital Devices

Types of Digital Devices

- **Digital devices:** Pieces of hardware that use computers or microcontrollers, found everywhere in our digital world.
- **Mainframe computer:** A large, powerful computer that can do a lot of complicated jobs quickly and can be used by a lot of people at the same time.
 - *Primary Uses:* Used by large organisations for complex processing tasks such as statistical analysis and bulk data processing.
- **Microprocessor:** The device that controls what a computer does; it takes data as input, does something with it and provides output.
 - *Primary Uses:* Used to control washing machines, televisions and other household appliances.

Personal Computers (PCs)

Personal computers (PCs) are common in homes and offices, coming in many different shapes and sizes.

- **Desktops:** Have more space for components than laptops and often provide users with the option to upgrade them or add additional components.
 - *Upgrade:* To make a computer better and able to do more things.
 - *Peripheral device:* Equipment that is connected to a computer and used with it.
 - *All-in-one:* Desktops that combine the monitor with the PC hardware.
- **Laptops:** Include a keyboard, a screen, a track pad and a rechargeable battery.
 - *Portable:* Easy to carry or move.
 - *Desktop replacements:* Laptops that tend to be larger than other laptops, as well as having a bigger and better-quality screen and higher performance levels.

Specialized & Educational Computers

- **Single-board computers (SBCs):** Affordable computers used in education, embedded computing projects and physical computing projects (e.g., Raspberry Pi Zero, which uses a cheap microSD card as its storage).
- **Embedded computing:** Computing hardware that is fixed into position and carries out a specialist task.
- **Physical computing:** Interactive systems that can sense and respond to the world around them.

Mobile Communication Devices

- **Mobile phones:** Use a SIM card to connect to a mobile phone network.
 - *SIM:* Subscriber identity module, used to identify the subscriber to a mobile phone network.
- **Specialist phones:** Provide users with functions that meet particular user needs.
 - *Emergency Button:* Linked to a list of emergency contacts. It calls each person on the list until someone answers.
 - *Accessibility:* A measure of how easy something is to use, especially by people with disabilities.
 - *Alto 2 'talking phone':* A specialist mobile phone for blind and partially sighted people; every feature and function is spoken aloud.
- **Smartphones:** Small computers with Wi-Fi and mobile phone connectivity to allow them to make phone calls and access the internet.
 - *Wi-Fi (wireless fidelity):* A way of connecting computers or other electronic machines to a network or the internet by using radio signals rather than wires.
 - *Application (App):* A program that allows a user to perform a task.
 - *Virtual keyboard:* A keyboard displayed on and used via a touch screen.
 - *Power Characteristics:* Because they combine features of cameras, media players, and handheld games consoles, they use more power, have a shorter battery life, and need to be charged more regularly.
- **Tablets:** Bigger than smartphones, but have similar features (touch screen, apps, Wi-Fi connectivity). Some include SIM card slots for cellular networks.

Media Capture & Display Devices

- **Cameras and camcorders:** Use light sensors to capture images formed by light passing through the device's lens.
 - *Resolution factors:* Image quality depends on the quality of the lens, the image processor, and the resolution of the sensor.
 - *Compensate:* To reduce or correct a negative effect (e.g., an image processor compensating for poor lighting).
 - *Pixel:* A small dot that helps to make up an image.
- **Televisions:** Display still and moving images on a screen.
 - *Resolution:* The number of pixels used by a screen to display an image.
 - *Pixelated:* An effect that creates an unclear image consisting of large individual pixels that are visible to the human eye.
 - *Smart televisions:* Use apps and can download and stream content from the internet.
 - *Stream:* Play a file on your computer while it is being downloaded from the internet, rather than saving it as a file and then playing it.

Home Entertainment Components

- **Sound systems:** Produce loud, rich sound using high-quality speakers and amplifiers. Can connect to personal devices via USB or Bluetooth.
 - *USB (Universal Serial Bus):* A standard for wired connectivity that can also supply electric power.

- *Bluetooth*: Wireless connectivity that allows devices to connect over short distances.
- **Personal video recorder (PVR)**: A device that records broadcasted content so that it can be watched at a later date.
- **Set-top box (STB)**: A device that sends video and audio received from a broadcaster to a television.
- **Blu-ray and DVD players**: Blu-ray players usually play DVD disks, but DVD players will not play Blu-ray disks.

Interactive & Navigation Electronics

- **Games console**: An electronic device that is used for playing computerised video games on a screen.
 - *Avatar*: A figure that represents a person in a computer game or other virtual environment.
 - *Handheld systems*: Mobile gaming versions with a built-in screen and less storage, requiring reduced image/sound quality and data complexity.
- **Media player**: An electronic device that can store and play digital music and videos and show digital photographs.
- **Navigation aids (Sat-Nav)**: Calculate the best route between two or more locations and provide updates.
 - *GPS*: The Global Positioning System that uses radio signals from satellites to show your exact position on the Earth on a special piece of equipment.
- **Smart home**: A home equipped with devices that can connect to the internet and be controlled remotely by a computer or smartphone app.

Device Relationships

- **Multifunctional device**: A device that can perform a range of different functions (e.g., a smartphone acting as a phone, camera, and navigation aid).
- **Convergence**: When the designs of devices change so that they become similar to one another.

Features of Digital Devices

Every teacher must ensure students can explain and analyze these nine core features:

- **Portability**: Directly related to size and weight; determines how easy a device is to carry and move around.
- **Performance**: The speed at which a device carries out instructions from its software.
 - *Processor*: One or more Central Processing Units (CPUs) that carry out software instructions.
 - *RAM*: Random Access Memory, which is the memory in a computer system that is used for running software.
 - *Virtual memory*: Storage used by the processor once the space in RAM has run out.

- **Storage:** Space where files and programs are held.
 - *File:* A collection of data that represents a document, image, database or similar.
 - *Program:* A set of instructions that are carried out by the processor.
- **User interface:** The system that allows a user to interact with a device.
- **Connectivity:** A device's ability to connect to networks and other devices.
- **Media support:** The ability of devices to read data from and write data to different types of media (e.g., memory cards, CDs, DVDs).
- **Energy consumption:** The requirement of electricity to work; lower consumption extends mobile battery life and reduces environmental impacts.
- **Expansion capability:** The ability to install additional internal components or connect peripheral devices via slots or ports (like USB).
- **Security features:** Automated options to keep private and valuable user data safe from unauthorised access.
 - *Lock:* To make a device inactive so that it requires a password or PIN to reactivate it.
 - *Wipe:* To delete data from a device or drive.

Comprehensive Breakdown of User Interfaces (UI)

1. **Command Line Interface (CLI):** Users enter text instructions precisely. It requires little memory and is ideal for limited storage.
2. **Menu-Driven Interface:** Displays a list of options as a menu. It is easier to use than a CLI but takes more time to navigate. Used in ATMs, televisions, and older mobile phones.
3. **Graphical User Interface (GUI):** Visually controlled by a pointer on the screen using **windows, icons, and menus** (WIMP). It uses more memory and storage.
4. **Voice Interface:** Allows users to give spoken commands using voice recognition software. Libraries of words are often stored online to save local storage, requiring internet access.
5. **Gesture Interface:** Allows the user to control a touch-screen device by swiping fingers or pinching to zoom.

Peripheral Devices

- **Input peripherals:** Devices that send data to the computer, allowing user control or data capture from outside the system.
- **Output peripherals:** Connected to a computer and output the results of processing in various human-understandable forms.

Input Peripherals

- **Keyboards:** Use keys to input text or interact with software.
 - *Shortcut:* A combination of key presses used to access common software commands quickly.

- **Pointing Devices:**
 - *Mouse*: Uses an optical sensor to recognize movement.
 - *Tracker ball*: A ball rolled directly by hand to steer a pointer; features no movable casing buttons.
 - *Track pad*: Surface that senses finger movements, touches, and presses to save space on laptops.
- **Joystick**: Commonly used on games controllers to move sprites.
 - *Sprites*: On-screen graphics in a computer game, used to represent characters, vehicles and objects.
- **Graphics tablet**: A flat pad used with a **stylus** (a pen-shaped device) to draw or write.
- **Scanners**: Use light sensors to record physical documents as images.
 - *OCR (Optical Character Recognition)*: Software that reads characters on a scanned document and stores the result in a text file.
 - *OMR (Optical Mark Recognition)*: Software used with scanners to detect simple marks, commonly used for recording responses to multiple-choice tests.
- **Barcode scanner**: Measures gaps and line widths to read patterns representing numbers or letters.
 - *Linear barcodes*: One-dimensional tracking lines.
 - *Matrix codes (QR codes)*: Two-dimensional tracking squares that hold more information and scan from any angle.
- **Webcam**: Specialized lower-quality cameras used for security, communication, or web streaming.
- **Microphone**: Captures sound waves using a flexible **diaphragm** that moves to produce an analogue electrical signal, which is converted to digital by a sound card.
- **Touch Screens**:
 - *Resistive*: Pressure-based screen where two underlying layers touch. It is durable but harder to read and lacks multi-touch capabilities.
 - *Capacitive*: Features a layer of charged material under glass. When a finger touches it, a small amount of charge flows away because humans are conductive, allowing precise multi-touch calculation.
- **Biometric scanner**: Measures unique physical human characteristics multiple times to perfection-match against an authorized sample. Includes *fingerprint recognition*, *facial recognition*, *voice recognition*, and *iris recognition* (which is 120 times more detailed and reliable than a fingerprint).
- **Card readers**: Read magnetic stripes (low security), programmable chips (high security Chip and PIN), or short-range wireless radio tags via **RFID** and **NFC**.
- **Sensor**: An input peripheral that inputs data about the physical environment automatically without human action (e.g., light, moisture, temperature, proximity, motion, pressure).

Output Peripherals

- **Monitor**: Electronic screen displays evaluated by diagonal size, pixel resolution, and energy standby efficiency.
- **Printers**:

- *Dot matrix*: An impact printer that prints noisily by striking paper through an ink ribbon; supports multi-layered carbon copies.
 - *Inkjet*: Sprays tiny dots from cartridges containing cyan, magenta, yellow, and black ink onto paper.
 - *Laser*: Sweeps a laser to discharge specific points on a charged rotating drum, attracting dry powder **toner** ink which is heated permanently onto paper.
 - *3D printer*: Builds real-life versions of digital models by adding sequential layers of heated material from bottom to top.
 - **Plotter**: Moves a physical pen across expansive sheets of paper to draw high-quality vector images created in Computer Aided Design (CAD) software.
 - **Data projector**: Projects magnified image outputs onto external projection screens using high-intensity light bulbs evaluated in lumens.
 - **Speaker**: Translates digital signals into physical sound outputs. Configured as single speakers for *mono sound* feedback alerts, pairs for *stereo sound* (left and right), or configurations for *surround sound* to provide environmental spatial awareness.
 - **Control device (Actuators)**: Mechanical or physical components of a system that make something happen in the real physical world when sent a command (e.g., valves, pistons, heaters, coolers, and motors).
-

Secondary Storage (Storage)

Key Storage Foundations

- **Secondary storage**: Non-volatile storage used to preserve data, documents, and programs securely when a system's power is turned off.
 - *Volatile*: Likely to change, especially quickly or unpredictably (e.g., RAM losing data when unpowered).
- **Device vs. Media Difference**: A storage device contains the storage media and accesses the data stored on it.
 - *Internal vs. External*: Internal devices connect directly to the computer's motherboard (the main internal circuit), while external devices attach via exterior ports.
 - *Backwards compatibility*: The ability to be used with older technology without having to be specially adapted.

Media Storage Formats

1. **Hard disks (HDDs)**: Storage structures containing concentric circular plates called **platters** that spin on a central spindle. A mechanical **read/write head** moves on an arm across tracks.
 - *Seek time*: The time it takes for a read/write head to locate the area on the disk where the data to be read is stored.

2. **Optical media:** Uses physical plastic layers under protection coatings to save data. Lasers read and write data to these discs.
 - *CD (Compact Disc):* Stores up to 700 MB.
 - *DVD:* Uses shorter wavelength red laser light to pack data tighter, storing 4.7 GB up to 18 GB.
 - *Blu-ray:* Uses a violet laser to store data at greater density, reaching capacities up to 50 GB.
3. **Flash media (Solid-State):** Energy-efficient media containing zero moving parts. It stores data as electrical charges held in tiny individual cells, making it highly robust for portable tracking.
4. **Magnetic tape:** Linear plastic ribbons coated with magnetic particles primarily used for large-scale whole-system backing up and archiving in enterprise environments.

Media Characteristics Comparison Table

Directly sourced from Table 1.4 of the textbook materials:

Data Access Speed	Maximum Capacity	Cost per GB	Primary Intended Use	Portability
Flash Media: Fastest	Varies, up to 128 GB	Very High	Laptops, mobile devices	Yes
Hard Disk: Fast	Varies, up to many TB	High	Servers, personal computers, backups	Not suitable
Optical Media: Slow	CD 700 MB, DVD 18 GB, Blu-ray 50 GB	Medium	Multimedia (music, games, films), file backups	Not suitable
Magnetic Tape: Slowest	Varies, up to 185 TB	Low	Whole system backups and archives	Not suitable

Units of Storage Capacity

Storage devices preserve all software tracks in binary form (using 1s and 0s). Scaling calculations apply base-1000 standard multipliers for International GCSE ICT criteria ($1\text{ KB} = 1000\text{ bytes}$):

- **Bit (binary digit):** A single unit of information with a value of either 0 or 1.
- **Byte:** A group of exactly eight bits.
- **Kilobyte (1 KB):** 1,000 bytes.
- **Megabyte (1 MB):** 1,000,000 bytes (1000 KB).
- **Gigabyte (1 GB):** 1,000,000,000 bytes (1000 MB).
- **Terabyte (1 TB):** 1,000,000,000,000 bytes (1000 GB).

SOFTWARE

Software Classification

Software is simply a collection of instructions executed by a computer's processor. It is broadly split into two distinct categories:

- **System Software:** Programs that manage, maintain, or operate the computer hardware itself (e.g., Operating Systems and Utilities).
- **Application Software:** Programs that allow you to do specific user tasks, like creating a document or playing a game.

Hardware Context

- **Single-Purpose Computers:** Built to run only one piece of software (e.g., a microwave controller).
 - **General-Purpose Computers:** Built to run a vast range of different software applications (e.g., tablets, smartphones, and laptops).
-

Utility Software

Utility software is a type of system software that carries out specific configuration, maintenance, and optimization tasks.

1. Backup Utilities

- Creates a duplicate copy of files and applications.
- **Exam Point:** They are usually scheduled to run automatically at off-peak times (like late at night) so they do not slow the system down while a user is working.

2. Defragmentation Utilities

- **The Problem:** Over time, files stored on a mechanical hard disk drive (HDD) become **fragmented**—broken into pieces and scattered out of order across the drive. The read/write head has to search multiple spots, which slows down file loading times.
- **The Solution:** Defragmentation utilities reorder these data blocks so that pieces of the same file are stored right next to each other. This speeds up file access time.

3. Compression Utilities

- Reduces the physical storage size of files.

- **How it works:** It spots repeating data blocks. For example, instead of writing out data as eight **a**s, six **b**s, and two **c**s, an algorithm like **Run-Length Encoding (RLE)** compresses it to a smaller notation: **a8b6c2**.
- **Decompressing:** Compressed files cannot be read directly by their original programs, so utility software must decompress (extract) them first.

4. Disk Formatting Utilities

- Prepares a storage drive (like a USB stick or hard drive) for its very first use by laying down a file system framework.
- **Warning:** Formatting an already used drive marks all existing data as unreadable. While formatting doesn't completely erase data from existence immediately (specialist recovery tools can sometimes find it), it makes the space available to be overwritten.

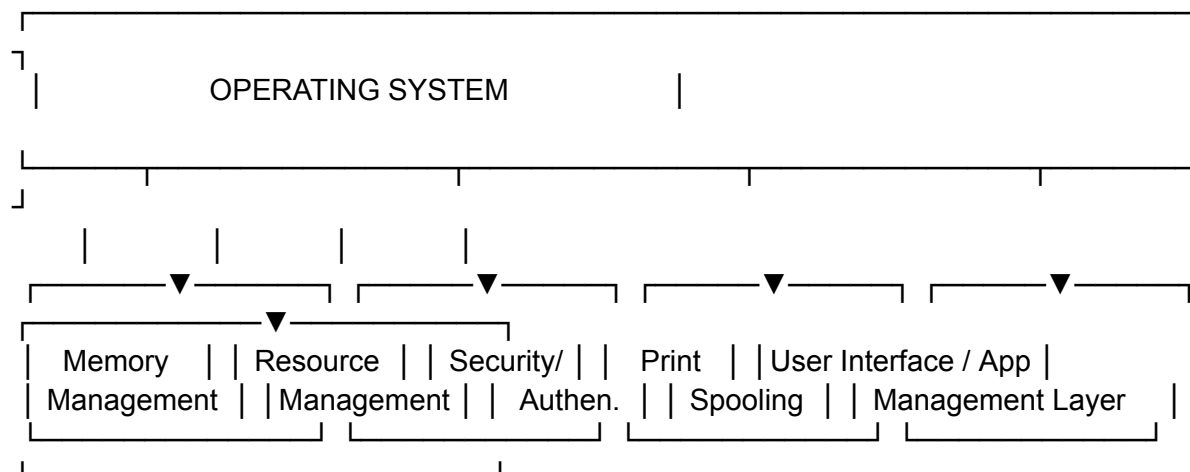
Operating Systems (OS)

The Operating System is the primary software layer that lets you control and manage the computer's underlying hardware.

Core Variations

- **Single-User OS:** Only supports a single general user profile. It cannot be customized for individual accounts and is common in consumer household appliances.
- **Network OS:** Manages multi-user login authentication, separates files so users cannot peek into each other's accounts, and coordinates access to shared network folders and network printers.

The 5 Essential Functions of an OS



1. **Memory Management:** Allocates the precise amount of temporary storage (RAM) an application needs to run. When you close the app, the OS reclaims that space for other programs.
2. **Resource Management:** Allocates processor (CPU) attention and internal hardware (like a graphics card) to various software programs. If two apps demand a printer at once, the OS queues the requests safely.
3. **Security (Authentication):** Controls machine access via usernames, passwords, biometrics (fingerprints), or smart cards. It also manages software firewalls to block unauthorized external network connections.
4. **Print Spooling:** Computers send documents to a printer much faster than the physical printer hardware can ink the pages. The OS puts these pages into an orderly queue (a buffer) so your computer doesn't freeze waiting for the printer to finish.
5. **Application Management & Interface:** Coordinates how application software requests talk to the physical hardware.

Application Software Classification

 **EXAM TIP:** When answering exam questions, always refer to an application by its generic **software type** (e.g., *spreadsheet software*) rather than using specific commercial brand names (e.g., *Microsoft Excel*).

Generic Software Type	Primary Focus & Core Features	Examples Mentioned
Word Processing	Documents that are primarily text-based. Includes spelling/grammar checkers, change tracking, and simple layout formatting.	LibreOffice Writer
Desktop Publishing (DTP)	Built for complex, precise page layouts (e.g., magazines or multi-page books).	Adobe InDesign
Spreadsheets	Automatic calculations using formulas and functions. Used for financial modeling and exploring "what-if" scenarios.	Microsoft Excel
Databases (DBMS)	Structured data storage. Features data entry forms, query editors for specific lookups, and clean report builders.	Oracle MySQL
Web Authoring	Creating web pages that export cleanly into HTML code. Often uses WYSIWYG ("What You See Is What You Get") visual environments.	Adobe Dreamweaver

Graphics Editors	Creating and adjusting Vector graphics (images made mathematically of shapes, points, and lines).	Serif Affinity Designer
Photo Editors	Modifying Bitmap images (digital photos made of grids of pixels). Features include cropping, red-eye removal, and brightness tuning.	GIMP
Sound Editors	Cutting/joining audio, adjusting track volumes, applying noise reduction, or altering the file's tempo (playback speed).	Audacity
Presentation Software	Combining text, images, video, and animation into sequential pathways or slides to support public speaking.	Prezi
Control Software	Reads inputs from physical electronic sensors, processes a logic choice, and sends a command out to physical mechanical actuators/motors.	Lift doors, robotic arms, smart heating
Project Management	Arranging dependent multi-stage tasks into timelines (Gantt charts) to pinpoint milestones and locate the Critical Path .	Gantt charting software
Communication Software	Providing remote network connection entry points to swap data. Includes Web Browsers, Email Clients/Webmail, SMS, MMS, and Instant Messaging.	Web browsers, TweetDeck

Software Licensing & Patches

Software Licences

A software licence is a legal contract giving an individual permission to install and activate an application on a computer processor. Using unlicensed software is a legal violation known as software piracy.

- **Free or Open-Source Software:** Grants users the legal freedom to copy, study, alter, and share the program. The underlying text instructions—the **Source Code**—is completely visible to the public.
 - *Note:* "Free" means free from restrictions on use, not necessarily free of cost. Authors can technically charge a fee to distribute it.
- **Proprietary Software:** Owned and copyrighted by an entity under a specific brand identity. The source code is locked away and kept secret. If a proprietary program is given away entirely for free, it is called **Freeware**.

Why Software Updates Are Vital

Developers push update patches through internet servers for five main reasons:

1. Fixing critical security exploits and code vulnerabilities.
2. Eliminating runtime system errors (bugs).
3. Ensuring compatibility with newly launched Operating Systems.
4. Improving computing performance and code execution efficiency.
5. Adding useful new tools or improving overall usability.

 **System Rule:** Always make a full data backup *before* running software updates in case an unexpected conflict causes software dependencies to break.

CHAPTER QUESTIONS

- 1 State what is meant by the term 'application software'. (1)
- 2 State three types of utility software. (3)
- 3 Describe the role of the operating system. (2)
- 4 Explain why utility software allows the user to schedule backup processes. (2)
- 5 Explain the benefit of using compression utilities to archive files. (2)
- 6 Describe how presentation software can help a presenter to give a talk. (2)
- 7 Describe one use of control software in a relevant industry. (2)
- 8 Discuss why a building company might use project management software. (8)
- 9 State three types of service provided by communication software. (3)
- 10 State two reasons for updating software. (2)

Memory & Processors

Primary Memory vs. Secondary Storage

Computers use different areas to hold data depending on how fast the processor needs to access it.

- **Main Memory (Primary Storage):** High-speed memory that the processor can access directly and quickly. It holds data and instructions currently being processed (e.g., RAM and ROM).
- **Secondary Storage:** Slower, long-term storage where your files, games, and operating system live when not in use (e.g., Hard Disks and SSDs). Programs must be copied into main memory before they can run.

Random Access Memory (RAM)

RAM stands for **Random Access Memory**. When you launch an app or open a file, it is loaded into RAM from secondary storage so the processor can execute it.

- **Volatile:** RAM cannot store data without electrical power. Turning off the computer completely erases everything inside RAM.
- **Capacity Impact:** The more RAM you have, the more programs and large files you can run smoothly at the exact same time.

The Danger of Virtual Memory

When your computer runs completely out of physical RAM space, the Operating System creates **Virtual Memory** by using an area of your secondary storage.

- Because secondary storage is significantly slower than RAM, your system performance will drop noticeably.
- If your system uses a Solid State Drive (SSD), constantly swapping files back and forth between RAM and Virtual Memory can drastically reduce the lifespan and performance of that SSD.

Read Only Memory (ROM)

ROM stands for **Read Only Memory**. It is designed to store data permanently and securely.

- **Non-Volatile:** Data is safe and is **not lost** when the computer power is switched off.
- **Uses:**
 - *Single-Purpose Devices:* Stores the only program needed to run appliances like calculators, digital watches, or washing machines.

- *General-Purpose Computers*: Stores the **boot** (start-up) program that tells the computer how to turn on its hardware components and load the main Operating System.

The 4 Types of ROM

1. **Mask Programmed ROM**: Standard ROM that cannot be edited or changed after it leaves the factory floor.
2. **PROM (Programmable ROM)**: Manufactured blank, allowing a device creator to write data to it exactly **once**.
3. **EPROM (Erasable Programmable ROM)**: The contents can be completely erased, but only by exposing the chip to strong **Ultraviolet (UV) light**.
4. **EEPROM (Electrically Erasable Programmable ROM)**: Can be erased and rewritten electronically by applying a voltage to a specific pin. This allows **firmware** (software that controls hardware) to be updated without taking the chip out of the machine.

Flash Memory

Flash memory is a highly advanced type of **EEPROM**. Like ROM, it is entirely non-volatile.

- **Key Features**: It has absolutely no moving parts, boasts incredibly fast access times, and has very low power consumption.
- **Common Uses**: Because it saves battery power, it is ideal for SSDs inside portable laptops and smartphones. It is also the core technology inside USB thumb drives and camera SD cards.

Summary Checklist: RAM vs. ROM

Feature	Random Access Memory (RAM)	Read Only Memory (ROM)
Saves data when power is cut?	✗ No (Volatile)	Yes (Non-volatile)
Can swap instructions in/out?	Yes	✗ No
What does it store?	Temporary data in current use	Permanent start-up data/firmware
Can the user physically upgrade it?	Yes (Adding more sticks)	✗ No

Processors (CPUs)

The processor is built from one or more **Central Processing Units (CPUs)** that execute software instructions.

- **The Processor Cycle:** The loop of **fetching** an instruction from memory, **decoding** what actions are required, and **executing** those tasks.
- **Cores:** Each individual CPU inside a processor is called a **core**. A quad-core processor has four cores working at once, meaning it can handle up to four times as much work as a single-core chip.
- **The Energy Advantage:** Multi-core processors allow individual cores to run slower. This uses less power, generates less heat, keeps laptops quieter by turning down fans, and extends mobile battery life.

Measuring Processor Speed

Processor speed is measured by its clock speed—the number of **clock cycles per second** it can carry out. It is measured in Hertz (Hz):

- **Hertz (Hz):** 1 cycle per second.
- **Kilohertz (kHz):** 1,000 cycles per second.
- **Megahertz (MHz):** 1,000,000 cycles per second.
- **Gigahertz (GHz):** 1,000,000,000 cycles per second.

 **Exam Trap:** A higher Gigahertz rating doesn't *always* mean a computer finishes a task first. Different processor designs allow some CPUs to complete significantly more work/instructions during a single clock cycle than others.

CHAPTER QUESTIONS

1. State what is meant by 'ROM'. (1)
2. State what is meant by 'RAM'. (1)
3. Explain why increasing the amount of RAM in a system improves the user experience. (3)
4. Describe what virtual memory is and why it is used. (2)
5. List three characteristics of RAM that make it different to ROM. (3)
6. Explain two reasons why the characteristics of flash memory make it suitable for use as secondary storage in smartphones. (4)
7. Which one of these describes the term 'clock cycle'? (1)
 - a. The number of times per second that a processor can carry out instructions.
 - b. The number of instructions that a processor can carry out each second.
 - c. The number of times that a processor can load instructions for processing.
 - d. The number of times that each process repeats in a processor.

Digital Communication

Core Data Transmission Concepts

The speed and user experience of any digital connection rely heavily on two core physical factors:

Bandwidth

- **Definition:** The number of bits that can be carried by a network connection in exactly one second.
- **Analogy:** Think of it as a water pipe; a larger diameter pipe transfers more water volume at once.
- **User Experience Impact:** Higher bandwidth accelerates file uploads/downloads and enables high-data real-time actions like multiplayer online gaming or HD video streaming.

Latency

- **Definition:** The amount of delay time it takes to send data package units between digital devices.
- **Measurement Technique:** Determined by "pinging" a web server domain name to get the round-trip return time.
- **User Experience Impact:** High latency introduces visible "lag" in online games (delayed actions) and causes a noticeable delay in live video playback feeds.

Video Streaming & Buffering

- **Buffer:** A temporary memory area holding video data so full downloads aren't required prior to playback launch.
- **Mechanism:** Continuous background downloads must fill the buffer faster than playback empties it to prevent pausing lag.

Factors Restricting Data Transfer Speed

Many environmental and physical factors disrupt or reduce available connection bandwidth:

- **Transfer Method:** Copper cable lines carry more physical wave frequencies than wireless bands, delivering higher absolute bandwidth.
- **Interference:** External electromagnetic signals from consumer appliances (microwaves, fridges) degrade signals.
- **Blockages:** Physical building walls and heavy furniture reduce wireless radio signal strength.

- **Distance:** Signal strength naturally degrades as the physical distance between data devices expands.

*Note: Cabled wires use thin metal layers called **shielding** to block electromagnetic signal interference.*

Four Major Network Classifications

When two or more computer devices connect together, they establish a network:

Network Type	Geographical Range	Primary Characteristics & Contexts
LAN (Local Area Network)	Small, restricted space (e.g., individual school, home, or office building).	High-speed local hardware data sharing.
WAN (Wide Area Network)	Large global/national space.	Links disparate corporate/government LANs; the Internet is the ultimate WAN.
PAN (Personal Area Network)	Immediately surrounding an individual person.	Syncs everyday personal accessories like a smartwatch, phone, and laptop.
Tethering	Short-range localized bridge. To share mobile data	Turns a mobile device into a host broadband modem for surrounding devices to use.

WAYS IN WHICH DIGITAL DEVICES COMMUNICATE

1. Satellite Communication

- Satellites transmit data to and receive data from digital devices.
- Digital devices use antennas to receive the radio signals that satellites transmit.
- **Benefits:** The number of satellites means that the system is always available. It also cannot be affected by power shortages.
- **Drawbacks:** Satellite signals do not pass through solid objects. This means that they will not work in areas with tall buildings or in tunnels. Signals can also be affected by atmospheric weather conditions such as heavy snow or rain.

Satellite Applications | Use

- **GPS:** Signals are sent from a network of 24 satellites orbiting the Earth. A view of only four satellites is required to calculate an accurate location.
- **Television (DVB-S):** Digital Video Broadcasting - Satellite transmits a video signal using a large antenna on Earth to one or more satellites, which then broadcasts the

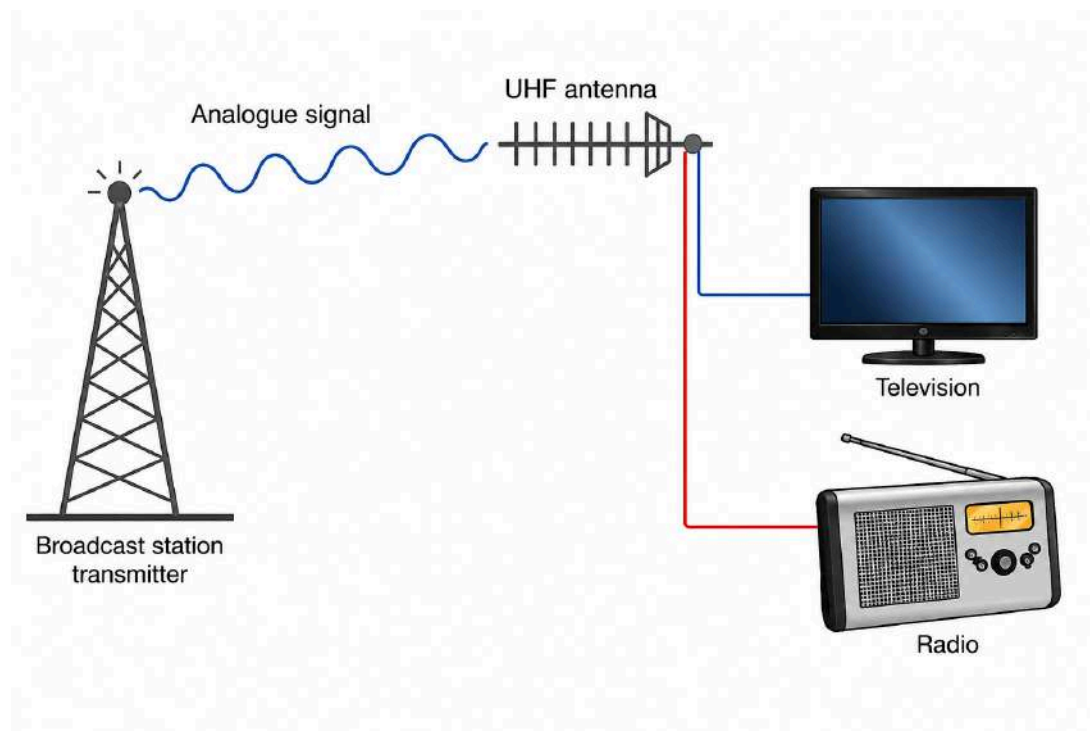
signal back down to Earth. A satellite television viewer requires an antenna and a set-top box (or an installed decoder) to decode the signal.

- **Telephone:** Used to allow people in remote areas to place voice calls using satellite telephones.
- **Military:** Used for communication systems, such as the Global Command and Control System.

2. Broadcast Communication

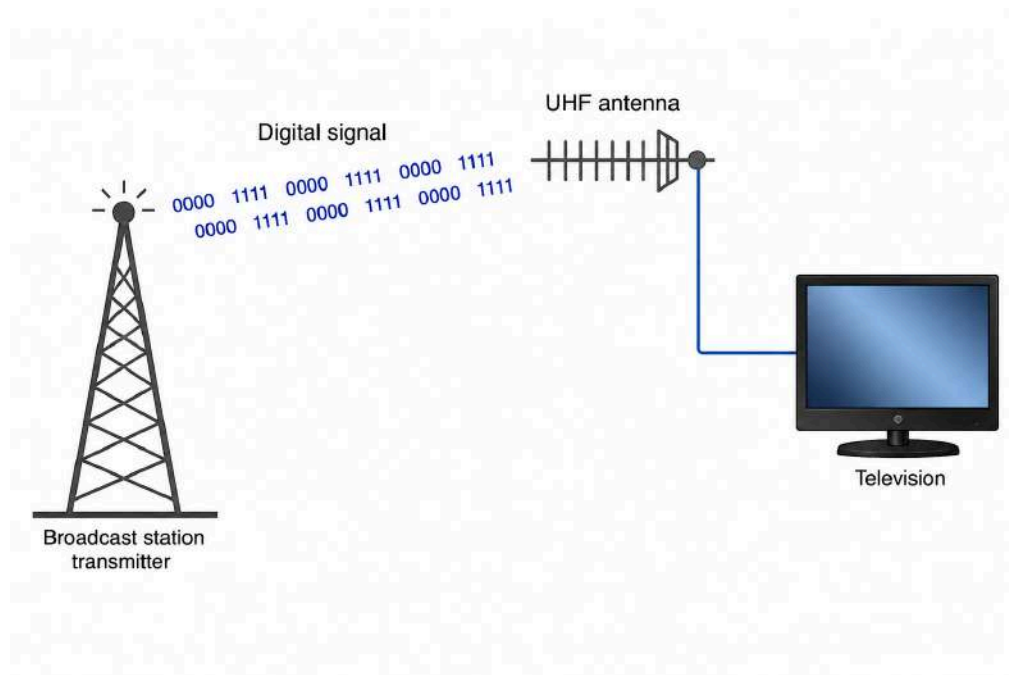
Analogue Television and Radio

- Transmitters broadcast television and radio signals that are received by a viewer's antenna.
- This antenna sends a signal through a wire to the television or radio receiver, which converts it into images and audio.



Digital Television (DVB-T)

- **DVB-T (Digital Video Broadcasting - Terrestrial)** is a method of DVB where the transmitters are based on Earth, rather than in orbit as they are in DVB-S.
- To receive digital television broadcasts transmitted by DVB-T, viewers can use the same antenna that they use to receive analogue broadcasts. They do not need a special antenna.
- **DVB-T2** is a newer standard that provides more functionality, such as High Definition Television (HDTV) and interactive services.



Digital Radio (DAB)

- **DAB (Digital Audio Broadcasting)** is broadcast in the same way as DVB.
- DAB provides more radio stations and can also carry text data (such as the time, name of the station, and details of the music being played) that DAB receivers can display.

3. Wired Communication

COMMON WIRED CONNECTION TYPES AND THEIR USES		
Connection Type	Visual Reference	Primary Use Case
HDMI Digital video connections		 Transmits digital video and audio between devices, such as a computer and monitor/TV.
S/PDIF Digital audio connections		 Transmits high-quality digital audio.
Minijack (3.5 mm) Personal headphones and audio		 Connects personal headphones, speakers, or microphones.
USB Storage and data transfer		 Transfers data and connects peripherals such as storage devices, keyboards, and mice.
Ethernet Networking connection		 Connects devices to a wired network or the internet.

4. Wireless Communication Standards

Wi-Fi

- Connects local devices to an overall building access point network.
- Follows the standardized **IEEE 802.11** protocol specification rules.

Bluetooth

- Built strictly for short-range personal device connections.
- Demands explicit device **pairing** and offers less bandwidth than Wi-Fi systems.

Feature / Criteria	WI-FI	BLUETOOTH
Range	Long ✓	Short ✗
Bandwidth	High ✓	Low ✗
Power	High ✗	Low ✓
Security	High ✓	Low ✗
Can connect multiple devices simultaneously	Yes ✓	Limited ✗ (usually have to be paired)

3G & 4G (Cellular Networks)

- Often termed **mobile broadband**, managed directly by mobile phone carriers.
- Supplies internet paths when external local Wi-Fi nodes are completely out of reach.

Infra-red (IR)

- Extremely short-range.
- Requires straight line-of-sight.
- Can be blocked by solid walls and sunlight.
- Extensively chosen for simple home appliances like television remote control units.

Near-Field Communication (NFC)

- Short-proximity system using local **RFID** chips.
- Powers contactless payment bank cards, travel cards, and smartphone payment wallets.

Direct Comparison: Wired vs. Wireless

Feature	Wired Connections (Ethernet/USB)	Wireless Connections (Wi-Fi/Bluetooth)
Speed	Highly optimal, much faster.	Generally slower than cable.
Stability	Highly reliable; minimal interference.	Easily disrupted by building walls/appliances.
Security	Highly secure; requires direct physical access.	Less secure; radio signals can be intercepted.
Portability	Fixed space; constrained strictly by cable length.	Fully mobile anywhere within active signal range.
Mess & Safety	Cable clutter; physical hazard for tripping.	Exceptionally clean and completely tidy workspace.

CHAPTER QUESTIONS

1. Explain one reason why some games console controllers use Bluetooth rather than infra-red to connect to the console. (1)
2. State one way in which using infra-red in games console controllers could affect the experience of the person playing the game. (1)
3. Describe how video streaming works. (3)
4. Explain why streaming is more convenient for the user than downloading. (3)
5. State three types of wired connection. (3)
6. Which one of these best describes the internet? (1)
 - i. LAN
 - ii. WAN
 - iii. PAN
 - iv. VLAN
7. Explain why global games companies use games servers in multiple countries to ensure that the experience of users is not negatively affected when playing online games. (3)

NETWORKS

A network is created when two or more computers are connected together. Using a network, a computer can communicate with others and share resources such as hardware, software and data. It does this by sending data in a packet, which is a unit of data packaged together so that it can travel across a network.

In order to connect to a network, computers need to fulfil certain requirements so that they all operate using standard protocols. Protocols are Rules that allow the exchange and transmission of data between devices.

NETWORK OPERATING SYSTEMS

A network operating system allows a computer to communicate on a network. It provides additional functionality to a stand-alone operating system, including:

- Passing usernames and passwords to a server for checking when a user logs in.
 - Separating user accounts and ensuring that users cannot access each other's files.
 - Providing access to network storage and shared resources such as networked printers.
-

HOW DEVICES ARE IDENTIFIED ON A NETWORK

There are three methods used to identify devices on a network:

1. Internet Protocol (IP)
2. MAC address
3. Device name

Internet Protocol (IP)

An IP address is a unique address that networked devices use to send data to each other. Each piece of data that is sent across a network carries the IP address of the destination, so that each device in the network knows where to send it.

- **IPv4:** Uses four groups of up to three numbers separated by full stops (for example, `192.168.1.1`). IPv4 can store over 4 billion addresses.
- **IPv6:** Uses eight groups of four hexadecimal numbers separated by colons (for example, `2001:0db8:0000:0042:0000:8a2e:0370:7334`).
- **Allocation:** IP addresses can either be assigned by a network administrator or allocated dynamically by a server running Dynamic Host Configuration Protocol (DHCP).

- **Hexadecimal:** A base-16 number system that uses the numbers 0–9 and the letters A–F.
- **Network administrator:** A person who manages an organisation's network.
- **Dynamically:** In a way that is open to change.
- **Dynamic Host Configuration Protocol (DHCP) server:** A networked computer that automatically assigns an IP address to other computers when they join the network.

MAC Address

Unlike IP addresses, which can be dynamically allocated by users or servers, media access control (MAC) addresses are universally unique identifiers given to the network interface card (NIC). MAC addresses are used in devices connected to local area networks (LANs) using Ethernet, Bluetooth or Wi-Fi. MAC addresses are assigned by the NIC manufacturer and are generally considered to be fixed addresses.

A MAC address is usually made up of six pairs of characters. The first three pairs identify the manufacturer and the remaining pairs are assigned by the manufacturer to uniquely identify the device.

Uses of MAC Addresses

- **Restricting or allowing access to a network:** MAC address filtering checks the MAC address of devices attempting to gain access to a network and only grants access to devices with specified MAC addresses.
- **Identifying a device on a network:** Some Wi-Fi hotspots only provide free access for a certain length of time, and they identify a device using its MAC address in order to work out whether it is trying to access the hotspot for longer than the permitted time.
- **Tracking a device:** Some companies and organisations track devices (and therefore their users) by checking which wireless access points have been accessed by specific MAC addresses.
- **Assigning 'static' or 'fixed' IP addresses:** Each time a device connects to a network, it is identified by a DHCP server (usually using its MAC) and given the same IP address as before.

Vocabulary

- **Universally unique:** The only one of its kind in the world.
- **Identifier:** A group of letters, numbers, or symbols that a computer has been programmed to recognise and use to process information.
- **Hotspot:** A place in a public building where there is a computer system with an access point, which allows people in the building with a wireless computer or Bluetooth mobile phone to connect to a service such as the internet.

Device Name

A device name is a descriptive name that helps users to identify computers on a network. Device names are not used by computers to communicate with each other as they are not always unique. This means that they could cause conflicts if data was sent to more than one device with the same name for processing. You can change a device name using tools in the device's operating system.

COMPONENTS OF WIRED AND WIRELESS SYSTEMS

Cables

- **Cat5e cables:** Used in homes and small businesses for Ethernet connections. Able to transfer data at 10 Mbit/s, 100 Mbit/s or 1 Gbit/s. Connects devices through their NICs.
- **Cat6 cables:** Used to transfer data at 10 Gbit/s. These are more expensive than Cat5e cables and are usually only used by businesses.
- **Fibre optic cables:** Flexible fibres containing a glass thread that bounces light signals between two devices faster and further than is possible with wire cables. Can carry data at 40 Gbit/s over many kilometres without affecting signal quality. They are expensive and used by telecommunications companies, science/engineering laboratories, hospitals, banks, schools and universities.

Wireless Access Point (WAP)

A wireless access point allows devices with Wi-Fi connectivity to connect to a wired network. They are often built into other hardware, such as routers, but they are also available as stand-alone devices that connect to a wired network using Ethernet cables.

Switch

A switch connects devices on a network. It has ports, each of which can be connected to a device using a cable. Connecting a wireless access point to a switch via a cable gives wireless devices access to the wired network. The switch makes sure that data sent from any device gets to the correct device on the network.

Gateway

A gateway connects two different types of network. For example, a LAN is connected to a WAN using a gateway.

Router

A router stores the addresses of all devices that are connected to it so that it can forward network traffic to its destination using the quickest route. Most routers used in homes include

a switch and a wireless access point. They also act as gateways, connecting the LAN to the internet, which is a type of WAN. Home routers are dynamically allocated an IP address by the internet service provider (ISP).

Booster

A booster is used to amplify the signal in a network so that its range can be extended. For homes and offices, wired Ethernet connections often have a maximum range of 100 m. Wireless signals have limited range, too. Boosters can be used for both wired and wireless connections. Wireless access points can be set to repeater mode in order to act as boosters for Wi-Fi signals.

SERVERS

A server is a computer that shares its resources with connected devices. Computers connected to a server are known as clients. A single computer can be used to perform multiple server functions, though performing multiple roles will affect its performance.

Server Types & Functions

- **Authentication Server:** Checks usernames and passwords. When a user successfully logs in, the client receives an electronic certificate that it can then use to access various resources, including applications and storage.
- **Print Server:** Manages multiple printers at a time, dealing with print requests from client computers and adding jobs to a queue so that individual printers are not overloaded with requests. They can also monitor and process print requests to invoice users or departments.
- **File Server:** Allows users to access shared and private storage.
- **Application Server:** Provides clients with access to applications that can be run directly from the server.
- **Web Server:** Processes requests for data made via Hypertext Transfer Protocol (HTTP). Together, all of the content stored on all web servers is known as the World Wide Web

Vocabulary

- **Client:** A computer connected to a server.
- **Hypertext Transfer Protocol (HTTP):** A set of standards that control how computer documents that are written in HTML connect to each other.

CONNECTING TO AND USING THE INTERNET

Internet Service Provider (ISP)

To connect to the internet, users need to subscribe to an ISP. ISPs provide connections to the telecommunications infrastructure that forms the framework for the internet. ISPs provide access via mobile telephone networks and landline telephone networks. Commercial ISPs charge subscription fees for access to the internet. Some ISPs provide free access as part of community schemes.

Web Browser

A web browser is a type of software application used to request and display information stored on web servers. Examples of web browsers are Mozilla Firefox, Google Chrome, Internet Explorer or Microsoft Edge, Opera, and Safari.

Search Engine

A search engine provides users with a way to find information in web pages stored on web servers. Users enter keywords that describe the information they want to find. The search engine then compares the keywords with those in its database of web pages and returns the results that are the closest match to the given keywords.

Hint: Many people confuse search engines with web browsers. A search engine allows users to search and find information using keywords. A web browser is an application that users can use to view online content. Users can use a web browser to access a search engine.

Filter Software

Filter software prevents users from accessing inappropriate information. When a user tries to access a web page, the address (URL) and/or the contents of the web page are compared against two lists of URLs and keywords stored in the filter software's database: the blacklist and the whitelist.

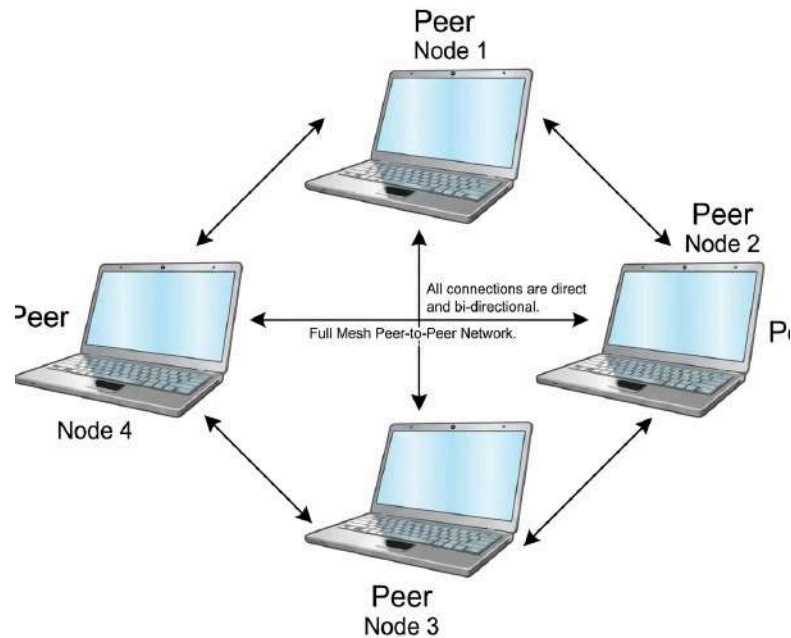
- If the results match anything in the **blacklist**, the user will be prevented from viewing the web page.
- If the result matches anything in the **whitelist**, then the user will be allowed to view the web page.
- If the result does not match anything in either list, the user will be allowed to view the information.
 - **URL (uniform resource locator):** A website address.
 - **Blacklist:** A list of unacceptable URLs.
 - **Whitelist:** A list of acceptable URLs.

LOCAL AREA NETWORKS (LANs)

A LAN is a network contained to a small area, such as a home or office network. Computers in a network can be connected using one of two different models:

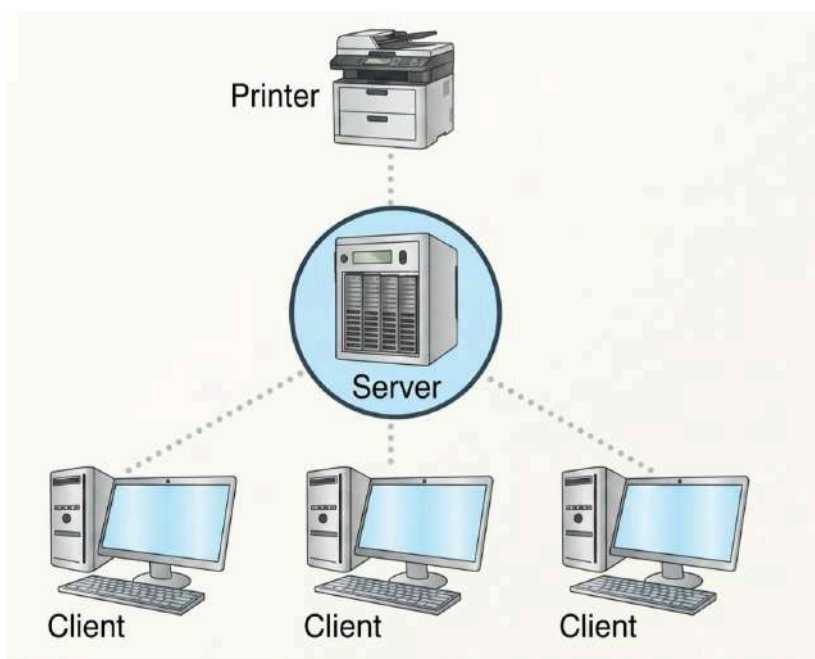
1. Peer-to-Peer Networks

Computers in a peer-to-peer network share their resources with other computers in the network, but they do not access servers.



2. Client-Server Networks

Some networks use servers. A network that uses servers and clients is called a client-server network.



Benefits of Using LANs

- Access to shared peripherals
- Access to shared storage and data
- Flexible access (being able to access peripherals, storage and data from any connected device)
- Media streaming (including movies, music and gaming)
- Communication (being able to send messages and files to others on the network)
- Shared access to the internet

Benefits of Using Client-Server Networks

- **Control of user access rights:** Users, or groups of users, can be given access to some resources (such as storage or printers) and restricted from accessing others.
- **Centralised administration:** Resources and user accounts can be managed by an individual, or individual group of servers and administrators. This ensures support is provided by people who have an overview of the network.
- **Centralised backup:** User data is protected from loss because backups can be automated for all users.
- **Shared software:** Application servers can provide access to shared software or operating systems.
- **Shared storage and file access:** Storage amounts can be managed centrally, allowing users to make files available to others with specific file permissions (read-only or read-and-write).
- **Roaming profiles:** This is the ability to log into any computer in an office and see your settings and files. This enables users to work from anywhere there is a client.

SECURING DATA ON NETWORKS

Security prevents unauthorised users from accessing network resources and data.

Logins and Passwords

Users log in to computers on a network to access centrally managed resources. Passwords are used to authenticate a user to the network. An authentication server can manage the authentication across a range of services so a user does not have to log in each time they access a resource.

- **Login:** User or account information, such as a user name or account name.
- **Authenticate:** Confirm that the user is who they say they are.

Firewalls

A firewall is used at the gateway to a network. It controls the network traffic to and from a network, particularly the traffic from the internet. Firewalls prevent unauthorised users from accessing network devices and resources, such as storage. Firewalls are available as hardware and software.

Encryption

Encryption is the process of encoding, scrambling or jumbling data so that unauthorised users are prevented from being able to understand it.

- **Symmetric Key Encryption:** Uses the same key at both ends of the process, meaning that the same key is used to encrypt and decrypt the data.
- **Public Key Encryption:** Uses two mathematically related keys called a key pair. One key (public key) is shared with other computers to encrypt the data, and a different, unshared key (private key) is used by the recipient to decrypt it.

Vocabulary

- **Cipher:** A code.

Wireless Protocols: WEP vs. WPA

It is easier to intercept data in a wireless network than in a wired network.

- **Wireless Encryption Protocol (WEP):** Used to secure the wireless transfer of data. It is the least secure wireless data encryption method because every device on the wireless network uses the same key for every transfer. Eavesdroppers studying enough packets can easily identify the key.
- **Wi-Fi Protected Access (WPA):** A security protocol designed to provide better encryption than WEP. WPA generates a new key for each device on the wireless network, and new keys are provided for each packet of data sent.
 - **Eavesdropper:** An unauthorised person or piece of software that intercepts data from a private communication.
 - **Packet:** A unit of data packaged to travel across a network.

Virtual Private Network (VPN)

A VPN provides access to a private LAN from a remote location. The connection to the LAN is created using the infrastructure of a public network like the internet. Data sent using a VPN is encrypted so that it is secure if it is intercepted. An individual might use a VPN to:

- Access their employer's network when working from home.
- Access computers in a different geographical location, perhaps to avoid local restrictions on access to web content (such as due to censorship or geolocation rights management).
- Make secure payments.
- Prevent surveillance of and access to their web activity.

Vocabulary

- **Geolocation rights management:** The use of a user's location to block access to services online.

File Access Rights, Logs, and Backups

- **File Access Rights:** Also known as file permissions. They can be set for individual files, folders or drives, and they ensure that users are either allowed to read only or allowed to read and write.
- **Transaction Logs:** All network activity can be recorded in a log file. A transaction log helps to identify which computers and network devices have been accessed, allowing administrators to identify unusual activity.
- **Backups:** A backup is a copy of one or more files. The backup or backups are usually stored on a different storage device to the original file.

Vocabulary

- **Log:** A record of events.
- **Transaction:** The exchange of data.

CHAPTER QUESTIONS

1. Which one of these connects a LAN to a WAN? (1)
 - A: Switch
 - B: Gateway
 - C: Modem
 - D: Server
2. Explain why public key encryption is more secure than symmetric key encryption. (2)
3. State which security device controls the traffic entering a network. (1)
4. State two methods by which devices are identified to each other on a network. (2)
5. Explain why IPv6 was used to replace IPv4. (2)
6. State two uses for a VPN. (2)
7. State the reason why a booster is used in a network. (1)
8. Describe how encryption secures data. (3)
9. List three methods of securing data on networks. (3)
10. State three benefits to users of using a local area network. (3)
11. Draw and label a network diagram for a home that includes: (8)
 - Two smartphones
 - One tablet device
 - Two desktop PCs
 - Internet access

RISKS TO DATA AND PERSONAL INFORMATION

Huge amounts of data are transmitted and stored digitally, and a lot of this data contains personal or financial information. Because of this, digital systems are targeted by criminals who try to access data so that they can use it to commit fraud or identity theft.

Threats to Data and Information

Unauthorised Access

Access to networks by users who are not permitted to access them is called **unauthorised access**.

Unauthorised users can attempt to gain access to networks directly by themselves. Alternatively, they may create software that runs thousands of times per second on devices, inputting multiple login details in order to attempt to gain access to networks with poor security.

Sometimes, devices on a network can be targeted by unauthorised users in order to be used as **botnets**. Botnets are groups of computers that have their resources used for harmful purposes, such as running and spreading malware.

Deliberate Damage by Malware

Malware (Malicious Software): Software that is created with the intention to do harm.

Malware can show messages, play sounds, delete files, or reprogram systems to perform tasks that will harm the system and the connected hardware.

Some malware, known as **ransomware**, threatens to delete a user's files or places restrictions on a user's access to software or resources until money is paid, usually to an anonymous account. These messages are usually very threatening and distressing for users. They are often written in a way that makes the user believe that they must pay quickly.

Accidental Deletion

Users can sometimes delete files or even the entire contents of a drive by mistake. This can happen if:

- They press a key on a keyboard by accident.
- They format media on the wrong storage device.
- Their device loses power unexpectedly.

Theft of Personal Data

Phishing

Phishing: The criminal activity of sending emails or having a website that is intended to trick someone into giving away personal information such as their bank account number or their computer password; this information is then used to get money or goods.

Phishing involves sending large numbers of messages that appear to be from real organisations, such as shops, banks, or charities. Phishing messages are often sent as emails. These emails ask the user to provide their information by replying to the message or following a hyperlink that opens a webpage into which the user is asked to type their personal details.

Sometimes, phishing messages are highly customised or personalised and are targeted at a smaller number of particular users. This technique is known as **spear phishing**. SMS phishing is sometimes referred to as **smishing**.

Pharming

Like phishing, **pharming** is a technique used by criminals to gain personal information and payment details from users. Criminals create fake versions of trusted websites to trick users into entering their login details, which are then used by the criminals to access users' accounts.

There are two main methods by which users are directed to a pharming site:

1. **DNS Alteration:** Internet traffic going to the real website is redirected to the fake website. Criminals do this by altering the domain name servers (DNS) to make internet traffic go to their fake site. They can also use malware to redirect web requests.
 2. **Typosquatting:** The URL of a pharming website is designed to be very similar to the URL of the real website. If a user misspells the URL when typing it into the address bar, they could go to the pharming site by mistake (e.g., visiting <http://moneybank.lk> instead of <http://moneybank.lk>).
- **Webform:** A data entry form on a web page.
 - **Internet Traffic:** Data transferred between computers connected to the internet.
 - **Domain Name Server:** A computer connected to the internet that translates domain names, such as pearson.com, into IP addresses.

Methods to Secure Data and Personal Information Online

1. Firewalls

Firewalls control the data travelling into and out of a network. They examine the network addresses and ports of the data, then compare those details to a list of rules that can be changed by network administrators. The list of rules determines what traffic should be allowed to travel into and out of the network. In this way, firewalls can prevent unauthorised access to a network and protect the network from malware.

2. Encryption

Encryption uses a key to scramble data into an unreadable form. If encrypted data is intercepted on the network, it is useless unless the interceptor has or can identify the decryption key.

3. Passwords, PINs, and Biometrics

These methods are used online to authenticate a user so that they can access an online system, such as webmail or an online bank account. When entering a password or a PIN, the characters are often **masked** (hidden) so that anyone watching the screen cannot see what is typed.

Students should ensure that their passwords follow these rules:

- More than eight characters long.
- A mix of letters, numbers, and symbols.
- A mix of uppercase and lowercase letters.
- Made up of random characters (not common words, names, or dates).
- Changed frequently.
- Something that they have not used before.

4. CAPTCHA Tests and Security Questions

- **CAPTCHA:** A computer program or system that can identify whether a user is a human or a computer. It stands for *Completely Automated Public Turing Test To Tell Computers and Humans Apart*.
- **Bot:** A computer program that can interact with systems or users.

CAPTCHA tests make sure that data is entered by a human and not by an automatic software program (a bot). They work by asking users to enter a randomly generated series of distorted

letters and numbers, select specific images (e.g., "Select all the trees"), or complete interactive tasks.

reCAPTCHA tests use extracts of text from scanned books or images. When a user solves a reCAPTCHA, their solution helps digitise books and annotate maps.

5. Anti-Malware Utilities

Anti-malware prevents malware from accessing or operating on computers. It scans computer files in **real-time** and allows users to scan specific files, folders, disks, or whole systems.

- **Anti-Virus:** Combats **viruses** (malware that uses networks to spread to connected devices via communication software like email or external devices like USBs). It constantly checks files for unique **virus definitions** (sequences of code found in viruses). If a match is found, it **quarantines** (isolates) the file in a protected area of storage so it cannot run.
- **Anti-Adware:** Detects, quarantines, and removes **adware** (software that displays unwanted advertisements).
- **Anti-Spyware:** Detects, quarantines, and removes **spyware** (software that secretly monitors and records computer data and user input, such as keyloggers that capture keystrokes).

6. Access Rights and File Permissions

- **Permissions:** Authorisation settings that provide the ability for a user or users to access files, folders, or drives.

Permissions can be set for access to files, folders, or drives, allowing users to have **read-only** access or **read-and-write** access.

7. Secure Websites (HTTPS)

- **HTTPS (Hypertext Transfer Protocol Secure):** A secure form of HTTP that authenticates payment servers and provides encryption using Secure Socket Layer (SSL) or Transport Layer Security (TLS).

HTTPS keeps communications private. Web browsers indicate a website is secure by displaying a **padlock icon** in the address bar.

8. Email Attachments and Web Links

Users must be careful when opening attachments or links. Anti-malware should be kept up to date, and students should remain alert if:

- They do not recognise the sender.
- The text is general, impersonal, or irrelevant.
- The text contains spelling or grammatical errors.
- The attached file is an executable file (such as `.exe` or `.zip` files).
- The text contains an urgent message demanding immediate action.
- The user does not recognise the URL.

9. Backup Procedures

A backup creates one or more copies of data, usually stored on an external storage device or online storage, to protect against theft, malware, flooding, fire, or power cuts.

Backup Type	Description
1. Full Backup	Creates a complete copy of all files.
2. Differential Backup	Copies all files that have changed since the last full backup .
3. Incremental Backup	Saves a copy of only the files that have changed since the last full or incremental backup .

Good backup strategies include:

- Setting up automatic backups (avoiding human error).
- Avoiding optical media (it deteriorates over time and is fragile).
- Scheduling backups for late in the evening to avoid file conflicts.
- Creating more than one copy and storing them at multiple geographical locations.
- Storing critical physical local backups inside a fireproof safe.

Online Payment Systems

1. Online Third-Party Payment Processors

Processors like PayPal or Skrill allow users to create an account to send and receive money securely using email accounts for identification.

2. Bank Cards

Bank cards allow customers to pay for items online. When making an online payment, a user must enter the:

- Card number (distinct from the bank account number)
- Expiry date

- Name on the card
- Three- or four-digit Card Security Code (CSC / CVV / CVC)

Online card systems often ask users to authenticate the transaction using secure verification steps like *Mastercard SecureCode*.

3. Contactless Cards Using NFC

Near Field Communication (NFC) transfers payment data wirelessly over short distances without requiring a PIN or user authentication. The transaction amount is strictly limited to prevent large-scale fraud if a card reader or app is used maliciously by criminals. NFC cards can be wrapped in foil to prevent weak signals from being intercepted.

Chapter Questions

1. **Which one of these is used to control internet traffic entering a network?**
 - A) Server
 - B) Backup
 - C) Firewall
 - D) Encryption
2. **Explain why CAPTCHA tests work.**
3. **State the purpose of pharming and phishing.**
4. **List three pieces of data from a bank card that a user is asked to enter when making an online payment.**
5. **Describe one way in which incremental backup differs from differential backup.**
6. **Describe how encryption secures data on a network.**
7. **State two methods used by an online system to authenticate a user.**
8. **State two methods used by criminals to get users to visit fake sites.**
9. **Explain one disadvantage of using online storage for backups rather than local storage.**

IMPACT OF THE INTERNET

THE INTERNET

- **Definition:** The internet is the interconnected network of computers that provides many online services to users. The World Wide Web is just one use of the internet.
 - **Protocols:** Rules that allow the exchange and transmission of data between devices.
 - **IMAP (internet message access protocol)** for email.
 - **FTP (file transfer protocol)** for sending files.
 - **SSH (secure shell)** for secure remote logins.
 - **VoIP (voice over internet protocol)**, used for voice call services like Skype.
 - **XMPP (extensible messaging presence protocol)**, used for messaging services like WhatsApp.
 - **HTTP (hypertext transfer protocol)**, used for the World Wide Web.
 - **Key Point:** The internet is connected devices. The World Wide Web is connected content.
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IMPACT OF THE INTERNET ON INDIVIDUALS

INFORMATION AND SERVICES

- The internet improves users with access to information from news, sport and weather services. Information can spread on the internet within seconds.
- Users can also use the internet to access booking systems for travel, leisure and entertainment; do their shopping and banking; study using virtual learning environments (VLEs).
- **Web 2.0:** The development of online interactivity, which allowed users not only to read webpages but also to add information to them.
- **Meme:** A photo, a piece of video, a joke or similar content that spreads quickly on the internet.

EMPLOYMENT

- **Skill Requirements:** Many employers now require employees to use the internet for work, meaning that employees need to learn new skills.
- **New Job Opportunities:** As the internet changes the work that people do, this provides new job opportunities. Skilled workers are needed to train and advise unskilled workers, create digital devices and software, and install and maintain systems.

- **Job Losses:** If employees do not keep updating their skills in the use of the internet, they may not have the skills required to carry out new work. If they are not willing or able to retrain, they could lose their jobs.

WORKING PRACTICES

- **Collaborative Working:** Working together. The internet enables people to connect with one another in real-time, making the process of sharing work more efficient.
 - *Split into individual tasks benefits:* Each employee can focus on one task and become experts through (narrow) experience or training.
 - *Split into individual tasks drawbacks:* Employees do not share skills and have reduced understanding of the whole project.
 - *More than one person on a task benefits:* Expertise is shared and employees can check each other's work.
 - *More than one person on a task drawbacks:* Difficult for lots of people to agree and difficult to co-ordinate the work.
- **Flexible Working:** A way of working that suits an employee's needs, allowing employees to decide the hours that they work or enables them to work from home.
 - *Benefits:* Employees can work at a time right for them, take breaks whenever needed, and fit work around family life.
 - *Drawbacks:* Employees do not meet face to face as much, may not manage work effectively, or cause exhaustion.
- **Mobile Working:** Allows employees to work when they are travelling or on the move to access up-to-date data when on-site.
 - *Benefits:* Access up-to-date information and work from anywhere with an internet connection.
 - *Drawbacks:* Employees could become exhausted if they work longer hours or may not have access to the same facilities.

SOCIAL IMPACT

- **Social Interaction:** Some people are concerned that our increasing use of virtual environments is replacing face-to-face social interaction. This is damaging people's ability to talk to each other and understand each other's emotions.
- **Cyberbullying:** Using the internet to send text, images or multimedia in order to upset or embarrass someone. Information can spread very quickly and to more people, and bullies often hide their identity.
- **Physical Activity:** Online goods and services have reduced the need to travel, and this reduced need for physical activity as part of daily life could have negative effects on people's health and wellbeing.

STAYING SAFE ONLINE

- **Zip It:** Keep your personal information private and do not share it with strangers (location, school name, phone number, real name, photos, mailing or email address).
- **Block It:** Always block offensive messages or friend requests from strangers on social networking sites, and do not open suspicious attachments or links.

- **Flag It:** You should always tell an adult if something online upsets you or if someone you don't know requests to meet you.

DIGITAL FOOTPRINT

- **Definition:** All the data that a user creates online, which is recorded and stored.
 - Much of what you share online can be recorded forever. Content can be easily viewed by people whom you did not expect to see it, copied, and shared easily.
-

ONLINE WORKING FROM HOME

- **Benefits to Individuals:** No need to travel to work (reduces stress, saves money, reduces travel emissions), can spend more time working or resting, no need to wear a uniform, can work in a comfortable environment, and organize work around family commitments.
 - **Benefits to Organisations:** Can attract a talented, motivated workforce, employ people who are located anywhere, do not need to buy or rent office space or furniture, and workers will not be delayed by weather or transport issues.
 - **Drawbacks to Individuals:** May be distractions at home, suffer from a lack of social interaction with colleagues, and feel disconnected from the company.
 - **Drawbacks to Organisations:** Concerns about data security, more difficult to manage and support employees, employees might not work as hard, and complicated to organise payments in different countries.
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IMPACT OF THE INTERNET ON ORGANISATIONS

POSITIVE IMPACTS

- **Improved Communication:** Communicate and interact with customers and employees more easily and in real-time using email, instant messaging and social media.
- **Access to Global Markets:** Made it easier for organisations to advertise and sell to customers in countries around the world without physical packaging or postage charges.
- **Access to Global Workforce:** Can employ people in other countries, make use of different skills and time zones to cover all 24 hours in one day, and reduce costs.
- **How Information is Managed and Used:** Organisations can make use of **big data** (the analysis of huge quantities of data collected from many sources) in order to find patterns of behaviour and interactions.

NEGATIVE IMPACTS

- **Security Issues:** Remote access means that data is more vulnerable to being accessed by an unauthorised user. Unauthorised users gain access by: authorised users revealing login details, intercepting data transferred over the internet, or hacking into systems.
 - **Greater Competition:** Organisations must now compete in a global marketplace. Entire industries have been put at risk: music and film retailers, DVD rental companies, record labels, estate agents, newspaper publishers, and postal services.
-

IMPACT OF THE INTERNET ON SOCIETY

THE CAUSES OF UNEQUAL ACCESS TO ICT

- **Digital Divide:** The gap between people who have access to digital devices and the internet and people who do not. Gaps are caused by:
 - *Economy and infrastructure:* Developing economies prioritise other needs like food and healthcare.
 - *Location:* Remote, rural, heavily forested and mountainous areas where wireless signals cannot reach.
 - *Politics:* Governments deliberately block, restrict, or disrupt internet access to control their citizens.
 - *Religion, Disability, and Social Factors:* Age, gender, education, and income affect access.

THE GAP BETWEEN INFORMATION RICH AND INFORMATION POOR

- **Information Rich:** People with good access to information, which improves their level of education and enhances their ability to make decisions.
- **Information Poor:** People with limited access to information, which decreases their education and reduces their ability to make decisions.

THE IMPACT OF LESS RESTRICTED ACCESS TO NETWORKS

- Social media and video streaming services have enabled people to broadcast their own messages and interact with global audiences.
- User-generated reference sites like Wikipedia have challenged society's ideas about expertise because truth, knowledge and accuracy are no longer checked by a limited number of experts.
- Networks have had an impact on law enforcement both negatively (rioters organising themselves via mobile networks) and positively (police using CCTV and smartphone footage to identify those responsible).

CHANGES IN WAYS OF SOCIALISING

- The internet has arguably affected the way that people think and their ability to form relationships with others.

- It can cause people to become isolated from their own families and local communities because they spend most of their time looking at the screens of their digital devices.
 - Having status updates sent directly means you no longer have to ask people what they are doing, leading some to argue that social media makes friendship meaningless.
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CHAPTER QUESTIONS

1. Which one of these activities is only possible when using the internet? (1)
 - a. Television
 - b. Radio
 - c. Streaming
 - d. Mobile telephone calls
 2. State two benefits to employees of working from home. (2)
 3. State two drawbacks to organisations of employees working from home. (2)
 4. Explain one reason why the use of the internet may affect the ability of individuals to interact with other people face-to-face. (3)
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 8. State three examples of services that are not available to those with limited or no access to digital technologies. (3)
 9. Describe two ways in which greater access to networks has affected society. (4)
 10. Describe the effect of the internet on the information gap. (2)
 11. Discuss the impact of the internet on how individuals socialise. (8)
 12. Describe two ways in which individuals can stay safe online. (4)
 13. Explain why organisations' use of the internet increases the risk to data security. (3)
-

ONLINE COMMUNITIES

- An online community is a group of people with a shared common interest who communicate online.
 - People use online communities to socialise, share information, play games, work, learn and create.
 - Each type of online community has a function (what it does for people who use it) and features (which enable or allow it to achieve its function).
 - Features that provide the function of one type of community now could be used for other types of community in the future, because the internet and its uses are always evolving.
-

Learning Objectives

- **Understand key features of online communities:**
 - social networking
 - online gaming
 - online work spaces
 - virtual learning environments (VLE)
 - user-generated reference sites: wikis, websites, forums
 - user-generated content: video sharing sites, blogs, websites
 - social bookmarking
 - **Know the functions and target audience** of different forms of online communities.
 - **Understand the ways** in which online communities are used to communicate and collaborate on a global scale.
 - **Be aware of the purpose** of responsible use and acceptable behaviour policies.
 - **Understand how to stay safe online:** cyber bullying, anonymity of others (misrepresentation), disclosure of personal information/location.
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Social Networking Communities

- **Function:** Online social network communities allow members to connect through shared interests or relationships. Examples include Facebook, Baidu® and Twitter.
- **Platform Convergence:** These communities include professional networks (LinkedIn®) and personal networks (Facebook). These different types of network started out with separate functions. However, the features and use of different types of social networks are starting to mix together. Personal networks are now used by professionals and businesses as well as for personal relationships.

Features list:

- **Profiles:** a collection of information about a user. Users can add information about themselves. Some services allow members to decide which info should be public, visible only to certain people, or invisible to anyone. Profiles can be personalised and include:
 - biographical details (name, gender, date of birth, location, language)
 - short 'about you' description
 - details about work and education
 - travel history
 - family details (relationship status, family members, pets)
 - contact info (telephone number, email address, website)
 - profile/background images, colour schemes, and designs.
- **Friend, follow and connect:** Features that allow users to add someone to their network. Some platforms restrict access to the full profile until a request is accepted, while others allow unrestricted access.
- **Stream, wall and timeline:** The place where posts from friends appear. It has changed how people communicate online, ensuring information is brought straight to the user rather than making the user look for it.
- **Status updates and posts:** Messages sent to other members in a network, publicly to all members, or to designated sub-groups.
- **Groups, lists and circles:** Allows creating named groups of members so posts are visible only to them (e.g., keeping family separate from work). Teams in organisations use this to communicate without sending info to everyone.
- **Tags:** Labels used to categorise content so other members can search for it.
- **User suggestions:** Suggests connections by analysing user interests, network interests, and friends of existing connections.
- **Reactions, ratings, likes, upvotes and downvotes:** Enables users to show reactions (liking or finding posts funny) or recommend posts. Downvotes show a dislike, though platforms try to avoid their negative message.
- **Share functionality:** Allows repeating others' posts to make them available to more people. A post that spreads widely far beyond its original network is described as 'viral'.
- **Comments and quotes:** Functional tools for members to write comments, facilitating online conversations.
- **Third-party integration:** Direct links on websites allowing people to share information to social networks easily.
- **Add content to posts:** Adding photos, videos, and URLs, with a preview tool before publishing.
- **Private or direct messages:** Allows sending private messages rather than posting publicly.
- **Notifications:** Alerts telling members about new interactions or content to keep them involved.
- **Analytics:** Services showing information about how people reacted to posts, helping users/organisations increase network size and followers.

Business and Monetization:

- **The Revenue Question:** Networks are often free to use, but their real cost depends on the value placed on personal information and how it is traded.
 - **Targeted Marketing Mechanism:** Platforms collect a huge amount of self-posted info, store it in databases, and analyse it. They sell this data to advertisers so they can target relevant sponsored posts to matched members.
 - **Promoted Posts:** Members pay a fee to elevate their posts. Targeted marketing allows organisations to aim promotions at matching attributes (e.g., specific age groups or genders) at the top of users' streams.
 - **Sponsored Posts:** Posts appearing at the top of all members' streams that do not use profile data for targeting.
 - **Market Scale:** As more members engage, organizations are more likely to pay for promotions because it provides a bigger market.
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Online Gaming Communities

- **Function:** Exist in order to allow members of the community to play multiplayer games together. Examples include PlayStation Network, Xbox Live, Steam®, and MMORPGs.
 - **Features list:**
 - links to social media
 - user profiles
 - information that allows users to find out how to complete games
 - experience points tracked and displayed on profiles, allowing users to choose to compete against others based on experience levels
 - discussion boards and forums to discuss tactics
 - statuses showing if other members are online
 - notifications about in-game events.
-

Online Work Spaces

- **Function:** Exist so that members of the community can collaborate together for the purposes of work. Examples include Adobe® Connect™, Slack®, Microsoft SharePoint®, and Workplace by Facebook.
- **Features list:**
 - cloud storage and web applications to work on documents via web browsers
 - comments left on documents for users to view and reply to
 - simultaneous multi-user document editing to develop files together
 - messaging systems for work discussions
 - shared calendars to view diaries and arrange meetings
 - shared contact lists
 - chat rooms to discuss work tasks
 - systems for booking resources (ICT equipment, meeting rooms, transport)

- virtual meeting spaces providing presentation viewing, VoIP communication, and video conferencing tools.
-

Virtual Learning Environments (VLEs)

- **Function:** Used to allow students and teachers to use learning and assessment materials. Examples include Pearson Active Teach, Google Classroom™, Moodle™, Schoology®, and Blackboard®.
 - **Features and Behavior:** VLE features are very similar to online work spaces. Many look and work like social networking communities, but they do not aim to connect users with people outside their network.
 - **Core Toolsets:**
 - a wall or timeline containing posts by teachers and students
 - a notice board for course announcements
 - audio, video, web link, or file sharing tools
 - quizzes or multiple choice tests that are often graded automatically
 - assignment submission portals
 - teacher-moderated communication tools (forums, chat rooms, wikis, blogs)
 - third-party integration features
 - **Single Sign-On (SSO):** Links log-ins to school systems to sign students in automatically when on the institution's network
 - **Document editors:** For teachers to create, edit, and collaborate on files with students
 - **Gradebooks:** Methods to monitor progress through courses and tasks
 - **Access statistics:** Tracking tools for teachers to see how frequently students use VLE facilities.
-

User-Generated Reference Sites

- Reference sites are information websites created and maintained by communities of members.

Wikis:

- **Function:** To allow members of the community to collaborate in order to build and edit web pages. Examples include Wikipedia and Discogs.
- **Features list:**
 - member accounts to track which edits were made by which members
 - an edit button taking editors directly to a text editor
 - structured language to format pages or add links
 - search tools.

Forums:

- **Function:** Provide members with online spaces for structured discussions arranged in topics or threads. Examples include Yahoo!®, Google Groups™, Stack Overflow®, Quora®, and The Student Room®.
 - **Features list:**
 - groups establishing different access rights for different members
 - **Moderators:** Members with the right to allow or block posts or members
 - **Administrators:** Members with moderator privileges plus rights to promote members to moderators and demote moderators
 - posts and date/topic sorted conversation threads
 - **Sticky notes (stickies):** Threads kept at the top of lists for easy access
 - user ratings to indicate helpful or genuine posts
 - private direct message functions (sometimes viewable by admins/moderators).
 - **Forum Safety Features:**
 - *Word or URL censoring:* Automated scanning software checks posts for inappropriate content; filters reject the post or send it to moderators.
 - *Ignore or block:* Features that mute members or stop forum access across the entire board.
 - *Rules and acceptable use policies:* Inform members of boundaries; violations result in private warnings or blocks.
 - *Report/flag user:* Allows members to report rule breakers to moderators.
-

User-Generated Content Platforms

Video-Sharing and Photo-Sharing Sites:

- **Function:** Allow people to access and share content created and uploaded by members. Examples include Vimeo®, Flickr®, 500px, Giphy®, and YouTube®.
- **Features list:** User accounts/profiles, Content Management Systems (CMS) to add/edit page content, tags, ratings, comments, and third-party social networking links.

Blogs and Vlogs:

- **Function:** Allow people to create online diaries of events or articles. Examples include Weibo®, Wordpress®, Tumblr®, and YouTube.
 - **The Community Distinction:** Individual blogs or vlogs are only called communities if two or more authors collaborate to create and update them, or if they allow interaction between the blogger and their readers.
 - **Features list:** Accounts, online text editors, media upload tools, tags, ratings, comments, and social sharing options.
-

Social Bookmarking Sites

- **Function:** Exist to allow people to share web documents and URLs with each other. Examples include Pinterest®, Digg®, Pocket™, reddit®, and StumbleUpon®.
 - **Features list:** Social networking features, third-party link integration buttons, categorization tags, and user accounts.
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ONLINE SAFETY & RISK PROTOCOLS

Anonymity and Misrepresentation

- Members can choose to stay anonymous online. Anonymity can protect users, but it also means some users can misrepresent themselves by pretending to be someone they are not.
- **The Safety Rule:** Make sure you know who you are talking to online. If you get to know someone online, never arrange to meet them in person on your own. If someone asks you to meet them, tell an adult whom you trust.

Personal Data and Location Disclosure

- Disclosing too much personal info is dangerous and can be exploited for grooming or to locate a person, their friends, and family.
- **Geotagging Leaks:** Location data can be given away, either on purpose or by accident, by uploading geotagged photographs to photo-sharing sites.
- **Mobile Tracking:** Community apps track coordinates via GPS data, Wi-Fi access points, or ISP data. If activated, any community member can find out exactly where that person is.
- **Defensive Actions:**
 1. Think carefully about how much personal information to make public.
 2. Do not post publicly any images containing location details.
 3. Check and update privacy settings regularly so only real-world contacts have access.
 4. Report anyone who is misrepresenting themselves or putting members at risk.

Cyberbullying

- Cyberbullying is the term used to describe the use of the internet to send text or images in order to upset or embarrass someone.
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Subject Vocabulary & General Vocabulary

- **online community**: a group of people with a shared common interest who communicate online
- **member**: someone who is part of an online community
- **profile**: a collection of information about a user
- **post (noun)**: a message sent to an internet discussion group so that all members of the group can read it
- **post (verb)**: to put a message or computer document on the internet so that other people can see it
- **targeted marketing**: advertising that is matched to internet users based on their attributes, such as their age group or their gender, or their internet browsing history
- **tag**: a label that you can add to a post
- **connections**: people or accounts to whom a user is connected
- **viral**: something that is circulated widely by being shared through networks to large numbers of internet users
- **notification**: an alert that tells a user about a new interaction or new content within a social networking community
- **analytics**: information that results from the analysis of data
- **Massively Multiplayer Online Role-playing Game (MMORPG)**: an online video game that allows large numbers of people to play together
- **multiplayer games**: games that are played by more than one person, usually online
- **experience points**: credits earned by a user for completing one part of a game
- **forum**: a website or web page where users can post comments and information and reply to other users' comments
- **cloud storage**: storage provided by servers that are connected to the internet
- **contact list**: a virtual address book that allows users to quickly access the contact details of friends and colleagues
- **chat room**: a place on the internet where users can write messages to other people and receive messages back from them immediately
- **virtual learning environments (VLEs)**: websites that contain teaching and learning tools
- **moderate**: decide whether content is appropriate; moderators decide whether posts should be removed or posted at all
- **single sign on (SSO)**: a system that allows users to log in once to access a number of related websites and systems
- **gradebook**: a virtual method of recording students' scores on assignments and tests
- **wiki**: a website or database that is developed by a number of collaborating users, all of whom can add and edit content
- **thread**: a series of messages concerning the same subject, written by members of an internet discussion group
- **blog (short for web log)**: a website or web page that is updated regularly, often written like a diary or a series of articles
- **vlog (short for video log)**: a video blog
- **blogger**: someone who creates or maintains a blog
- **social bookmarking**: using tags to categorise web documents and URLs so that other people can find content by using the tags in a search
- **geotag (verb)**: to add location data to a piece of content
- **professional**: someone who works in a job that needs special education and training, such as a doctor, lawyer or architect

- **promote**: publicise
 - **categorise**: to put things into groups or categories according to the type of thing that they are
 - **cluster**: a group of things that are arranged very close together
 - **preview**: see a document, page or film before it is produced in its final form and published
 - **tactics**: the methods used to achieve a goal
 - **demote**: make someone a lower rank or remove their privileges
 - **mute**: silence someone or something
 - **anonymous**: unidentified and unnamed
 - **anonymity**: the state of being anonymous; when other people do not know who you are or what your name is
 - **misrepresentation**: pretending to be someone or something that you are not
 - **in person**: face-to-face; in the real world
 - **disclose**: make information known to other people
 - **grooming**: pretending to befriend someone (often a child or young adult) with the intention of harming or abusing that person
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End of chapter questions

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IMPLICATIONS OF DIGITAL TECHNOLOGIES

Digital technologies have fundamentally changed how society shares and uses information, music, images, and films. While they provide powerful tools for development, they introduce significant societal challenges that force us to rethink:

- Ownership and Control
- Privacy and Identity
- Censorship
- Health, Safety, and Environmental Impacts

Data Protection and Rights

Legal Frameworks

Some countries enact specific laws to ensure that personal data stored by organizations remains protected.

- **Singapore (Personal Data Protection Act):** Requires organizations to obtain an individual's explicit consent to collect, use, or disclose personal data. Data must only be used in an appropriate manner for informed, specific purposes that a reasonable person would consider appropriate under the circumstances.
- **United Kingdom (Data Protection Act):** Restricts data use to strict data protection principles. Organizations must ensure personal data is:
 - Used fairly and lawfully.
 - Used for limited, specifically stated purposes.
 - Adequate, relevant, and not excessive.
 - Accurate
 - Kept no longer than necessary.
 - Handled according to individual data rights
 - Kept safe and secure.
 - Not transferred outside the European Economic Area (EEA) without adequate protection.

Individual Legal Rights

Under legislation like the UK Data Protection Act, individuals have the right to:

1. Access a copy of their stored personal data.
2. Object to data processing that causes damage or distress.

3. Prevent data processing used for direct marketing (e.g., newsletters, automated promotional emails).
4. Object to decisions made entirely by automated systems.
5. Have inaccurate personal data rectified, blocked, erased, or destroyed.
6. Claim financial compensation for damages caused by an organizational breach of the Act.

International Agreements

Because data is often stored across international border lines, agreements are crucial to ensure organizations act lawfully.

- **The Challenge:** Depending on the online services used, user data might be stored in countries with weaker or non-existent data protection laws, leaving individuals with no legal recourse if their rights are abused.
- **The Solution:** Agreements like the **EU-US Privacy Shield Framework** aim to protect data transferred across borders (e.g., between the US and the EU), which is vital for multinational technology companies with massive data operations.

Copyright Legislation

Concept of Copyright

Copyright is a legal power providing creators of original work with the sole authority to use and distribute it. If users download digital assets (software, books, music, images, or videos) without permission in a jurisdiction protected by copyright law, they face potential legal action from the copyright owner.

Distribution Licences

Owners can choose how they share their work:

- **Paid Licences:** Users pay for the right to use the media (e.g., buying a track or app from an online store), but this does not grant the right to distribute it to others.
- **Free Licences (Open Source / Creative Commons):** Creative Commons licences allow owners to distribute their work freely without payment, while still protecting their rights by explicitly specifying how other people can use, share, or alter the work. Free software licences lower the barrier to technology access, helping narrow the **digital divide** and the **information gap**.

Copyright Exemptions (Fair Use)

In many countries, certain conditions allow exceptions where copying materials without the explicit permission of the owner is lawful:

- Digital or physical copies made by libraries, educational establishments, museums, or archives.
- Record keeping of broadcasts intended specifically for archive purposes.
- Alternative formats developed directly for the benefit of people with a disability.
- Non-commercial research or private study.
- Material utilized for the purpose of a parody (exaggerated humorous imitation).

Digital Rights Management (DRM) and Geoblocking

To prevent illegal distribution and check whether a device is authorized to open a file, owners use copy protection and **Digital Rights Management (DRM)** systems (e.g., files downloaded on Apple iTunes). Additionally, organizations implement **geoblocking** to limit access to internet content based on the user's geographical location, preventing users in different legal frameworks from accessing copyrighted work.

Monitoring Individuals

Monitoring Movements

Governments and organizations can track individual locations using various mechanisms:

- **CCTV:** Closed-circuit television networks, often augmented with facial recognition software.
- **ANPR:** Automatic Number Plate Recognition cameras used by law enforcement.
- **Card Transactions & IDs:** Monitoring border crossings via passports, travel cards, identification cards, and bank transactions.
- **Network Tracing:** Identifying device footprints on networks via IP or MAC addresses.
- **GPS Technology:** Utilizing real-time GPS data shared by mobile applications or tracker devices clipped to clothing.

Benefits vs. Drawbacks of Movement Monitoring

- **Benefits:**
 - Locating lost individuals (such as young children),
 - organizing social gatherings via friend-finder apps,
 - identifying people on network ,
 - tracking down potential criminals at public events or airports,
 - verifying identities to lower financial crime rates.
- **Drawbacks:**
 - Severe compromises to personal privacy,
 - high financial expenses to establish and maintain systems,
 - heavy energy consumption impacting the environment,
 - causing citizens to feel distrusted.

Countermeasures (Anti-Surveillance)

To fight back against tracking, some modern fashion designs function as camouflage:

- **Anti-Surveillance Clothing:** Patterns featuring shapes that facial recognition software misinterprets as a face, overloading the system with data and hiding the true face of the individual.
- **Anti-Surveillance Eyewear:** Glasses specifically built to reflect the infra-red light used by CCTV cameras during low-light capture, blocking them from recording facial dimensions.

Monitoring Communications

Digital interactions—such as text messages, emails, social media posts, blog comments, and web history—can be systematically logged. Parents often use parental control software to check their children's activities. In certain jurisdictions, laws like the UK's *Investigatory Powers Act 2016* legally permit police and security agencies to access digital communications, sparking major civil liberties and human rights controversies regarding the right to privacy.

Sustainability Issues

The growing infrastructure of Information and Communication Technology (ICT) places immense strain on global ecology.

- **Sustainability:** Not using up natural resources until they run out; maintaining the balance of nature
- **Mitigation:** Reducing severity of a problem

Key Sustainability Challenges and Mitigations

Cause of Issue	Sustainability Problem	Mitigation Strategies
Power Requirements of Digital Devices	Electricity generated using non-renewable energy sources (coal, gas) depletes the planet's natural resources.	• Transitioning to renewable energy sources. • Manufacturing and using more energy-efficient hardware.
Cooling Systems in Data Centres	Massive server farms require high-power artificial cooling setups that consume non-renewable fuel.	• Building data centres next to rivers to harness natural hydroelectric power. • Constructing data centres in naturally cold climates (e.g., Northern Sweden) to eliminate the need for high-energy air conditioning units. • Using recycled canal water or seawater for cooling systems.

Use of Poisonous Substances	Digital components contain harmful chemicals like bromine, mercury, and chlorine. When improperly discarded as e-waste, these leak into soil and water systems, causing wildlife destruction and severe human health conditions (cancer, lung disease, central nervous system damage).	• Promoting public education and recycling initiatives. • Adhering to strict international directives (e.g., the EU's Waste Electrical and Electronic Equipment Directive). • Engineering components using harmless alternative materials.
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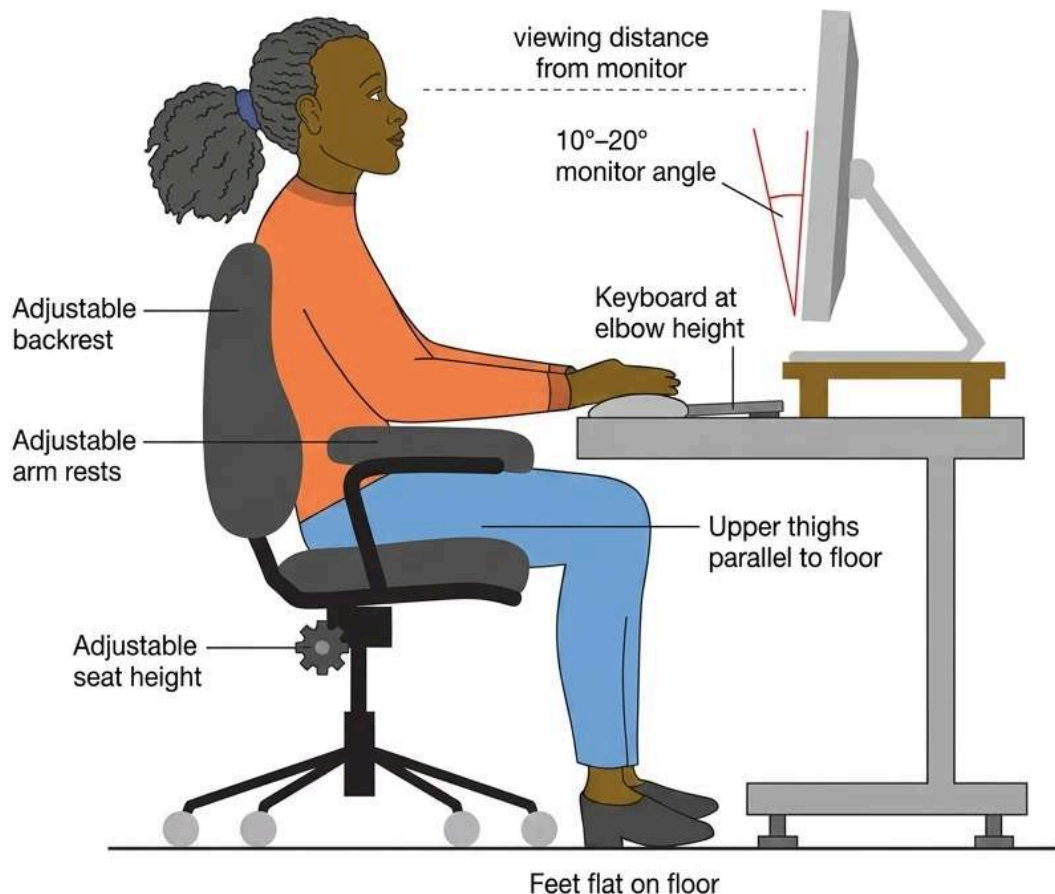
Health and Safety Issues

Prolonged or improper use of digital devices can lead to distinct occupational hazards.

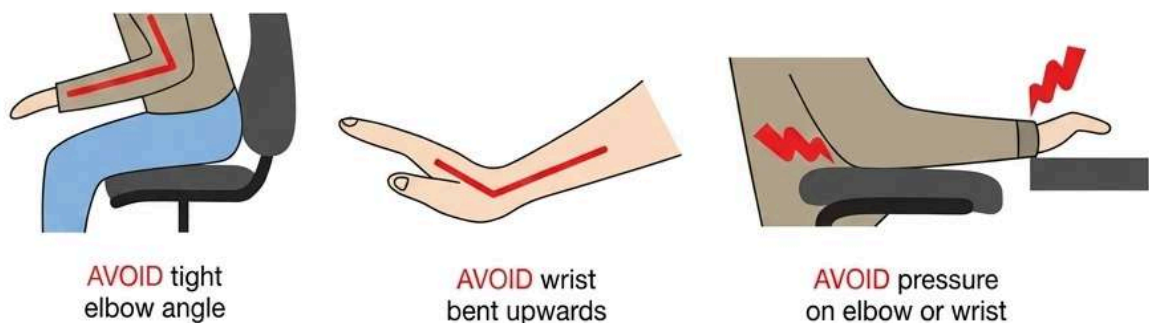
Risks, Causes, and Minimization Methods

- **Eye Dryness and Fatigue**
 - *Cause:* Looking at screens for extended periods without blinking. Human eyes are sensitive to short-wavelength blue light emitted by LEDs, which can disrupt sleep cycles.
 - *Minimization:* Take frequent breaks to look into the distance, ensure screens are kept at an appropriate viewing distance, use adequately sized monitors, apply blue light filters, and use glare-reducing lighting.
- **Repetitive Strain Injury (RSI) / Carpal Tunnel Syndrome**
 - *Cause:* Repeating the same physical actions continuously while using devices incorrectly or maintaining poor posture, compressing the wrist nerve.
 - *Minimization:* Use ergonomically designed devices and supportive peripherals, such as cushioned wrist pads.
- **Back and Neck Ache**
 - *Cause:* Slouching or maintaining an incorrect sitting posture over extended periods.
 - *Minimization:* Sit with a proper posture: upper thighs parallel to the floor, feet flat, keyboard positioned at elbow height, and an adjustable backrest.
- **Trip Hazards**
 - *Cause:* Trailing or loose power and data cables across walkways.
 - *Minimization:* Establish structured cable management by securing wires inside defensive plastic trunking channels.
- **Electric Shock**
 - *Cause:* Damaged external wire insulation or spilling liquids directly onto active electronic devices.
 - *Minimization:* Perform regular inspection of cables, immediately repair/replace worn wires, keep liquids away from workstations, and use Residual Current Devices (RCDs) to interrupt faulty circuits.
- **Fire Hazards**
 - *Cause:* Hardware overheating due to blocked ventilation or overloading electrical plug sockets with too many high-draw adapters.

- *Minimization*: Install dedicated active cooling devices, conduct scheduled maintenance, fit sockets with appropriate internal fuses, and keep fire extinguishers accessible nearby.
- **Physical Injury or Death**
 - *Cause*: Sudden lack of environmental awareness due to severe mobile distractions (e.g., texting while walking or driving).
 - *Minimization*: Strictly refrain from interacting with digital screens when interacting with roads or navigating immediate physical hazards.



▲ **Figure 9.7** Maintaining good posture when sitting at a computer



▲ **Figure 9.8** Incorrect posture can cause health problems

Artificial Intelligence (AI) and Risk

As artificial intelligence automation takes over daily human decisions (e.g., autopilot systems in airplanes or self-driving cars applying brakes, steering away from lane drift, and monitoring road speed limits), it introduces a critical ethical dilemma.

AI applications operate on programmatic rules rather than human common sense or moral judgment. This creates complex issues:

- **The Speed vs. Judgment Paradox:** While computers react significantly faster than human drivers—potentially reducing the 94% of accidents caused by human error—they cannot flexibly break a rule to avoid an unexpected hazard (e.g., safely exceeding a speed limit to escape danger).
 - **The Ethical Dilemma:** If an automated car must swerve to avoid a collision but accidentally strikes a pedestrian in the process, a machine cannot be held ethically or morally responsible for that action.
 - **The Attention Lapse:** When automated systems take long-term control, human operators experience degraded reaction times or may even fall asleep, leaving them unable to react during critical emergencies.
-

End of chapter questions

- 1) Which one of these is used to reduce the risk to health and safety? (1)
 - a) Using a small monitor
 - b) Increasing the amount of blue light emitted by a screen
 - c) Using an RCD
 - d) Working in a room with low-level lighting
- 2) Explain why free software licences have helped to reduce the digital divide. (2)
- 3) Explain why users should be able to give their consent before an app shares their location. (2)
- 4) State three types of digital information that can be protected by copyright. (3)
- 5) State one method of monitoring an individual's movements and describe how it is used. (2)
- 6) Explain why owners of digital information may decide to use copyright. (2)
- 7) State one benefit to individuals of having their location monitored. (1)
- 8) State one benefit to individuals of having their communications monitored. (1)
- 9) State one drawback to individuals of monitoring their location and communications. (1)
- 10) State three ways in which organisations should act responsibly to make sure that they protect the data they hold about individuals. (3)
- 11) Explain why the large-scale use of online storage and applications can have a negative effect on the environment. (4)
- 12) Describe one way in which individuals can reduce the effect of their use of digital devices on the environment. (3)

Online Information

- "Living in a connected world means that there is a lot of information available for individuals to find".
- "However, it can be difficult to find good quality information".
- "When you find quality information, it is important that you know how to use it appropriately".
- "Getting information off the Internet is like taking a drink from a fire hydrant" – Mitchell Kapor.

Information Sources

- **Primary source:** "something that you have created yourself".
- **Secondary source:** "something that has been created by someone else".
- **Primary Examples:**
 - "Photographs that you have taken yourself".
 - "Interviews or questionnaires conducted by you".
 - "Your own blogs, social media posts or emails".
 - "Your own sound or video recordings".
- **Secondary Examples:**
 - "Newspapers, books and maps".
 - "CDs, DVDs or Blu-rays created by others".
 - "Television and radio broadcasts".
 - "Websites created by other people".

Search Engines & Operation

- "Search engines compare the words entered by a user with words in a database of web pages".
- "They then show the user the results that are the closest match to their original keywords".
- **Keywords:** "the words or search terms that a user types into a search engine in order to look for matching information".
- **Autofill:** "the automatic suggestion of a completed word or phrase that is provided as the user types".
- **Browsing history:** "the URLs that an individual user has visited, which are stored in a file".

Search Syntax Rules

- **Syntax:** "the rules that describe how words and phrases are used in a language".
- **AND (+):** "Adding + between words will return only results that match both words".
- **NOT (-):** "Adding - before a word will return only results that do not include that word".

- **Phrase matching (""):** "Placing speech marks around a group of words will return only results that include the whole phrase with the words in that specific order".
- **Usage rights:** "the way in which a piece of information is permitted to be used".

Evaluating Information

- **Fitness for purpose:** "whether the information is suitable or good enough for the intended use".
- **Accuracy checks:**
 - "Checking that the information is from a trustworthy source that has good systems for checking facts before publication".
 - "Comparing it against other trustworthy sources".
 - "Seeing if the information uses evidence to support its claims".
- **Age checks:**
 - "Checking the publication date".
- **Relevance checks:**
 - **Relevance:** "whether the information relates directly to what you want to know".
 - "Checking that the information's topic matches the topic for which you intend to use it".
- **Reliability checks:**
 - "Comparing it against other unrelated sources to see if it matches".
- **Bias checks:**
 - **Bias:** "unfairly favouring one point of view, or not considering other points of view".
 - "Checking to see if it provides a range of viewpoints".
 - "Considering whether you could use the same information to tell a story from a different perspective".

Plagiarism & Paraphrasing

- "The internet has made information easily available to many people".
- "However, this means that it is becoming more common for people to copy and paste information, sometimes without even checking that it is suitable or true".
- **Plagiarism:** "using someone else's work and pretending it is yours".
- **Paraphrasing:** "expressing what someone else has said in a shorter, clearer or different way".
- "If you use another person's work but fail to state that it is theirs and not yours, you are violating their rights and committing plagiarism".
- "If you do this in your assignments for school, college or university, you will be punished and your marks could be reduced or you could fail the assignment completely".
- "One way to avoid plagiarism is to rewrite or paraphrase information".

CHAPTER QUESTIONS

1. Which one of these is used to match a full phrase in a search? (1)
 - a. +
 - b. -
 - c. @
 - d. ""
2. State two ways in which an image search can be refined. (2)
3. State one reason why you should check whether information is biased before using it. (1)
4. Describe how a search engine works. (4)
5. Explain why it is important to check whether information is protected by copyright before using it in your own work. (2)
6. State one benefit to students of rewriting information rather than copying it word-for-word. (1)
7. State three examples of search syntax. (3)
8. Complete Table 10.3 by stating whether each source is primary or secondary. (8)

SOURCE	PRIMARY OR SECONDARY?
A podcast that you have created	
A video of a discussion that you have with an expert	
Your own copy of a film	
Notes that you take while visiting a museum	
A book that you borrow from someone	
An image that you download from the internet	
Your own recording of a radio broadcast	

Online Goods and Services

Online Services Overview

- **Access Evolution:** Before the development of the internet, remote access was possible using the telephone or postal systems. Making services available online has enabled us to access them anywhere with any connected device.

Types of Online Service

- **Shopping Sites:** When a user shops online, they use a product catalogue.
 - **Product Catalogue Contents:** Can store an image of the product, a product description, the product's price, ratings by customers, shop staff or other people on review sites, the popularity of the product, a product code, and the number of items in stock.
 - **Categorisation:** Products are often grouped into types or categories displayed in menus with sub-categories appearing in sub-menus.
 - **Basket & Checkout:** Baskets keep a record of the items that the customer selects. At the checkout, they can enter their delivery details and pay. Sometimes, customers can use voucher codes in order to receive a discount.
 - **Security & Accounts:** Shopping sites often offer secure payment systems. Customers can often shop as a "guest" without creating an account.
- **Booking Systems:** Train tickets, airline tickets and bus tickets can all be purchased online.
 - **Features:** Customers choose travel times, the number of tickets, and the type of tickets.
 - **Delivery:** Tickets are often sent via email. Companies use e-tickets linked to the customer's passport, or QR codes that can be scanned.
- **Banks:** Offer online services including checking balances, setting balance alerts, making payments and transfers, applying for loans and other financial products, managing alerts and notifications, accessing customer service, and changing personal information.
- **Education and Training Providers:** Online training courses allow learners to access libraries of online journals, online support from teachers and other learners, and learning and assessment materials. This access is provided through virtual learning environments (VLEs).
- **Gaming Sites:** Online applications that allow users to play games that are hosted on a remote server, including real-time multiplayer games.
- **News and Other Information Providers:** Give up-to-date access to stories using blogs and notifications sent to subscribers. Provide real-time notifications about weather, sports, stock markets, and currency values.

- **Auction Sites:** Allow sellers to list items for sale for a certain period of time and allow buyers to offer money. Items can also be listed with a fixed price.
 - **Ratings:** Provide ratings for buyers (for being reliable and paying on time) and ratings for sellers (for being reliable and sending items promptly).
-

Impact on Lifestyle and Behaviour

Benefits of Online Goods and Services

- **Time Management:** Gives people more time to do important things such as spending time with family, socialising with friends or working.
- **Online Grocery Shopping:** Allows customers to easily reorder the same products, avoid having to travel to the supermarket, check whether items are in stock, and avoid having to wait in queues to pay.
- **Online Booking Systems (Rail):** Allows customers to avoid having to go to the station to buy a ticket, avoid having to queue, buy tickets in advance, and be notified if there is a delay.
- **Online Banking:** Allows customers to avoid travelling to the bank to check balances and transfer funds, and avoid carrying cash, which is safer for the individual.
- **Online Learning Services:** Users avoid having to travel or move closer to colleges or universities.
- **Online Information Providers:** People receive weather alerts or traffic advice, and can be rerouted automatically based on traffic conditions.
- **Lifestyle Summary:** More time can be spent with family and friends, a safer lifestyle as there is no need to carry cash, and individuals are more engaged with creative media like art, photography, videos and drama.

Drawbacks of Online Goods and Services

- **Patience:** The immediate availability of online services could make people less patient and more demanding of others.
 - **Isolation:** People become more isolated as individuals and detached from their real-world communities.
 - **Communication Skills:** Checking online communications constantly means spending less time connecting face-to-face, which reduces face-to-face communication skills.
 - **Memory & Attention:** Less need to memorise facts causes a negative impact on memory. People become accustomed to skim-reading online information, making them less likely to pay attention to details.
 - **Lifestyle Summary:** More time is spent using online services, individuals can be inconvenienced if no connectivity is available, and individuals may become less active.
-

Impact on the Ways that Organisations Do Business

Changes in the way organisations operate include:

- Improved communication.
 - Access to global markets.
 - Access to a global workforce.
 - Use of big data to understand customer behaviour.
 - The way in which the organisation secures business information.
 - Increased competition.
-

Transactional Data and Cookies

- **Definition:** An online transaction occurs whenever data is sent between devices on the internet. It relates to the movement of any data between connected devices, not just financial transactions.
- **Cookies:** Data tracking stores information in cookies on a user's computer so that the website will recognise that user when they use it again.
- **Data Stored in Cookies:**
 - The time and date of the transaction.
 - IP addresses.
 - The referring website.
 - The products, services, or categories viewed or bought.
 - Items added to the online shopping basket.
 - Buttons pressed by the user.
 - Data entered into web forms (name, address, credit card information).

Types of Cookies

- **Session Cookies:** Only stored until a web page is closed in the browser.
 - **Persistent Cookies:** Used to record information about the user's use of the internet over time. Can keep a user logged into an online account or allow advertisers to track internet use.
 - **Third-Party Cookies:** Used to personalise users' online experience or to send targeted adverts. They are used when web pages contain content such as banner advertisements from an ad server.
 - **Targeted Marketing:** Advertising that is matched to people based on their attributes or their browsing history using third-party cookies.
-

End of chapter questions

- 1) Which one of these is not provided by banks as an online service? (1)
 - a) Paying in cheques
 - b) Transferring money
 - c) Viewing an account's balance
 - d) Applying for a loan
- 2) State two ways in which education providers can use online services to help learners. (2)
- 3) State three features of shopping sites. (3)
- 4) Explain why auction sites provide ratings for buyers. (2)
- 5) Explain why banks use HTTPS for their online services. (3)
- 6) Explain the benefit of e-ticketing for airline flights. (2)
- 7) State one drawback of the use of third-party cookies. (1)
- 8) State three examples of data that can be stored in cookies. (3)
- 9) State the difference between session cookies and persistent cookies. (1)
- 10) State the way in which news services can use a feature of smartphone apps to keep users up to date. (1)
- 11) Discuss how the use of cookies by organisations impacts on individuals. (8)

The Cloud

Introduction to the Cloud

The term cloud describes the internet when it is used to provide software or space for storing information, rather than relying on local storage.

- **Local Hardware:** When you use software stored directly on your computer's hard drive to work with files, you are using local hardware. Because there is no network connectivity required to transfer data, this method generally provides the best performance speed.
- **Cloud Computing:** When you use the cloud, you use the internet to access applications and files that are stored on remote servers housed in data centers. Examples include using web applications like Twitter or Prezi, or syncing local folders with online storage systems.

Hosted Applications Software

Web applications (web apps)—also known as **hosted applications**—are programs stored and run on online servers. The user interface is displayed to you through a standard web browser. This setup relies on the client-server model.

Performance Factors

Because hosted applications are only available through an active internet connection, their performance is heavily dependent on **latency** (delay) and **bandwidth** (data capacity). Low bandwidth or high latency can result in:

- Slow application loading times.
- A noticeable delay between user input and browser activity, making the application seem **unresponsive**.

To minimize these performance drops, hosted software usually offers fewer complex features than locally installed software. Processing less data keeps the user experience smoother.

Collaboration

Because hosted software operates on a client-server model, multiple users can access the exact same application or file simultaneously. This enables real-time collaboration, often supported by integrated chat functions and tracking tools to add or resolve edits and comments.

Benefits and Drawbacks of Hosted Applications

Benefits	Drawbacks
Accessible from many types of connected devices.	Only accessible when you have an active internet connection.
Often includes integrated online data storage to save files online.	User experience can be poorly affected by high latency or low bandwidth.
Features powerful built-in tools that enable group collaboration.	Generally has fewer advanced features compared to local software.

Online Data Storage

Just like hosted applications, online data storage allows you to access files from any device in any location, provided you have an internet connection.

Benefits and Drawbacks of Online Data Storage

Benefits	Drawbacks
Files are frequently backed up automatically by the provider.	Potential data protection issues if data is stored on servers in another country.
The storage is scalable (it can expand or shrink based on demand).	Security is entirely dependent on the online storage provider.
Sharing data with other users is much easier.	File transfer speeds are significantly slower than local storage.
Many providers offer unlimited storage packages, reducing local hardware needs.	Only accessible when you have an internet connection.
Enables real-time document collaboration.	Some file formats cannot be opened online without downloading them first.
Frees up local storage space for local applications and files.	Users often have to pay subscription fees for additional storage space.

The Impact of the Cloud

On Technology and the Digital Divide

The growth of cloud services means devices no longer require massive internal storage drives or ultra-fast processors. This has led to the manufacturing of inexpensive computers like Chromebooks and netbooks. Lower hardware costs allow more people to access modern technology, helping to reduce the digital divide.

On Organizations

Cloud services provide businesses with highly scalable infrastructure that adjusts to their changing needs. It eliminates the massive overhead expenses associated with purchasing, housing, and maintaining physical in-house servers.

On Individuals (Online Services)

The cloud underpins many essential digital services used daily by individuals, including:

- Shopping and online auction platforms
- Travel, leisure, and entertainment booking systems
- Digital banking
- Virtual Learning Environments (VLEs) and Massive Open Online Courses (MOOCs)
- Online gaming
- Digital news and information portals
- On-demand entertainment services (streaming television and radio)

On Online Communities

Cloud infrastructure connects people globally by hosting interactive communities such as:

- Social networking sites
- Online workspaces for remote employment
- User-generated reference portals, including wikis and forums
- User-generated content platforms, such as blogs and video-sharing sites
- Social bookmarking sites

Vocabulary Reference

- **The Cloud:** The internet when it is used to provide software or space for storing information, rather than relying on local storage.
- **Application Loading Time:** The total time it takes for an application to open and become ready for use.
- **Unresponsive:** Not reacting to user input or commands, often caused by lag.
- **Edit:** A modification or change made to a document.
- **Scalable:** The capacity of a system (like storage or processing) to grow or shrink flexibly to perfectly meet varying demands.

- **Data Protection Issues:** Security problems or compliance risks associated with maintaining the privacy and safety of information.
 - **Netbook:** A compact computer designed primarily for internet access that lacks large internal storage or high-speed processing components.
-

Review Questions

1. **Multiple Choice:** Which one of these options is used to store and run hosted applications?
 - A) Laptop
 - B) Smartwatch
 - C) Server
 - D) Tablet
 2. **Short Answer:** State three distinct benefits of utilizing online data storage.
 3. **Short Answer:** Identify three types of everyday online services made possible by cloud-based services.
 4. **Definition:** In your own words, state what is meant by the term "the cloud".
-

Applying Information and Communication Technology (ICT)

This unit focuses on the practical application of ICT tools to solve problems and meet specific user requirements.

Core Competencies (Assessment Objectives)

- **AO1 (Knowledge & Understanding):** Grasping the fundamental concepts of ICT.
- **AO2 (Application):** Utilizing software to edit, format, and organize information into functional products.
- **AO3 (Analysis & Evaluation):** Critically reviewing outcomes against original project requirements to identify strengths and weaknesses.

2. Theoretical Framework: Data and Context

- **Definitions:**
 - **Data:** Raw, unprocessed elements (numbers, text, sound, images, or animation). In this raw state, it is meaningless.
 - **Information:** Data that has been processed to include context and meaning.
- **Contextual Necessity:** Data often requires specific context to become information.
 - *Example:* An animation of a sphere circling another could be an Earth-Sun model, a company logo, or something else entirely; it is only "information" once the context is established.
 - *Example:* A simple "beep" might be an alarm, a game sound, or an app notification, depending on the context.

3. Data Types and Meaningful Processing

Data Type	Characteristics	Methods to Produce Meaning
Text	Characters (letters/symbols).	Syntax/structure of language.
Numbers	Integers, decimals, currency, percentages.	Calculation, graphs, and charts.
Images	Bitmap or vector graphics.	Surrounding text, metaphor, and semiotics.

Animation	Files like SWF or animated GIFs.	Storytelling, surrounding information, text.
Sound	Audio clips.	Hardware (sound card/speaker) and supplementary data.

4. Software Application Purposes

- **Word Processor:** Used for letters, menus, reports, and essays. Features include pagination, spell check, and paragraph styling.
 - **Database Management:** Used for entering, filtering, sorting, and searching data.
 - **Graphics:** Used for creating, manipulating, and editing images.
 - **Web Authoring:** Used for creating websites and pages.
 - **Presentation:** Used for displaying multimedia content to an audience.
 - **Spreadsheet:** Used for financial budgeting, modelling, and performing calculations.
-

5. Design, Layout, and House Style

- **House Style:** A formal standard for layout and writing within an organization, used to maintain a consistent "corporate identity".
 - **Typography:**
 - **Serif:** Includes a small line at the ends of letters (e.g., *Times New Roman*). Best for printed media as the serif helps guide the reader's eye.
 - **Sans Serif:** Without serifs (e.g., *Verdana*). Often better for low-resolution digital screens.
 - **Colour & Contrast:**
 - **Colour Wheel:** Used to find complementary (opposite) colours for high contrast or analogous (adjacent) colours for thematic harmony.
 - **Accessibility:** Designers must account for colour-blindness. For example, never use red and green together, as some users cannot distinguish between them.
 - **White Space:** Balancing empty space is critical. Too much makes content feel isolated; too little makes it appear cluttered and confusing.
-

6. Review, Iteration, and Evaluation

Reviewing and Refinement

- **Automated Checking:** Software provides tools like spell and grammar checkers. These may not catch contextual errors (e.g., "there" instead of "their").
- **Proofreading:** Essential as automated tools may suggest incorrect changes.
- **Iterative Development:** Reviewing products against requirements during development to improve quality.

- **Evaluation:** Involves identifying strengths and weaknesses, considering alternative applications, reflecting on skill usage, and assessing if the product met user needs.
- **Evaluation Criteria:**
 - Did the software selection match the task needs?
 - Could the process be more efficient (e.g., through keyboard shortcuts)?
 - Did the product meet the intended audience/purpose requirements?

Chapter Questions

1. Which one of these data types is used to store a photograph?
 - ☐ (A) Image,
 - ☐ (B) Number,
 - ☐ (C) Text,
 - ☐ (D) Sound.
2. Explain one reason why it is important to use white space well when designing a digital product.
3. State one type of data and describe how it can be turned into information.
4. Identify three errors in this text: "How was I supposed to know Mr Singh".
5. Explain why using a house style can improve finished products when multiple documents are created for one organisation.
6. State what is meant by the term 'serif font'.
7. State one drawback of using serif fonts on devices with low-resolution screens.
8. Using the colour wheel, identify and state the name of the colour that contrasts most with red.
9. State one accessibility need relating to the use of colour in product designs.
10. State why it is important to proofread your work to check for errors.

File Management

File management involves the systematic organization, naming, storage, security, and version tracking of digital files. Proper file management saves time, prevents data loss, ensures you are working on the correct version of a product, and allows digital assets to be safely shared with others across various platform environments.

1. Directory Architecture & Folder Structures

A **folder structure** (or directory tree) is a hierarchical system used to organize files into logical, nested compartments on secondary storage media.

Hierarchical Folder Architecture

- **Root Directory:** The top-level folder on a storage device (e.g., Local Disk C:). All other folders and sub-folders branch out from this root.
- **Sub-folders:** Folders nested inside another folder. Grouping files into specialized sub-folders prevents clutter and streamlines retrieval.
- **Pathname:** The exact, sequential address pointing to a file's location within the directory tree.
 - *Example Path:* C:\School\ICT\Spreadsheets\PracticeActivity1.xlsx

2. Professional Naming Conventions & Version Control

Choosing vague, default filenames (such as Document1 or Logo-1) makes it difficult to find files later. Implementing a rigorous naming convention is essential for collaborative environments and long-term project management.

Sensible Filename Rules

A professional filename must be descriptive and consistent. It should ideally contain:

- The core **subject matter** or client name.
- The specific **asset description**.
- A standardized **version number** or date stamp.
 - *Poor Filename:* FinalLogo.png
 - *Sensible Filename:* School_Logo_v1.02.png

Advanced Version Management

To prevent accidentally overwriting previous iterations of a digital asset during development, users employ structured versioning codes:

- **MAJOR.REVISION.BUILDdate / BUILDnumber format:** A system where the

filename reflects the scale of project modifications.

- *Example:* E_Commerce_Project_v1.3_20260531
- **Major changes** increase the first number (v1.0 to v2.0).
- **Minor revisions** increase the second number (v1.2 to v1.3).
- **Date stamps** allow users to quickly find the most recent file when sorting chronologically.

3. Data Protection: Saving Practices & Security

Digital files are vulnerable to hardware failures, power cuts, software crashes, and unauthorized access. Protecting files requires proactive habits and security configurations.

The Power of Regular Saving

- **Volatile Memory (RAM):** While you are actively working on a file, your modifications are held temporarily in RAM. If your computer suddenly loses power or crashes, all unsaved data in RAM is completely lost.
- **Secondary Storage:** Saving transfers data from volatile RAM to non-volatile secondary storage (like an internal SSD, external hard drive, or cloud storage).
- *Best Practice:* Use the universal keyboard shortcut **Ctrl + S** (Windows) or **Command + S** (Mac) frequently while working to commit changes to disk.

File Security Methods

To protect files from unauthorized modifications or exposure, apply these foundational security restrictions:

Security Method	Functional Mechanics	Practical Application
Password Protection	Encrypts the file contents, requiring a secret passcode string to decrypt and open it.	Protecting private company payroll spreadsheets or student grade lists.
Read-Only Status	Allows unauthorized users to safely open and view a file, but blocks them from making or saving any modifications to the original file.	Distributing a master school policy document or an employee handbook.
Access Rights / Permissions	Network rules that grant specific user groups different levels of access (Full Control, Modify, Read & Execute, or Deny).	Allowing managers to edit files while blocking general staff from seeing them.

4. Application vs. Published File Formats

File extensions (the three or four letters following the dot at the end of a filename, e.g., .docx, .pdf) declare a file's format. Formats are divided into two primary structural categories:

1. Application File Formats

These are the native, default working formats used by specific software packages while a digital product is actively being created or edited. They preserve specialized background data layers, formula paths, and development undo histories.

2. Published / Exported File Formats

When a digital product is completed, it is exported or "published" into a universal format. This ensures that anyone can view the finished file exactly as intended, even if they do not own the premium editing software. Published files are compressed and highly secure, but they lock content into a static layer, making it very difficult for others to modify.

Common Format Mappings

Application Type	Source Application	Native Application Format	Universal Published Format
Word Processor	Microsoft Word	.docx	.pdf (Portable Document Format)
Presentation	LibreOffice Impress	.odp	.ppsx (PowerPoint Show) / .pdf
Graphics Editor	GNU Image Manipulation (GIMP)	.xcf	.jpeg (Compressed) / .png (Supports transparency)
Web Authoring	Serif WebPlus	.wpp	.html (HyperText Markup Language)

Student Activities

Activity 1: Folder Tree Conception

1. Sketch a hierarchical folder layout designed to manage an entire academic year of schoolwork.
2. Ensure you establish logical sub-folders for separate subjects, and nested sub-folders within those subjects for different units (e.g., separate folders for Spreadsheets, Databases, and Word Processing).
3. Write down the complete system pathname for a file named ExamRevision.pdf saved within your ICT unit folder.

Activity 2: File Format Analysis

1. Create a basic text document in your native word processor.
2. Insert a shapes asset and a custom text heading, then save it normally as an editable application format (.docx or .odt).

3. Export or "Save As" the file into a Portable Document Format (.pdf).
4. Open the .pdf file using a web browser or basic viewer tool. Attempt to select a single word, move a shape asset, or delete a paragraph. Note down how published formats protect content layouts from unauthorized editing compared to editable application formats.

Chapter Questions

1. Which one of these is the most sensible filename format for a logo asset designed for a school?
 - A) Logo-1
 - B) FinalLogo
 - C) School_Logo_v1.02
 - D) 102 School
2. Explain why it is vital for a computer user to save their active project files to secondary storage frequently.
3. State the reason why including versions in filenames is useful when developing a digital product.
4. State three distinct universal published or exported file formats.
5. Describe how an organization can use the MAJOR.REVISION.BUILDdate filename format to help employees quickly locate a specific asset iteration at a later date.
6. State one software method used to secure a sensitive spreadsheet from being edited by unauthorized personnel while still allowing them to view the data.
7. Define the term "Root Directory" within a hierarchical folder tree.
8. Explain why a graphics designer should export a company logo as a .png file format instead of a .jpeg file format if it needs to be placed over a dark website banner background.
9. Explain what happens to data held in RAM if a computer abruptly loses electrical power before the user hits the save command.
10. Which file extension represents a universal published layout format used to share text-heavy reports reliably across different operating system devices?
 - A) .docx
 - B) .xlsx
 - C) .pdf
 - D) .wpp

Word Processing

Word processing software is used to produce a wide variety of professional documents, such as letters, reports, newsletters, posters, leaflets, information sheets, fact sheets, and memos. Popular examples include *Microsoft Word*, *LibreOffice Writer*, and *Google Docs*. While software interfaces vary, they all share common features designed to enter, edit, and format text and assets into a polished document that is appropriate for its intended audience and purpose.

1. Text Entry, Editing, and Font Formatting

The foundation of word processing is entering or editing text using accurate spelling, punctuation, and grammar to ensure clarity and professionalism.

Text Wrap

When text reaches the right page margin at the end of a line, the software automatically moves the next word down to the beginning of the subsequent line if it is too long to fit. This is known as **text wrap**. It ensures that text automatically conforms to the page margins without requiring you to press the Enter key at the end of every line.

Font Formatting Features

Font formatting alters the appearance of text to create hierarchy, emphasize key points, and improve readability:

- **Font Style / Family:** The design of the characters (e.g., Arial, Times New Roman, Calibri).
- **Font Size:** The scale of the text, typically measured in points (pt). Headings require a larger font size than body text to establish structural hierarchy.
- **Font Weight and Style:**
 - **Bold:** Used to heavily emphasize keywords or apply strong formatting to headings.
 - *Italics:* Used for book titles, foreign words, captions, or mild emphasis.
 - Underline: Used to draw attention to specific standalone text strings.
- **Colour:** Used to draw visual attention or match an organization's branding. Text color must contrast strongly against the background to remain readable.

2. Paragraph and List Organization

Formatting at the paragraph level changes how blocks of text align and flow across the page canvas.

Text Alignment

Alignment determines how text lines distribute relative to the left and right margins:

Alignment Style	Visual Structure	Primary Practical Use Cases
Left Aligned	Flushed flat against the left margin; ragged right edge.	The default standard for business letters, reports, and essays because it is easiest to read.
Right Aligned	Flushed flat against the right margin; ragged left edge.	Used for page headers, dates, or contact addresses at the top of formal letters.
Centred	Aligned equally between both margins; ragged left and right edges.	Used exclusively for primary titles, document covers, certificates, and poster headlines.
Justified	Flushed flat against both the left and right margins simultaneously.	Used in newspapers, books, newsletters, and formal reports to create clean, block-like columns.

Structural Lists and Line Spacing

- **Bullets:** Used to create unordered lists when the sequence of items does not matter (e.g., a list of required equipment).
- **Numbered Lists:** Used to create ordered lists when the exact sequence is critical (e.g., step-by-step instructions or recipe steps).
- **Sub-numbering / Multilevel Lists:** Nests secondary lists within a main list to show sub-categories or dependent steps.
- **Line Spacing:** The vertical distance between lines of text within a paragraph. Increasing line spacing (such as using 1.5 or double spacing) improves readability for proofreading and grading.

3. Structural Layouts: Columns and Tables

Tables and columns are used to structure complex data or create complex layouts like those found in brochures and newsletters.

Multi-Column Layouts

Columns break a continuous body of text into parallel vertical strips. When text reaches the bottom of one column, it flows automatically to the top of the next column on the same page. This is ideal for maximizing space in newsletters, leaflets, and newspapers.

Table Geometry and Formatting

A **table** organizes data into an intersecting grid of horizontal **rows** and vertical **columns**. The intersections form individual data spaces called **cells**.

```
      Column 1      Column 2
Row 1 [ Merged Cell Header Block ] <-- Two cells combined into
one
Row 2 [ Left Aligned ] [ Right Align] <-- Text alignment styles
within cells
```

- **Horizontal & Vertical Text Alignment:** Text inside a cell can be aligned horizontally (Left, Centred, Right) and vertically (Top, Middle, Bottom) relative to the cell's boundaries.
- **Merge Cells:** Combines multiple adjacent cells into a single, larger cell block. This is frequently used to create an overriding title row across the top of a table.
- **Split Cells:** Divides a single selected cell into multiple smaller rows or columns.
- **Borders and Shading:**
 - * **Borders:** The lines that define the boundaries of the cells. They can be thickened, colored, or hidden completely.
 - **Shading / Background Fill:** Appears as a solid background color inside cells. Shading the top header row helps distinguish column titles from data records.
- **Gridlines:** The light background lines that map out cell boundaries on screen. They help you align data while editing and can be turned off so they do not show on printed copies.

4. Master Page Layout and Navigation Components

Page layout settings apply background structural rules to the entire document or entire sections.

Orientation and Margins

- **Orientation:** Dictates which way the page faces:
 - **Portrait:** The page is oriented vertically (height is greater than width). This is the standard choice for letters and formal reports.
 - **Landscape:** The page is oriented horizontally (width is greater than height). This is ideal for wide tables, landscape images, certificates, or multi-column brochures.
- **Margins:** The blank workspace buffers that run along the top, bottom, left, and right borders of a page, keeping text from running off the physical edges.

Page Elements and Navigation

- **Headings and Sub-headings:** Apply consistent structural styles across a document. This makes the document easy for readers to scan and allows the software to automatically generate a dynamic **Table of Contents**.
- **Page Breaks:** Forces the word processor to immediately end the current page and move any text after the cursor to the top of a completely new page. This keeps new chapters or sections from starting halfway down a page.
- **Headers & Footers:** Dedicated margin spaces at the top (Header) and bottom (Footer) of every page. Information entered here repeats automatically across every page in the document.
 - *Common Header/Footer Content:* Page numbers, total page counts, author names, document titles, creation dates, or corporate logos.
- **Page Numbering:** An automated field utility usually placed within the footer. Rather than typing manual numbers, using an automated page number field guarantees that

numbers update sequentially even if pages are added or removed later.

5. Visual Integration: Multi-Asset Documents

Professional documents often require you to combine, or integrate, text with visual assets to make information clearer and more engaging.

- **Images:** Inserting photos or illustrations helps clarify complex text. Images can be resized and positioned using text wrap settings (such as *Square*, *Tight*, or *In Line with Text*) so that text flows smoothly around the image borders.
- **Charts & Tables:** Complex numeric data can be imported from spreadsheet software and embedded directly into a report document.
- **Callouts and Autoshapes:** * **Autoshapes:** Geometric figures (arrows, banners, rectangles) used to separate information visually.
 - **Callouts:** Text boxes with an attached pointer arrow. They are used to point out specific data points on charts, label diagrams, or highlight pulled quotes from an article.
- **Text Boxes:** Standalone text blocks that can be placed anywhere on the page canvas. They allow you to pull out key summaries or position sidebars independently of the main paragraph flow.

Student Activities

Activity 1: Creating Hierarchy and Table Formats

1. Open a blank document in your word processing application.
2. Type the title text: Quarterly Department Report
3. Set the font size of this title text to **24pt**, change the weight to **Bold**, and select a **Sansom font** like Arial or Calibri. Centre-align the title line.
4. Insert a table directly below the title containing **3 Columns** and **4 Rows**.
5. Select the entire first row of your table, right-click, and select **Merge Cells**. Type Department Performance Financials inside this newly merged cell and apply a light gray background **shading fill**.
6. In Row 2, enter column headers: Department Name, Target Value, Actual Outcome. Set these labels to **Bold** and center-align them.
7. Fill the remaining rows with placeholder numbers. Highlight the numbers in the target and actual columns and **Right Align** them. (Note why right-aligning numbers makes them easier to compare down a column).

Activity 2: Investigating Page Layout Breaks

1. Type a short paragraph on page 1 of a new document.
2. Press the Spacebar or Enter key repeatedly until a second page is created. Note down how many times you had to press the key.
3. Delete the empty spaces to bring the cursor back to the end of your initial paragraph.
4. Insert a formal **Page Break** via the *Insert* menu (or by pressing Ctrl + Enter).
5. Type a second paragraph on this new page. Note down how this method is cleaner

and faster than using the Enter key repeatedly, and why it prevents layout shifting if you edit text on page 1 later.

Chapter Questions

1. Which layout style ensures that text lines align flat against both the left and right page margins simultaneously?
 - ☐ A) Centred
 - ☐ B) Justified
 - ☐ C) Left Aligned
 - ☐ D) Right Aligned
2. Explain the purpose of using automated page number fields in a footer instead of manually typing numbers on each page.
3. Describe what happens to text when it hits the right margin edge, referencing the concept of text wrap.
4. Which type of document break should you insert if you want to force any text following the cursor to start instantly at the top of the next page?
 - ☐ A) Column Break
 - ☐ B) Line Break
 - ☐ C) Page Break
 - ☐ D) Section Break
5. List three distinct types of content items that are commonly inserted into headers or footers so they repeat automatically across all pages of a formal corporate report.
6. Describe how combining a table cell merge with cell shading can improve the readability of an extensive data table.
7. State the most appropriate page orientation to use if you need to display a very wide structural financial table containing 15 separate data columns.
8. State the feature used to wrap text smoothly around an imported image so that the text follows the contours of the graphic frame.
9. Define what an autoshape callout box achieves when analyzing a chart embedded inside a word-processed document.
10. Which text alignment configuration is standard for typing contact addresses or dates at the very top of a formal business letter layout?
 - A) Left Aligned
 - B) Centred
 - C) Justified
 - D) Right Aligned

Digital Graphics

Graphics software is used to create, edit, and manipulate visual images. These digital graphics are frequently integrated into other ICT products, such as multimedia presentations, desktop published documents, or web pages. Popular application examples include *Adobe Photoshop*, *GIMP* (for bitmap images), *Adobe Illustrator*, *Inkscape*, and *Google Drawings* (for vector graphics).

1. Features of Image Types: Bitmap vs. Vector

All digital graphics fall into one of two primary architectural categories. Understanding their differences is crucial for choosing the correct tool for a given project.

Bitmap Images

Bitmap images (sometimes called raster graphics) are typically used for photographs and scanned documents.

- **Composition:** A bitmap image is made up of a grid of tiny squares called **pixels** (picture elements). Each individual pixel can display only one single color at a time.
- **Resolution:** The quality of a bitmap is determined by its resolution, which refers to the number of pixels packed into a given area. Higher-resolution images contain more pixels, creating a sharper, more detailed picture.
- **Pixelation:** When a bitmap image is enlarged (stretched), the individual pixels become visible as blocky squares, causing the image to lose clarity and look jagged. This effect is known as **pixelation**.

Vector Graphics

Vector graphics are typically used for logos, text typography, icons, and technical illustrations.

- **Composition:** Instead of using pixels, vector graphics are built from mathematical equations that calculate the paths, lines, curves, shapes, and coordinates of objects relative to one another.
- **Scalability:** Because vectors rely on math rather than a fixed grid of squares, they can be scaled or enlarged infinitely without ever losing quality or pixelating. The software simply recalculates the mathematics to redraw the lines perfectly sharp at any size.
- **Realism Limitation:** A major drawback of vector graphics is that they struggle to represent natural objects or complex scenes realistically. They cannot easily capture the subtle color gradients, shadows, and textures found in real-world photographs.

2. Structural Image Composition: Working with Shapes, Text, and Layers

Creating or editing graphics efficiently requires working with structured object assets and organizing them cleanly on your canvas.

Basic Shape Integration

Graphics applications provide vector-based geometric tools to construct compound images by combining primitive assets:

- **Rectangles & Squares:** Form foundational layout containers or solid background borders.
- **Circles & Ovals:** Used for round design elements. Increasing the pixel grid or resolution allows curves to look smooth rather than jagged.
- **Lines, Triangles, and Arrows:** Used extensively to build technical schematics, flowcharts, or user interface components.
- **Text Boxes:** Allow you to insert typography, captions, and labels directly over graphic layouts.

Organizing Layouts with Layers

Advanced graphics software uses a stacked system called **layers** to organize components.

- **Independent Editing:** Each element or asset can be placed on its own transparent layer. This allows you to edit, transform, recolor, or delete one specific part of an image without affecting any of the other surrounding assets.
- **Stacking Order:** Layers can be dragged up or down to change which items appear in front of or behind other elements on the canvas (e.g., bringing a text layer to the absolute front so it sits clearly over a solid background rectangle).

3. Image Editing and Transformation Techniques

Modifying existing graphics involves changing their compositions or erasing flaws using dedicated editing tools.

Aspect Ratio Control

When resizing any image, maintaining its **Aspect Ratio** is vital. The aspect ratio is the proportional relationship between an image's width and its height.

- **Distortion:** If you pull an image's resizing handles from only one side or top edge, you alter its width without scaling its height, causing the image to look stretched, squashed, or **distorted**.
- **Proportional Scaling:** To prevent distortion, always scale images from their corner adjustment handles, or hold down the Shift key while resizing to lock the aspect ratio.

Original [$\odot$]	Distorted (Side Drag) [$\odot$]	Proportional
(Corner Drag) [$\odot$]		
(Correct)	(Wrong)	
(Correct)		

Transformation Operations

- **Cropping:** Trimming away unwanted outer edges or background areas of an image to change its canvas boundaries, focus on a specific subject, or alter its layout proportions.
- **Grouping:** Locking multiple individual shapes or layers together so they act as a single, unified object. Grouped items can be moved, resized, or rotated together, preserving their layout spacing.
- **Rotation:** Pivoting an object or an entire canvas around a central axis to change its angle of orientation.
- **Deletion of Unwanted Aspects:** Using clone stamps, erasers, or selection masks to paint over or erase unwanted elements from a photograph (such as removing a background power line or blemish).

4. Legality in Graphics: Understanding Copyright

When sourcing images from the internet to build your digital products, you must respect intellectual property regulations.

- **Copyright Infringement / Breach:** Copying, altering, or publishing an image without obtaining explicit permission or a valid license from the original creator is a breach of copyright law.
- **Safe Sourcing Practices:** To avoid legal breaches, use images that are explicitly marked as *Public Domain*, distributed under *Creative Commons* licensing (which often requires attribution), or sourced from royalty-free stock repositories.

Student Activities

Activity 1: Tracking Pixelation and Resolution

1. Open a bitmap image (such as a .jpeg or .png photograph) inside a web browser or image viewer.
2. Zoom in repeatedly (to 400\% or higher) on a sharp boundary line within the photograph.
3. Document your observations: Describe how the line breaks down into small, distinct squares of solid color, and explain how this links to the concept of **pixelation**.

Activity 2: Maintaining Aspect Ratio

1. Insert any circular shape asset into a presentation slide or word processing document.
2. Select the shape, click the handle on the far-right edge, and drag it horizontally

- outward. Note down how the circle distorts into an oval.
3. Undo your change. Now, click and drag the handle located at the exact bottom-right corner while holding down the **Shift key**.
 4. Explain why locking the aspect ratio is critical when scaling image assets in professional presentations and documents.

Chapter Questions

1. Which one of these options must be strictly maintained when resizing a graphic asset to prevent it from becoming visually distorted?
 - ☐ A) Z axis
 - ☐ B) Aspect ratio
 - ☐ C) Rotation
 - ☐ D) Grouping
2. Explain why vector graphics do not pixelate or lose quality when they are enlarged to extreme sizes.
3. Describe how layers can be used to construct a complex digital graphic image.
4. State the name of the tiny, single-colored square elements that combine to make up a standard bitmap image.
5. Give one distinct drawback of choosing to use vector graphics instead of bitmap graphics when representing real-world objects.
6. State the term used to describe the action of cutting away the outer edges of an image to remove unwanted background space.
7. Explain what occurs if a digital designer breaches copyright laws when sourcing graphics for a commercial web page.
8. State the feature used to lock multiple separate shapes or text items together so they can be moved or rotated as a single entity.
9. Define the term "resolution" in the context of bitmap photographs.
10. Which type of graphic file format would you choose to preserve a logo's transparent background when exporting it for a website?
 - ☐ A) .jpeg
 - ☐ B) .bmp
 - ☐ C) .png
 - ☐ D) .vector

Multimedia Presentations

A **multimedia presentation** is a digital product composed of a series of structured slides used to support a speaker delivering information on a specific topic to an audience. Presentations are an exceptional way to add visual context to a speech, highlight key data, and actively engage listeners. Popular application software engines include *Microsoft PowerPoint* and *LibreOffice Impress*.

1. Structural Blueprinting: Master Slides & Templates

A professional presentation must maintain a consistent layout, style, and design framework across all slides to avoid distracting the audience. This structural uniformity is achieved using master slides and templates.

The Master Slide

The **Master Slide** is a central design blueprint that stores information about the underlying theme and layout rules of a presentation. Modifying a master slide automatically applies those exact style changes (such as background colors, logo placement, or font choices) to every single slide in the presentation.

Component Geography

- **Placeholders:** Pre-formatted bounding containers embedded into slide layouts. They dictate precisely where text paragraphs, bulleted lists, images, charts, and video files will sit on the canvas.
- **Font Enhancements:** Standardizing typography constraints on the master slide (e.g., using **Bold** for primary titles, *Italics* for supporting annotations, or Underlines for standalone links) establishes an instant visual hierarchy.
- **Headers and Footers:** Dedicated margin spaces positioned at the absolute top (Header) and bottom (Footer) of the slide canvas.
- **Slide Numbering:** An automated text field commonly nested within the footer. Rather than typing numbers manually on each page, using an automated slide number utility guarantees that the software tracks and displays the correct, sequential page index dynamically.

2. Interactive Navigation: Action Buttons & Hyperlinks

While presentations are often delivered linearly, advanced multimedia products incorporate interactive elements that allow for non-linear navigation, allowing the speaker or user to jump

across sub-topics dynamically.

- **Hyperlinks:** Active hot-links applied to text strings or graphic items. Clicking a hyperlink can instantly launch an external website, open an attached file, or jump directly to a different slide within the same deck.
- **Action Buttons:** Built-in interactive geometric shape buttons (such as arrows, home icons, or information symbols) placed on a slide canvas. These buttons are programmed to execute specific navigation actions when clicked or hovered over during a slide show.
 - *Common Action Mapping:* Moving to the immediate next slide, returning to the previous slide, jumping back to the home slide index, or terminating the presentation delivery.

3. Dynamic Motion: Animations vs. Transitions

Adding movement can draw an audience's attention to key milestones or control the flow of information. However, designers must use motion features sparingly; overusing them can look unprofessional and diminish their impact.

Slide Transitions

Transitions are the visual motion effects that occur when the presentation moves from one slide to the subsequent slide during a show.

- **Control Settings:** Applications allow designers to customize transitions by altering their execution triggers:
 - **On Mouse Click:** The transition pauses and only advances when the speaker manually clicks the mouse or presses a keyboard arrow key.
 - **After (Timed Duration):** The slide automatically advances onto the screen after a specified, pre-programmed length of time has passed.
 - **Apply To All:** A function that instantly maps the active slide's transition settings onto every slide in the workbook, ensuring visual consistency.

Object Animations

Unlike transitions, **Animations** are motion effects applied to individual component assets (such as a specific text bullet, a shape, or an image) *within* a single slide.

- *The Power of Bullet Reveal:* Animating a bulleted list to appear line-by-line allows a speaker to introduce points one at a time. This prevents the audience from reading ahead and keeps them focused on the current point being spoken.

4. Production Output & Hardcopy Distribution

Presentations can be viewed digitally or printed out in various layout styles depending on who will be reading them.

Output Layout Style	Structural Blueprint	Primary Practical Use Case
Full Page Slides	Prints one single slide across an entire physical sheet of	Used by the speaker during presentation practice or as

Output Layout Style	Structural Blueprint	Primary Practical Use Case
	paper.	large visual posters.
Handouts	Combines multiple slides scaled down onto a single printed page (commonly configured as two to a page or three to a page).	Distributed directly to the audience prior to a speech so they can follow along and review the content later.
Notes Pages	Prints a single scaled-down slide at the top of the page, leaving a wide, blank text workspace underneath it.	Used exclusively by the speaker to write out their spoken script or talking points for that specific slide.

Student Activities

Activity 1: Harnessing the Power of Master Slides

1. Launch a blank presentation inside your slide authoring application.
2. Select the *View* menu tab and switch into the **Slide Master** view layout.
3. Select the primary master slide thumbnail at the very top of the hierarchy tree.
4. Insert a small geometric autoshape (such as a solid blue square) into the absolute top-right corner of this master layout.
5. Exit the Slide Master view and return to your standard editing workspace.
6. Insert 4 completely new blank slides into your presentation deck.
7. Examine each slide canvas and document how the master blue shape responds.
Explain why utilizing a master slide is far more efficient than copying and pasting an asset onto 50 individual slides manually.

Activity 2: Investigating Transition Automation

1. Create a presentation containing 3 slides. Place large numbers (1, 2, 3) inside the center text placeholders of each slide.
2. Navigate to the *Transitions* options panel. Select a clean transition style (such as a fade).
3. Uncheck the manual configuration box for **On Mouse Click**.
4. Check the box for **After** and set the automated countdown timer duration to exactly 00:03.00 (3 seconds).
5. Click the **Apply To All** button.
6. Launch your slide show presentation from slide 1. Observe how the system advances across the entire slide structure automatically without any physical input.

Chapter Questions

1. Which layout utility should you select if you want to print slides alongside a blank text zone to host a speaker's written spoken scripts?
 - a. A) Handouts (three to a page)
 - b. B) Full Page Slides

- c. C) Notes Pages
 - d. D) Master Slide View
2. Explain why a presenter should configure bulleted lists to reveal lines sequentially using animation rather than displaying all text simultaneously.
 3. Describe the structural difference between a slide transition effect and an object animation effect.
 4. Which operational button configuration allows a designer to instantly copy a single transition style across every single slide in a deck?
 - ☐ A) Preview
 - ☐ B) Duration
 - ☐ C) Set Timing
 - ☐ D) Apply To All
 5. State three distinct pieces of information or automated fields that are commonly placed inside slide footer layout panels.
 6. Explain how editing the background layout properties on a master slide affects the rest of the presentation file.
 7. State the purpose of printing audience handouts configured to display three slides to a single page.
 8. State the feature used to inject interactive clickable elements that jump across different slides or launch external web URLs.
 9. Define what a slide "placeholder" represents within a multimedia design template.
 10. Which manual trigger event is the default method used to advance from one slide to the next during a live speech delivery?
 - ☐ A) After Timer
 - ☐ B) On Mouse Click
 - ☐ C) Master Slide Update
 - ☐ D) Layout Reset

Web Authoring

Web authoring software is used to design, create, and edit web pages and complete websites. Web pages are essentially documents written in **HTML (Hypertext Markup Language)**, which is the universal structural language of the World Wide Web.

Web browser software (such as Google Chrome, Microsoft Edge, or Mozilla Firefox) reads the HTML code, interprets the instructions or "markup tags," and displays the resulting visual page. To author websites, designers either write raw code in a plain-text editor or use specialized **WYSIWYG** (What You See Is What You Get) applications like Adobe Dreamweaver, KompoZer, or Amaya, which generate the underlying HTML automatically.

1. Website Infrastructure: Templates, Navigation, and Layouts

A professional website must maintain visual consistency and clear usability across all its pages. This is achieved using structural layout templates.

The Purpose of Templates

A **template** is a master page layout template used to create multiple web pages with a matching look, feel, and structure. Templates save development time and ensure that branding elements remain identical across the entire site.

Standard Page Features

Web pages are typically broken down into distinct structural zones using HTML layout blocks or tables:

- **Banner / Header:** Positioned at the very top of the page canvas. It typically contains the company logo, the website title, and primary corporate branding graphics.
- **Navigation Menu:** A structured set of buttons or text hyperlinks clustered together (either horizontally below the banner or vertically down the left sidebar). The navigation menu must be clear, intuitive, and appear in the exact same position on every single page so users can navigate the site easily.
- **Sidebar:** A narrow vertical column used to host secondary links, search bars, or social media widgets.
- **Main Body Content:** The primary space reserved for the unique text, images, and videos of that specific web page.
- **Footer:** Positioned at the absolute bottom of the page. It contains copyright notices, privacy policy links, and contact information.

2. Page Content & Asset Integration

Building an engaging web page requires integrating different digital assets directly into the

layout.

- **Text Formatting:** Web text is structured hierarchically using strict formatting styles to make the content easy to scan:
 - **Headings (<h1>):** Large, prominent text used for the main page title.
 - **Sub-headings (<h2>, <h3>):** Medium text used to break content into logical sub-sections.
 - **Body Text (<p>):** Standard scale text optimized for reading paragraphs.
- **Images:** Graphic files (such as .png or .jpeg) inserted to support the text. Images require an **Alt Text (Alternative Text)** description embedded in the code. Alt text is read aloud by screen readers for visually impaired visitors and displays on screen if the image file fails to load.
- **Buttons & Interactivity:** Interactive graphic blocks or text elements designed to trigger actions when clicked, such as submitting a form or jumping to another section.
- **Animations:** Embedded moving elements (such as .gif loops or HTML5 assets) used to draw attention to specific announcements or banners.

3. Mastering Core HTML Code & Tags

HTML uses **tags** enclosed in angle brackets (< and >) to instruct web browsers how to handle and display content. Most markup tags work in pairs: an opening tag to start an effect and a closing tag containing a forward slash (/) to end it.

Essential Text Enhancement Tags

- **:** Instructs the browser to display text in **bold** to signify strong importance.
 - *Syntax:* Bold Text
- **:** Applies *emphasis* to text by rendering it in italics.
 - *Syntax:* Italicized Text
- **<u>:** Displays text with a structural <u>underline</u>.
 - *Syntax:* <u>Underlined Text</u>

Structural Layout Tags

- **<p>:** Defines a standard paragraph. The browser automatically injects a wider vertical space buffer between paragraph blocks to ensure readability.
- **
:** Inserts a direct **line break**. Unlike paragraph tags, a line break forces text to drop down to the next line immediately *without* adding extra vertical spacing. **Note:**
 is a standalone or "empty" tag; it does not require a closing </br> tag.

Hyperlinks and Asset Insertion

- **Inserting Images ():** Links an external image file into the page layout. It utilizes key attributes to specify the file source location, alt text, and dimensions:
 - *Syntax:*
- **Inserting Hyperlinks (<a>):** Creates an active link connecting the page to another asset or external website using the href (Hypertext Reference) attribute.

- *Syntax:* `<a href="http://www.pearson.com" target="_blank">Visit Pearson</a>`
- *The Target Attribute:* Adding `target="_blank"` instructs the web browser to open the clicked link in a completely new window or tab, preventing the user from navigating away from your original website.

Student Activities

Activity 1: Visualizing Page Structures

1. Open any major educational or news website in your web browser.
2. Carefully sketch the structural layout framework of the homepage on a sheet of paper.
3. Label the boundaries of the **Banner**, the **Navigation Menu**, the **Main Body Content**, and the **Footer**.
4. Navigate to a different sub-page on the same website. Note down whether the banner and navigation menu remain in the exact same positions, and explain how this consistency benefits the user experience.

Activity 2: Writing and Interpreting Raw HTML

1. Open a basic plain-text editor (such as *Notepad* on Windows or *TextEdit* on Mac configured to plain text mode).
2. Carefully type out the structural HTML template code block displayed below:

```
<!DOCTYPE html>
<html>
<head>
  <title>My First ICT Web Page</title>
</head>
<body>
  <h1>Welcome to Web Authoring</h1>
  <p>This is a standard body paragraph containing
<strong>important bold words</strong>.</p>
  <p>This line ends here<br>and drops down immediately using a
line break tag.</p>
  <p><a href="https://www.wikipedia.org" target="_blank">Click
here to visit Wikipedia in a new tab</a></p>
</body>
</html>
```

1. Save the file with the exact name `index.html` (ensuring the extension is `.html` and not `.txt`).
2. Locate the saved file in your file explorer and double-click it to open it in a web browser. Verify that the font enhancements, line breaks, and hyperlinks render and function correctly.

Chapter Questions

1. Which HTML tag is used to force text to drop down to the very next line instantly without adding a wide paragraph space buffer? A) <p> B) <href> C)
 D)
2. Explain why web designers utilize layout templates when building large websites containing multiple pages.
3. Describe the role of a web browser application when interacting with an HTML file.
4. Which attribute must be included inside an <a> anchor tag to specify the exact web address destination of a hyperlink?
 - ☐ A) src
 - ☐ B) alt
 - ☐ C) target
 - ☐ D) href
5. State the purpose of embedding Alt Text (alt) descriptions within an image tag structure.
6. Explain the structural difference on screen between wrapping text inside paragraph tags (<p>...</p>) versus separating text lines using a line break tag (
).
7. State the exact HTML code snippet required to display the text "Revision Master" in bold styling.
8. State the purpose of adding the target="_blank" attribute configuration into a web page hyperlink anchor tag.
9. Identify two standard design features that are typically locked into a website's master header banner zone.
10. Which character is used inside an HTML tag to signify to the web browser that a specific markup effect is ending?
 - ☐ A) ?
 - ☐ B) !
 - ☐ C) /
 - ☐ D) *

Spreadsheet

Spreadsheets (also known as workbooks) are software applications used to store structured numeric and text data, perform automatic mathematical calculations, and transform raw numbers into informative charts to uncover patterns or trends. Popular engine examples include *Microsoft Excel* and *LibreOffice Calc*.

1. Workbook Architecture & Cell Geography

The Matrix Layout

A workbook is a collection of one or more distinct sheets called **worksheets**. Each worksheet is organized into a grid made up of columns and rows:

- **Columns:** Vertical sections identified by letters of the alphabet (A, B, C, D, E, F...).
- **Rows:** Horizontal sections identified by consecutive numbers (1, 2, 3, 4, 5...).

	A	B	C	D	E	F
1	[]	[]	[]	[]	[]	[]
2	[]	[]	[]	[]	[]	[]
3	[]	[]	[]	[]	[]	[]
4	[]	[=====]		[]	[]	[]
5	[]	[Active Cell B4]		[]	[]	[]
6	[]	[=====]		[]	[]	[]
7	[]	[]	[]	[]	[]	[]

Cell References and Selection

- **Cell:** The precise rectangular space formed where a column and a row cross each other.
- **Cell Reference:** The unique coordinate identifier of a cell, created by combining its column letter and row number. For example, the very first cell is referred to as **A1** because it is in Column A and Row 1.
- **Active Cell:** The specific cell that has been selected by a user. It is highlighted by a green box, its column and row headings are highlighted, and its coordinate appears in the **Name Box** in the top-left corner.
- **Formula Bar:** Located directly next to the Name Box, this bar displays the underlying data, text, or equation nested inside the active cell. If the cell is empty, the Formula Bar remains blank.

Defining Ranges

- **Range:** A collective group of adjacent cells in a worksheet.
- **Syntax:** A range reference is declared by typing the first cell in the top-left corner, followed by a colon (:), and then the final cell in the bottom-right corner.

- *Example:* The cell range reference **B4:D7** encompasses every cell from Column B to Column D, and down from Row 4 to Row 7.

2. Data Types & Structural Formatting Rules

Formatting your worksheet effectively guarantees that your data is readable and acts exactly the way you expect it to during analysis.

Number vs. Text Data

Cells interpret entries differently based on how the data is classified:

- **Number Data:** Defined as data on which you can perform mathematical calculations.
- **Text Data:** Alphanumeric characters, labels, or numbers that will never undergo math operations.
- *Critical Concept:* Identification figures like telephone numbers **must** be stored as text data. Because phone numbers often start with a 0, a spreadsheet treating it as numeric data will automatically omit the leading zero, ruining the entry.
- *Quick Trick:* You can format a value as text instantly by typing a leading single quote or apostrophe (') directly before the value.

Common Number Formats

The **Format Cells** window gives you precise control over how numbers look on-screen without changing their actual background value:

Format	Example	Purpose & Notes
Currency	£25.00	Used to display financial money values. It provides choices for automatic formatting of negative numbers (such as using red text or brackets). <i>Note: The Accounting format can also be used here.</i>
Percentage	25%	Used to display percentage values as proportions of 1. To display 20%, you must enter the decimal value 0.2. Entering a whole 20 and switching to percentage will show 2000%.
Number	0.25	Controls the number of decimal places used to display numbers to different levels of accuracy. Adjusted quickly using the Increase Decimal and Decrease Decimal tools on the toolbar.
Date & Time	24/06/2010	Maps numeric entries to custom calendar layouts. Dates can toggle between Short Date (01/01/2019) and Long Date (Monday, 24 June 2019) format options.

Layout Control Tools

- **Merge and Centre:** A spreadsheet tool that combines multiple selected cells into a single, large cell and center-aligns the text within it.

- **Text Wrap:** Prevents long text labels from overlapping into adjacent empty rows by forcing the text to wrap into multiple lines within the same cell block.
- **Column Width & Row Height:** Adjusting these paths ensures data is fully visible. If a column is too narrow to display numeric data, the application will display placeholder characters instead, such as #####. You can change width or height by dragging or clicking the divider lines between column headers.
- **Gridlines:** The vertical and horizontal lines that define cell boundaries. They do not print by default unless explicitly chosen using an option found in the *Page Layout* menu.
- **Hiding Rows/Columns:** Allows you to remove temporary clutter from the screen or printouts. Hidden data remains fully active within background worksheet calculations.

3. Building Formulae & Order of Operations

A **formula** allows you to carry out arithmetic operations on numeric data stored in your spreadsheet cells.

Essential Formula Rules

- **The Equals Sign Rule:** When using a spreadsheet application, **all formulae must start with an = sign**. If you omit it, the program handles your input as a text label.
- **Mathematical Operators:** Spreadsheet calculations use specific operational symbols:
 - Add: +
 - Subtract: -
 - Multiply: * (The asterisk symbol is used; never use the letter x)
 - Divide: /

Single vs. Multiple Operator Configurations

- **Single Operator Formula:** A formula utilizing only one operational action step. For example, multiplying call costs per minute by usage minutes: =B4 * B12.
- **Multiple Operator Formula:** Combines multiple different operators together to solve complex mathematical strings. For example, adding monthly metrics together before multiplying them over an annual period: =B3 + (B17 + B18 + B19) * 12.

Mastering BIDMAS Order of Operations

If you use multiple operators in a single formula, the spreadsheet program computes your math string from left to right using a strict priority sequence:

1. **Brackets** (Calculations inside parentheses are always solved first)
2. **Indices** (Powers, exponents, or square roots)
3. **Division and Multiplication** (Evaluated sequentially as they appear from left to right)
4. **Addition and Subtraction** (Evaluated last)

Why Formulae Trump Manual Calculation: Using automated cell references means you

never have to redo your work if your raw numbers change. If an input cell updates, the dependent formula automatically recalculates the result instantly.

4. Replication Mechanics & Cell Referencing

Replicating via the Fill Handle

You can copy a formula down an entire column or across a row instantly by using the **Fill handle**—the tiny square displayed at the bottom-right corner of the active cell selection box.

[Cell B17: =B4*B12]
 [] [■] <-- Click and drag this Fill Handle
 downwards

Relative Cell Referencing

By default, spreadsheet formulae use **relative cell referencing**. This means when you copy or drag a formula to a new cell, the formula maintains the exact structure of the original equation but automatically updates its cell references to point to cells in the same relative position to the new cell.

- *Example:* If cell E2 contains the formula =A2 * C2, dragging it down to row 3 replicates the formula as =A3 * C3.

Absolute Cell Referencing

An **absolute reference** is used when you do not want a cell reference in a formula to change when the formula is copied or replicated from one cell to another. It fixes the references to a particular column or row.

- **The Syntax:** You freeze a coordinate by adding a dollar sign (\$) in front of either the row or column part, or in front of both parts of the cell reference.
- *Example:* In the multiplication layout below, cell C3 references cell A3 relatively, but references the multiplier cell D2 absolutely as D\$2. When replicated down to row 4, the formula becomes =A4 * D\$2.

	A	B	C		D
1	[]	[]	[]		[]
2	[]	[]	[]		[3] <-- Fixed number to multiply
3	[1]	[]	[=A3 * D\$2]		[]
4	[2]	[]	[=A4 * D\$2]		[]

5. Data Summarization via Core Functions

A **function** is a built-in mathematical expression that represents a complex calculation or process automatically, preventing manual syntax errors.

Mathematical and Statistical Functions

- **SUM:** Calculates the total combined sum of a designated range of values.
 - *Example:* =SUM(C5:E5)
- **AVERAGE:** Calculates the mathematical mean of a range of values.
 - *Example:* =AVERAGE(C5:E5)
- **MINIMUM (MIN):** Scans a range of cells and returns the smallest value.
 - *Example:* =MIN(C5:E5)
- **MAXIMUM (MAX):** Scans a range of cells and returns the largest value.
 - *Example:* =MAX(C5:E5)
- **PRODUCT:** Multiplies a range of values together. This is highly efficient for multiplying large adjacent cell blocks, such as =PRODUCT(B1:B3, C1:C3), which performs the same calculation as =B1*B2*B3*C1*C2*C3.

Counting Functions

- **COUNT:** Counts how many cells in a specified range contain numbers.
- **COUNTA:** Counts how many cells in a specified range are not empty (counting both text and numbers).
- **COUNTIF:** Counts how many cells in a range meet specific designated criteria or conditions.
 - *Example:* =COUNTIF(K4:K9, G12) checks the range K4:K9 and counts the number of times it finds a value matching the item in cell G12.

Logical Analysis: The IF Function

The IF function performs a logical test of a condition using conditional operators (such as =, >, <, >=, <=). It returns one value if the test condition is met (**True**) and another value if the condition is not met (**False**).

- *Example:* =IF(J6=3, VLOOKUP(G6, boundaries, 2), "Incomplete") checks if the value in cell J6 is equal to 3. If true, it performs a lookup; if false, it outputs the text string "Incomplete".

Cross-Reference Indexing: VLOOKUP & LOOKUP

- **LOOKUP:** Finds a value in a row or column range and then moves to another row or column range to return an associated value.
 - *Example:* =LOOKUP(G5, F12:F15, G12:G15)
- **VLOOKUP:** Looks down the first column of an array vertically until it locates a specified value, then moves across that row to return the value of a cell from a designated column number.

- **Sorting Rule:** The values in the first column of an array used in a LOOKUP or VLOOKUP function **must be in ascending order** for these functions to work correctly.
- **Example:** =VLOOKUP(G6, boundaries, 2) searches for the value of cell G6 in column 1 of the array named boundaries, then returns the grade text from column 2.

6. Visual Data Analytics & Formatting Charts

Data can be represented visually using graphs and charts to make trends, relationships, and patterns within a dataset much easier to see, analyze, and interpret.

Choosing the Correct Chart Type

- **Pie Chart:** Used to represent categorical data as a proportion or percentage of a complete whole.
- **Bar / Column Chart:** Used to compare discrete values between different categories or groups using horizontal bars (bar chart) or vertical bars (column chart). Categories are placed on the horizontal x-axis and values are placed on the vertical y-axis.
- **Line Chart:** Used to track and show how data values change over time, making it easy to identify upward or downward trajectories and spot long-term trends.
- **Scattergram:** Used to show the visual relationship or level of correlation between two distinct sets of numerical data values plotted on the x-axis and y-axis.

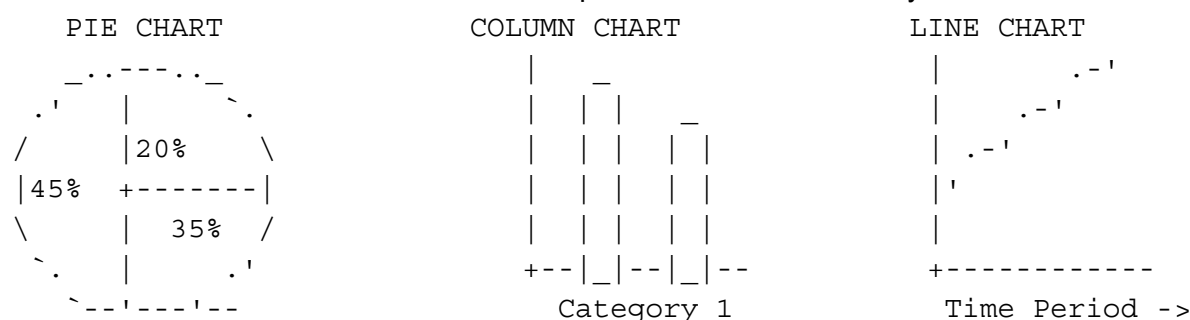


Chart Formatting Elements

Formatting adds clear context so that an audience can read and interpret your chart quickly:

- **Title:** Explanatory text positioned at the top stating exactly what the chart represents.
- **Axis Labels:** Clear text descriptions detailing the measurement units and categories mapped along the x-axis and y-axis lines.
- **Legends:** A color-coded key index that identifies what different colors or series represent in a chart.
- **Scale:** The exact mathematical intervals and boundary limits chosen between numbers on an axis.
- **Trend Line:** A straight or curved line of best fit drawn through data points to show the general direction your numbers are heading.

7. Sheet Output & Printing Operations

Worksheets can be viewed in two distinct operational modes:

- **Data View:** The default view which displays the computed visual results of formulae contained by the worksheet.
- **Formula View:** A manual diagnostic mode that shows the actual background formulae used in the worksheet, rather than their results. Toggled via **Show Formulas** under the *Formulas* tab in Excel.

Production Layout Controls

- **Set Print Range:** Allows users to highlight a specific cell selection block and print only those selected cells via the print dialogue menu, rather than printing the entire worksheet.
- **Orientation:** Allows users to adjust the output layout between **Portrait** format (vertical height orientation) or **Landscape** format (horizontal width orientation) to accommodate wide tables.
- **Headers & Footers:** Places repeating textual data (such as page numbers, author names, or document titles) at the very top and bottom margins of every single printed sheet.
- **Row and Column Headers to Repeat:** If a table spans multiple pages, column headers will only appear on the first page. By utilizing the **Print Titles** feature, you can choose specific rows or columns (e.g., Row 1 containing column labels) to automatically repeat at the top of every single printed page.

Student Activities

Activity 1: Cell Formatting Exploration

1. Open a new, blank worksheet in your spreadsheet software.
2. In cell A1, type the telephone number 07700900123 and press Enter. Note down exactly what happens to the leading zero character.
3. Select cell A2, right-click, select *Format Cells*, and set its category type to **Text**.
4. Type the same telephone number 07700900123 into cell A2. Observe and describe what happens to the initial zero now.
5. In cell A3, type '07700900123 (using a leading single quote apostrophe). Note whether this functions similarly to step 4.

Activity 2: Relative vs. Absolute Referencing Setup

1. Enter the values 10, 20, and 30 down cells A3, A4, and A5.
2. In cell D2, enter the number 5.

3. In cell C3, enter the formula =A3 * D2.
4. Use the cell Fill Handle to click and drag this formula down across cells C4 and C5.
Note why cells C4 and C5 return unexpected zero results or errors.
5. Double-click cell C3 and modify the formula to utilize absolute row locking syntax: =A3 * D\$2.
6. Replicate this updated formula downward using the Fill Handle to cells C4 and C5.
Note down the newly calculated correct outcomes.

Chapter Questions

1. Which one of these is a valid cell range reference style?
 - A) G1-J1
 - B) G1 to J1
 - C) G1 → J1
 - D) G1:J1
2. Explain why named ranges are useful in a spreadsheet when creating calculations.
3. Explain why absolute referencing is used in spreadsheets, referencing how it alters during formula replication.
4. Which one of these functions would you use to calculate the total addition sum of a range of cells?
 - A) SUM
 - B) COUNTIF
 - C) COUNT
 - D) PRODUCT
5. List three distinct types of charts that you could choose to represent different forms of spreadsheet data values visually.
6. Describe the background mechanics of how a VLOOKUP function operates to retrieve relevant data.
7. State the most appropriate chart type to choose if you want to display the level of correlation between two distinct sets of data values.
8. State the most appropriate chart type to choose if you want to represent data values as a proportional segment of an entire whole.
9. State the most appropriate chart type to choose if you want to visually compare individual value metrics across different categorical groups.
10. Describe how adding an automated trend line over a series of plotted chart data values can be utilized to make forward business predictions.
11. State the critical structural advantage of using dynamic formulas containing cell references to perform math operations over manually entering static numeric values.
12. Which one of these logical test statements will successfully evaluate to a TRUE condition?
 - A) 9 > 11
 - B) 9 = (5 + 4)
 - C) 9 < 5
 - D) 9 >= 10

Database Management

A **database** is a structured, organized collection of data stored electronically in a computer system. Databases allow users to input, store, organize, manipulate, search (query), and output data efficiently in either a printed or electronic format. Common database applications include *Microsoft Access* and *LibreOffice Base*.

Database Architecture & Hierarchy of Data

Databases organize data into a strict hierarchy, moving from the smallest structural element (a single attribute) to the largest container (the entire relational system).

- **Field:** A single piece of information or a specific characteristic about an object, person, or event. It represents a column in a database table. Examples include VisitorID, Surname, or DateOfBirth.
- **Record:** A complete set of data relating to a single item, person, or entity. A record is made up of multiple fields and represents a horizontal row in a database table.
- **Table:** A collection of related records structured into rows (records) and columns (fields).
- **Relational Database:** A database structure that contains multiple tables linked together by common fields to share information and eliminate data redundancy.

Fundamental Data Types

When a field is created in a database table, it must be assigned a specific data type. This determines what kind of data can be entered into the field and ensures the computer handles and stores it correctly.

Data Type	Description	Practical Examples
Alphanumeric / Text	Accepts letters, numbers, punctuation marks, and symbols. It is also used for numeric strings that do not require mathematical calculations.	Jon, HR12 4QQ (Postcode), 07700900123 (Telephone)
Numeric / Number	Used exclusively for numbers that will be used in mathematical calculations. Can be whole integers or decimals.	15 (Quantity), 1.85 (Height in meters)
Date / Time	Stores calendar dates and specific times. Automatically enforces standard calendar rules.	24/06/2010, 14:30
Currency	Automatically attaches a monetary symbol and fixes decimal places. Used for financial records.	£25.00, \$120.50
Logical / Boolean	A state that can only hold one of two possible mutually exclusive values.	Yes / No, True / False, M / F

Relational Infrastructure & Key Keys

To connect multiple tables together in a relational database, specific key fields must be established to ensure data integrity.

Primary Key

A **Primary Key** is a field (or a combination of fields) that uniquely identifies each individual record in a database table. No two records in the same table can share the exact same primary key value.

- *Examples:* StudentID, NationalInsuranceNumber, ProductID.

Foreign Key

A **Foreign Key** is a field in one table that links to the primary key field in another table. It is the structural mechanism used to establish and enforce a relationship between the two tables.

Types of Relationships

- **One-to-Many Relationship (1:\infty):** The most common relational structure. A single record in Table A can match multiple records in Table B, but a record in Table B can only match one record in Table A.
 - *Example:* One Customer can place many Orders, but each individual order belongs to only one specific customer.

Enforcing Data Integrity: Validation Checks

Validation is an automatic computer check performed by the software when data is entered into a database. Its purpose is to ensure that the data is sensible, reasonable, and valid before it is accepted into storage.

Important Distinction: Validation checks if data *could* be correct (sensible), but it cannot check if data is *actually* correct (accurate). If a user enters a valid but incorrect postcode, validation will accept it.

Standard Validation Types

- **Presence Check:** Ensures that a field cannot be left completely blank during data entry.
 - *Application:* Making a Surname or EmailAddress field mandatory on a registration form.
- **Range Check:** Checks that a numeric or date value falls within a specified upper boundary and lower boundary.
 - *Application:* Ensuring an examination percentage field only accepts values ≥ 0 and ≤ 100 .
- **Length Check:** Verifies that the data entered contains a specific number of

characters, or stays within a strict character limit.

- *Application:* Setting a Gender field to accept a maximum length of 1 character (M or F), or enforcing a fixed length for a password.
- **Type Check:** Verifies that the entered data matches the data type assigned to that field.
 - *Application:* Preventing a user from accidentally typing alphabetical letters like "abc" into a numeric Age field.

Information Retrieval: Queries & Search Logic

A **query** is a search extraction method used to interrogate a database table and display only the records and fields that meet specific, designated search conditions (criteria).

Relational and Logical Operators

Queries use operators to filter records precisely:

- **Relational Operators:** Used to compare values against one another:
 - = (Equal to)
 - < (Less than)
 - > (Greater than)
 - <= (Less than or equal to)
 - >= (Greater than or equal to)
 - <> (Not equal to)
- **Logical Operators:** Used to combine multiple search conditions together:
 - **AND:** Returns records *only* if **all** specified search criteria are simultaneously met.
 - **OR:** Returns records if **at least one** of the specified search criteria is met.
 - **NOT:** Inverts the search criteria to return everything *except* the specified value.

Multi-Field Sorting

Databases allow you to sort query results using single or multiple fields in either **Ascending** order (A to Z, 0 to 9) or **Descending** order (Z to A, 9 to 0). When sorting by multiple fields, the second field is used to break ties if duplicate values exist in the first sorted field.

Input & Output Operations: Forms, Reports, and Mail Merge

Database interactions are divided into operational interfaces designed for clear inputting and structured outputting.

Data Entry Forms

A **Form** is an onscreen user interface designed to simplify the process of inputting, editing, and deleting records in a database table. Instead of typing data directly into a massive grid

table, forms present clear input text boxes, drop-down menus, and radio buttons for a single record at a time, making data entry less prone to human error.

Database Reports

A **Report** is a formatted, structured presentation of data extracted from a database query or table, specifically designed to be printed or distributed electronically. Unlike tables, reports allow you to group related data, format layouts with custom headers and footers, and calculate summary values (such as grand totals or averages).

Mail Merge Documents

A **Mail Merge** is a process that automatically combines a single word-processed template document (such as a customer notification letter or email) with a structured database table or query address list.

- **Placeholders / Merge Fields:** Special markers (e.g., <<FirstName>>, <<Surname>>) are inserted into the document template.
- When the mail merge is run, the software automatically loops through the database table, replacing the placeholders with unique data values from each record to generate personalized individual letters.

Student Activities

Activity 1: Conceptualizing the Hierarchy of Data

1. Look at the data table provided below:

VisitorID	FirstName	Surname	Gender	Age
1	Jon	Smith	M	24
2	Sarah	Jones	F	31
3	Amanda	Patel	F	19

1. Using the data table above, identify and write down:
 - The total number of **fields** present in this table structure.
 - The total number of data **records** saved into this table.
 - A suitable, data-safe primary key field.

Activity 2: Designing Validation Strategies

Imagine you are setting up a database table to store details for a school's sports club enrollment. For each of the fields listed below, determine the most suitable **Data Type** and select a highly appropriate **Validation Check** that should be implemented to prevent incorrect data entries:

1. StudentID (e.g., STU12345)
2. DateJoined
3. EmergencyContactPhoneNumber
4. PaidMembershipFee (Yes/No status)

Chapter Questions

1. Which database component is represented by a vertical column containing a single type of information across an entire table?
 - A) Record
 - B) Field
 - C) Primary Key
 - D) Relationship
2. Explain why a telephone number or a postal area code must be configured as an Alphanumeric/Text data type instead of a Numeric data type.
3. Describe the distinct roles of a Primary Key and a Foreign Key in establishing a secure link between two database tables.
4. Which one of these validation checks is specifically designed to prevent a critical field on an online database form from being left blank by a user?
 - A) Range Check
 - B) Length Check
 - C) Type Check
 - D) Presence Check
5. A theme park database query needs to extract a list of all visitors who are female (F) AND are aged 18 or older. Write out the logical search criteria statement required to execute this query.
6. Explain the operational difference between running an automated validation check on a database field versus manually proofreading the entered data.
7. State the name of the database feature that is specifically designed to provide an organized, visually polished output layout of query results suitable for printing.
8. State the type of relationship created when a single record in a Customers table can link to multiple separate records in an Orders table.
9. Define what an automated "Mail Merge" process achieves and describe the role that database placeholders play within it.
10. Which relational operator represents the mathematical concept of "Not equal to" within a database query environment?
 - A) <=
 - B) >=
 - C) <>
 - D) ==

ANSWERS

CHAPTER 1

- 1 ▶ Barcodes save time (1).
OR
Barcodes ensure that fewer errors are made when entering product ID numbers (1).
- 2 ▶ These devices are designed to be portable (1).
- 3 ▶ The charge cannot be reduced on the capacitive layer (1) because wool is not a conductive material (1). (The charge is usually drawn away by touch, thereby **reducing** the charge held in the capacitive layer.)
- 4 ▶ Any **three** from: a burglar alarm (1); a web server (1); an engine management system (1); a home heating system (1); many other examples.
- 5 ▶ Laser printers are faster (1) and produce each copy more cheaply (1) than inkjet printers.
- 6 ▶ Any **two** from: architecture (1); product design (1); cartography (1); engineering (1).
- 7 ▶ To reduce the impact (1) on the environment (1) when the finished models are discarded (1).
- 8 ▶ Several answers. For example: An accelerometer (1), used to detect 'tilt' in a game (1), and a light sensor (1), used to alter the brightness of a screen to match the environmental light conditions (1).
- 9 ▶ Several answers. For example: Sensors are used in cars for many reasons. Some examples include detecting whether doors are open, whether seatbelts are secured, whether passengers are in their seats and whether it is dark outside. This type of feedback on the environment can be useful to the driver, so that they know what is around the car. For example, the car could use distance sensors to help drivers to park or avoid collisions. Similarly, cars could use light sensors under the car to help drivers know if they are moving out of a lane on a road and the car's computer could trigger an alarm to alter the driver's course. The use of sensors can take some of the need to think away from the driver, allowing them to concentrate on the driving. Sensors can also produce reactions that are quicker than a human's, therefore increasing road safety. However, sensors (and the computers to which they are linked) are not good at making ethical decisions and so, if they sensed a person in the road ahead, they may not be able to decide whether to steer off the road and maybe crash into a wall (potentially injuring the driver) or crash into the person crossing the road, potentially injuring the pedestrian. In summary, sensors can be very useful, but humans will always need to take responsibility for their actions and should not become complacent when in charge of a vehicle.
- 10 ▶ Console games use detailed graphics (1), which use a lot of data and therefore data storage (1).
OR:
Optical media is required because download sizes would be too big (1) and Blu-ray media has the greatest capacity of all optical formats (1).

CHAPTER 2

- 1 ▶ A piece of software you can use to carry out specific tasks that are not related to the operating system (1).
- 2 ▶ Any **three** from: backup (1); anti-virus (1); defragmentation (1); compression (1); disk management (1).
- 3 ▶ The operating system allows the user to control and manage (1) the computer's hardware (1).
- 4 ▶ So that backup processes can be completed when the system is not in use (1) to avoid file conflicts (1).
- 5 ▶ Compression utilities can free up storage space (1) by reducing the amount of data (1) needed to store a file.
- 6 ▶ By allowing the user to create slides that contain content (1) that support (1) their talk.
- 7 ▶ Several answers. For example: Control software could be used in manufacturing factories (1) to control robots on a production line (1).
- 8 ▶ Several answers. For example: Building companies must adhere to scheduled timelines because multiple resources such as materials and personnel such as electricians need to be coordinated in sequential phases which are dependent upon each other. This ensures that the project hits targets, and forecasts for completion can be provided, with progression monitored so that stakeholders (such as homeowners) can be updated with the progress of the project.
- 9 ▶ Any **three** from: web browser (1); social media (1); SMS (1); email (1); MMS (1); instant messaging (1).
- 10 ▶ Any **two** from: fixing security vulnerabilities or bugs (1); increasing compatibility with newer operating systems (1); improving performance and efficiency (1); introducing new features (1); improving usability (1).

CHAPTER 3

- 1 ▶ Read Only Memory (1).
- 2 ▶ Random Access Memory (1).
- 3 ▶ Increasing the amount of RAM means that more programs can be open at the same time (1) without needing to use virtual memory from storage (1), which has slower access times than RAM (1).
- 4 ▶ Virtual memory is an area of secondary storage (1) used to store data when the computer system does not have enough space in RAM (1).
- 5 ▶ Any **three** from: volatile (1); instructions can be swapped in and out or temporarily stored (1); used for temporary storage (1); upgradable (1).
- 6 ▶ Flash memory does not have any moving parts (1), so it has low power consumption (1). It also means that it is less susceptible to data loss (1) if the device is knocked or shaken (1).
- 7 ▶ A The number of times per second that a processor can carry out instructions (1).

CHAPTER 4

- 1 ▶ a Infra-red signals cannot travel as far as Bluetooth signals (1) and the console is sometimes not in range (1).

OR

Infra-red requires line of sight (1) and the controller is often pointed away from the console when in use (1).

- b Loss of connection would give the user a bad experience (1).
- 2 ▶ Parts of a video file are downloaded to a buffer (1), which is then topped up (1) as the video plays (1).
- 3 ▶ The user does not have to wait (1) for the full file to be downloaded (1) before it can be played (1).
- 4 ▶ Several answers. For example: USB (1), Ethernet (1) and HDMI (1).
- 5 ▶ B WAN (1).
- 6 ▶ People play games with others from all round the world (1) and they want to avoid lag (1) caused by latency (1).

CHAPTER 5

- 1 ▶ B Gateway (1).
- 2 ▶ Public key encryption does not use the same key (1) to encrypt and decrypt the data (1).
- 3 ▶ Firewall (1).
- 4 ▶ IP address (1) and MAC address (1).
- 5 ▶ IPv4 did not provide enough unique addresses (1) to service the number of devices being connected to the internet (1).
- 6 ▶ Any **two** from: working from home (1); avoiding geolocation rights management (1); making secure payments (1).
- 7 ▶ A booster is used if the network signal has a limited range (1).
- 8 ▶ Data is scrambled (1) using a key (1), meaning that it can only be accessed by other users with the key (1).
- 9 ▶ Any **three** from: logins/passwords (1); firewalls (1); encryption/VPN (1); file access rights (1).
- 10 ▶ Any **three** from: access to shared peripherals (1); access to shared storage and data (1); ability to use any device on the network (1); streaming media such as movies, music and games (1); ability to send messages and files to others on the network (1); shared access to the internet (1).
- 11 ▶ Several answers. Ask your teacher to check your diagram.

CHAPTER 6

- 1 ▶ C Firewall (1).
- 2 ▶ Characters cannot be read or interpreted (1) by an automated computer program (1).
- 3 ▶ To gain users' personal information without their permission (1).
- 4 ▶ Any **three** from: card number (1); expiry date (1); start date (1); name of the cardholder (1); CSC (1).
- 5 ▶ A differential backup saves a copy of all files that have changed since the last full backup (1). An incremental backup saves a copy of only the files that have changed since the last full **or incremental** backup (1).
- 6 ▶ Encryption ensures that unauthorised users cannot read data (1) without the key (1).

- 7 ▶ Any **two** from: passwords (1); PINs (1); biometrics (1).
- 8 ▶ Domain name spoofing (1) and redirecting traffic (for example, using malware) (1).
- 9 ▶ It can be slower to backup (1) because the connection to the internet provides less bandwidth than a local data connection (1).

CHAPTER 7

- 1 ▶ C Streaming (1).
- 2 ▶ Any two from: no need to travel to work (1); no stress caused by traffic or public transport delays and overcrowding (1); saves money (1); reduces impact on the environment (1); employees can spend more time working because they spend less time travelling to work (1); no need to wear uniform or business clothes, reducing employees' costs (1); employees can work at a time that suits them (1); employees can work on tasks for longer periods of time without distractions from colleagues or scheduled meetings (1); employees can work in a comfortable environment (1); employees can organise their work around social or family commitments (1).
- 3 ▶ Any **two** from: there may be concerns about data security (1); it can be more difficult to manage and support employees who are not in the office (1); employees working at home might not work as hard as employees in an office (1); it can be complicated to organise payments and permissions for employees in different countries (1).
- 4 ▶ Body language (1) is not always visible online (1) so people do not practise using and reading body language and other social and emotional cues (1).
- 5 ▶ Bullies can remain anonymous (1), making the bully feel as though they can behave more inappropriately (1).

OR:

Bullies can remain anonymous (1), which removes victims' ability to control the situation (1).

- 6 ▶ Working together (1).
- 7 ▶ Any **three** from: economy and infrastructure (1); location (1); politics (1); religion (1); disability (1); social factors such as age, gender, education and income (1).
- 8 ▶ Any **three** from: entertainment and leisure (1); shopping (1); education (1); banking (1); tax (1); applications for driving licences (1).
- 9 ▶ It has helped build strong communities (1) using social media (1).
It has increased access to the global society (1) at the expense of their local societies (1).
- 10 ▶ The gap has been made wider (1) because the internet has increased the amount of information available to the information rich (1).
- 11 ▶ Several answers. For example: The internet provides more opportunities for us to socialise by allowing us to connect with each other using global communication services such as instant messaging and social networks. We can communicate at any time of the day with friends from around the world and make use of the ability to connect data from different account profiles to create

new connections with people we have never met. We use social features in many services, such as when we are watching television, using television as a 'back-channel' line of communication to comment upon, rate and discuss live television programmes and news. This allows us to access more than one point of view about the issues being shown on the news. We also use social features of the internet when shopping and learning in order to get more than one perspective, keep informed of new developments and gain a wider perspective on the world. However, some people's use of the internet means that they are not in touch with the real-world communities in their local geographical area, leading to the disintegration of those communities. Some people become so engrossed in online communication that they lose the skills that they need to communicate with people in the real world. Some people also argue that, despite the wide variety of social groups available to individuals, the ones you tend to join are those which share your views, as that is often how suggestions are made based on users' profiles. This means that individuals who think they are joining in with a wider community are simply reaffirming their own opinions, so they do not actually come to understand and integrate with the wider social communities available to them via the internet. In conclusion, there is a wide variety of social interaction available thanks to the internet, but people need to understand the ways in which they are connected to others to truly understand how to make the most of the opportunities that are available to them.

- 12 ▶ Keep personal information private (1) by being careful what is shared and by using privacy settings to control access to personal profiles (1); block people, especially strangers (1), who are offensive or whose behaviour is concerning (1); report (1) inappropriate behaviour to relevant authorities (1).
- 13 ▶ The use of the internet provides another method (1) of unauthorised access (1) to the data held by the organisation (1).

CHAPTER 8

- 1 ▶ D Acceptable use policies (1).
- 2 ▶ Tags are used to categorise (1) information so that it can be located more easily (1).
- 3 ▶ Sharing increases the reach of content (1) and encourages members to be more active users of the service (1). This means that they provide the service with more information (1), which the service can then use to sell marketing opportunities to advertisers (1).
- 4 ▶ Any **three** points from: Different levels of user responsibility can be set (1) so that people can be responsible for quality checking members and their posts (1) to ensure that the policies are being followed (1), block posts or enforce sanctions such as banning members from the community (1).
- 5 ▶ Any **three** from: biographical details such as name, gender, date of birth, location and language (1); an 'about you' or short description of the user (1); details about the

user's work and education (1); travel history (1); family details such as relationship status, family members and pets (1); contact information such as telephone number, email address and website (1); profile and background images, colour schemes and designs (1).

- 6 ▶ To allow users to share URLs (1).
- 7 ▶ Any **three** points from: Because the organisation's workers will be located in different countries (1), which would make communicating and collaborating on work more difficult or expensive to organise and time consuming (1) due to differences in time zones and the costs and time associated with travelling (1). They would also benefit from having centralised resources such as virtual meeting spaces, shared documents and shared online calendars (1).
- 8 ▶ By matching (1) information shared by members (1).
- 9 ▶ By matching (1) information shared by members (1).
- 10 ▶ Any **four** from: students can access resources (1); students can interact with teachers (1); students can get notifications of assignments and grades (1); students can chat with classmates and share ideas and research (1); students can get immediate feedback from automatically marked quizzes (1); students can collaborate on assignments with other students and teachers (1).
- 11 ▶ Rate/like/upvote (1); share/repost (1); comment/quote (1).
- 12 ▶ Answers should discuss: cost of and need to protect their safety; companies sharing users' personal information; targeted marketing; cost to users' privacy.

CHAPTER 9

- 1 ▶ C Using an RCD (1).
- 2 ▶ They have reduced the cost of technology (1) so that more people can afford digital devices (1).
- 3 ▶ Location sharing has an impact upon their privacy (1) because their location could be shared with other people with whom they do not actually want to share their location (1).
- 4 ▶ Any **three** from: music (1), e-books (1), video (1), apps (1).
- 5 ▶ Several answers. For example: CCTV (1) can be used to watch people (1); ANPR (1) can trigger alerts for police (1); ID, travel or bank cards (1) can be tracked when scanned at terminals or used in transactions (1); IP or MAC addresses (1) can be identified on networks (1); GPS position (1) can be broadcast to apps (1).
- 6 ▶ To make them the only people who can legally use and distribute their work (1) and to prevent people from making a profit by using their work (1).
- 7 ▶ Several answers. For example: It can help parents keep their children safe (1); It can help friends to find each other (1).
- 8 ▶ Several answers. For example: It can help parents see what their children are saying online to keep them safe (1); It can help governments protect individuals by identifying potential criminals (1).
- 9 ▶ It reduces their privacy (1).
- 10 ▶ Any **three** from: ask people for consent (1); use data appropriately (1); use data only for the specified purpose (1); use data fairly and lawfully (1); use data for limited,

specifically stated purposes (1); use data in a way that is adequate, relevant and not excessive (1); ensure that data is accurate (1); keep data for no longer than is absolutely necessary (1); handle data according to people's data protection rights (1); keep data safe and secure (1); if in the EU, not transferring data outside the European Economic Area without adequate protection (1).

- 11 ► Several answers. Should include up to four linked points from: online storage and apps are based in large data centres (1) which use lots of energy (1) to cool the servers (1) and power the servers (1). This energy could be created by using non-renewable energy (1), in order to avoid depleting the planet's natural resources and contributing to climate change (1).
- 12 ► Several answers. For example: Recycle their devices (1) so they do not leak toxins (1) which can cause health risks to plants, wildlife and people (1).

CHAPTER 10

- 1 ► D "" (1).
- 2 ► Any **two** from: size (1), colour (1), type (1), time and date (1), usage rights (1).
- 3 ► It might not represent the issue in full (1).
- 4 ► A search engine compares (1) words or phrases entered by a user (1) against a database of pages (1) and returns the URLs or descriptions of those that match (1).
- 5 ► You might infringe others' rights (1), leading to ethical and legal issues (1).
- 6 ► It shows that the student understands the meaning of the information (1).
- 7 ► AND (+) (1), NOT (-) (1), phrase matching ("") (1).
- 8 ► **Table 10.3** Identifying primary and secondary sources

Source	Primary or secondary?
A podcast that you have created	Primary (1)
A video of a discussion that you have with an expert	Primary (1)
Your own copy of a film	Secondary (1)
Notes that you take while visiting a museum	Primary (1)
A book that you borrow from someone	Secondary (1)
An image that you download from the internet	Secondary (1)
Your own recording of a radio broadcast	Secondary (1)

CHAPTER 11

- 1 ► **A** Paying in cheques (1).
- 2 ► Any **two** from: VLEs (1), online support (1), remote access to journals and books (1).
- 3 ► Several answers. For example: basket (1), product catalogue (1), checkout (1).
- 4 ► EITHER: It helps sellers know which buyers they can trust (1) to pay for items (1).
OR: It stops sellers having to relist (1) unpaid-for bids

because they can restrict access to bidders with good ratings who are more likely to complete the purchase (1).

- 5 ► It uses encryption (1) which stops data, such as people's financial information, being read (1) by unauthorised users (1).
- 6 ► There is no need to check in at a check-in desk at the airport (1), which saves passengers time (1).
- 7 ► EITHER: Their use affects users' privacy (1).
OR: Users' experience is based on decisions made by computer code rather than on choices that they make (1).
- 8 ► Any **three** from: time and date of the transaction (1); IP addresses (1); referring website (1); the products and services previously viewed or bought (1); categories of products and services previously viewed or bought (1); items added to online shopping basket (1); buttons pressed (1); data entered into web forms, such as name and address (1); any other sensible response.
- 9 ► Session cookies are only stored while the browser has the site open or until the browser closes the site. Persistent cookies are stored after the browser closes the site.
- 10 ► The use of notifications or alerts (1).
- 11 ► Several answers. Relevant points include: Organisations use cookies to track the transactions that individuals carry out online, such as the websites that they visit. This browsing history can then be used by organisations to target their marketing and create a personalised experience for users. Although some people find this useful, other people do not like the fact that their online activity is tracked and monitored; as they believe this to be intrusive and to have a negative effect on their privacy. They do not like the ways in which these systems are used to decide the content people are shown and say that they would prefer to decide this themselves as it may limit or narrow their experience.

CHAPTER 12

- 1 ► **C** Server (1).
- 2 ► Any **three** from: automatic backup (1); greater scalability (1); enables sharing data (1); cheaper than local storage or even free (1); makes collaboration possible (1); reduces use of local storage making more space for locally installed software (1).
- 3 ► Any **three** from: shopping or auction sites (1); booking systems (1); banking services (1); education services such as VLEs and MOOCs (1); gaming (1); news and information (1); entertainment services such as on-demand television (1).
- 4 ► Online storage and/or applications that are hosted on online servers (1).

CHAPTER 13

- 1 ► **A** Image (1).
- 2 ► If there is not enough white space (1), the information could appear cramped (1).
- 3 ► Several answers. For example: number data (1) is turned into information through the creation of graphs or charts (1).

- 4 ▶ Spelling error because should be spelled 'supposed' (1); grammar error because the sentence should end with a question mark (?) (1); contextual error because 'two' should be given as 'to' (1).
- 5 ▶ A house style ensures that documents are consistent (1), which helps to create a brand or corporate identity (1).
- 6 ▶ A typeface that adds a short line, known as a serif, to the ends of some letters (1).
- 7 ▶ A low-resolution screen may not be able to display the serif details and make the text difficult to read (1).
- 8 ▶ Green (1).
- 9 ▶ **One** from: colour-blindness (1), partial-sightedness (1).
- 10 ▶ Spell checkers do not always pick up all errors (1).

CHAPTER 14

- 1 ▶ **C** School_Logo_v1.02 (1).
- 2 ▶ Any **three** marks from: To prevent data loss (1) if the computer crashes or loses power (1), because until data is saved in secondary storage (1) it is stored in RAM (1), which is volatile (1).
- 3 ▶ It enables you to go back to previous versions (1).
- 4 ▶ Several answers. For example: .pdf (1), .ppsx (1), .jpeg (1).
- 5 ▶ It allows the user to set the version (1), date (1) and build number for that date (1).
- 6 ▶ Add a password (1).
- 7 ▶ Exporting files saves them in a format that cannot be edited (1).

CHAPTER 15

- 1 ▶ **D** Newsletter (1).
- 2 ▶ If words are used out of context (1) but spelled correctly (1), they will not be highlighted by the spell checker.
- 3 ▶ Portrait (1) and landscape (1).
- 4 ▶ Any **three** from: Page title (1); Author's name (1); Page numbers (1); Date/time document created/modified (1); File name/location (1). Award one mark for 'document/file information' if no other mark awarded.
- 5 ▶ A data source is selected (1) and fields from it are placed into the document (1), which is then populated (1) from selected records (1).
- 6 ▶ Several answers. For example: Dear [Title Surname] (1).
- 7 ▶ Several answers. For example: Yours sincerely (1); Kind regards (1).
- 8 ▶ Any two from: Spreadsheet (1); Delimited file such as a .csv file (1); A table in a (word-processed) document (1); A contact list such as a mail application contact list (1).

CHAPTER 16

- 1 ▶ **B** Aspect ratio (1).
- 2 ▶ Because they are made up of co-ordinates and lines (1), which are recalculated (1) when the image is redrawn (1).
- 3 ▶ Layers allow elements to be placed (1) on top of or

beneath each other on the z axis (1).

- 4 ▶ Pixels (1).
- 5 ▶ Vector graphics often look stylised or unnatural (1).
- 6 ▶ Any **two** from: Erase tool (1); Cropping (1); Recolouring or airbrushing (1).

CHAPTER 17

- 1 ▶ **D** High contrast between background and text (1).
- 2 ▶ The slide master layout is applied to all slides in the presentation (1).
- 3 ▶ **One** from: It may distract the audience (1); The audience might not listen to the speaker (1).
- 4 ▶ Bold (1), underline (1) and italic (1).
- 5 ▶ To emphasise the changes (1) between slides (1).
- 6 ▶ **One** from: Use images that represent the text (1); Summarise points into bullet points (1).
- 7 ▶ **One** from: To provide navigation (1) because some users who will not have access to keyboard and mouse (1); To provide navigation (1) because presentations are often published in kiosk mode for use with a touchscreen interface (1).
- 8 ▶ Any **two** from: URLs (1); Slides (1); Local files (1); Email addresses (1).
- 9 ▶ Handouts (1), full page slides (1) and notes pages (1).
- 10 ▶ Any **two** of: It can help a speaker to time their presentation (1) by displaying timings (1); Speakers can prepare for the next slide (1), which is displayed as a thumbnail (1); Notes for the slide are displayed (1), allowing the speaker to use them to introduce animations at the correct point (1).

CHAPTER 18

- 1 ▶ **A** <a> (1).
- 2 ▶ Image files are referenced, not embedded (1), so the web browser (1) will not be able to find them if their location is altered (1). This will result in a broken link and will make the browser display a placeholder image (1).
- 3 ▶ To provide the layout into which content can be added (1).
- 4 ▶ To draw attention to important information (1).
- 5 ▶ **A** index.html (1).
- 6 ▶ Any **three** from: Animation (1); Image (1); Text (1); Buttons (1); Video (1).

CHAPTER 19

- 1 ▶ **D** G1:J1 (1).
- 2 ▶ **EITHER:** As a way of documenting the spreadsheet (1) with meaningful names (1) that others/you can understand (1)
OR: You don't have to check the cell range every time you refer to it (1), saving time when referencing ranges in formulae or functions (1).
- 3 ▶ Unlike relative references, absolute references do not change (1) when copied to other cells (1), meaning that a

cell reference can be fixed (1) and reducing the likelihood of errors and the work required to avoid errors (1).

- 4 ▶ **A** SUM (1).
- 5 ▶ Any **three** from: Pie chart (1); Bar chart (1); Column chart (1); Line chart (1); Scattergram (1).
- 6 ▶ It finds a given value (1) in the first column of a given array (1) and returns the corresponding value (1) from a given column (1).
- 7 ▶ Scattergram (1).
- 8 ▶ Pie chart (1).
- 9 ▶ Bar or column chart (1).
- 10 ▶ A trend line shows the general direction of data values or line of best fit (1), which can be extended beyond the range of the latest date (1).
- 11 ▶ The formulae will automatically update when values in referenced cells are changed (1).
- 12 ▶ **B** $9 = (5+4)$ (1).

CHAPTER 20

- 1 ▶ **B** Alphanumeric/text (1).
- 2 ▶ To ensure that each record is uniquely identifiable (1).
- 3 ▶ Any **two** from: To ensure that the data entered is acceptable (1); To reduce data entry errors (1); To enforce referential integrity (1).
- 4 ▶ Any **three** from: Title (1); ISBN (1); Number of pages (1); Author (1); Publisher (1); Published Date (1); Format (1).
- 5 ▶ Any **three** from: Referential integrity (1); Presence (1); Type (1); Length (1); Range (1).
- 6 ▶ The database will only allow data to be entered (1) from a foreign table (1) into a linked table (1) if the related record exists (1).
- 7 ▶ Range check (1).
- 8 ▶ 1,3,4,6,7 (1).
- 9 ▶ False (1).
- 10 ▶ **C** NOT Boy (1).

GLOSSARY

absolute reference in a spreadsheet, used to fix a cell reference in a formula so that does not change when it is copied from one cell to another

active cell the selected cell in a worksheet

ad server a web server that stores, delivers and often tracks banner advertisements

adware software that displays unwanted adverts

alphanumeric containing both letters and numbers

alt text alternative text that is displayed if an image cannot be loaded

analogue representing information with continuously variable electrical signals that digital signals approximate

analytics information that results from the analysis of data

application a program that allows a user to perform a task; applications are often called apps

application loading time how long it takes an application to load

application suite a collection of application software that share the same look and user interface; often, they can share data between each other and share some functionality

array a series of items

artificial intelligence (AI) the ability of a computer program to make decisions that would otherwise be made by humans

aspect ratio the ratio of an image's width and height

attribute something that provides additional information about an HTML element; in a database, a characteristic, feature, or quality of an entity

authenticate confirm that the user is who they say they are

authentication the process of confirming that a user is permitted to access certain files, hardware and software

autofill the automatic suggestion of a completed word or phrase that is provided as the user types

avatar a figure that represents a person in a computer game or other virtual environment

back up to make a copy of information stored on a computer in case the original is lost or damaged

backwards compatibility the ability to be used with older technology without having to be specially adapted

balance alert a service that automatically informs a customer, by text message or email, when the balance of their account reaches a specified amount

bandwidth the number of bits that can be carried by a connection in one second

banner advertisement an image, animation or video that displays an advertisement for a product or service in a section of a web page

barcode a pattern of lines and gaps that can be read by a device

big data the analysis of huge quantities of data collected from many sources (such as smartphones, online applications, social media and payment systems) in order to find patterns of behaviour and interactions

bit (binary digit) a single unit of information with a value of either 0 or 1; eight bits equal one byte

bitmap a computer image that is stored or printed as an arrangement of bits

blacklist a list of unacceptable URLs

blog (short for web log) a website or web page that is updated regularly, often written like a diary or a series of articles

blogger someone who creates or maintains a blog

Bluetooth wireless connectivity that allows devices to connect over short distances

boot start up

borders lines that surround the cells in a table

bot a computer program that can interact with systems or users

browsing history the URLs that an individual user has visited, which are stored in a file

buffer an area of memory used to temporarily store data, especially when streaming video

bugs errors in the program's source code

build date the date on which a file was created

build number the number within a series of versions from the build date

capacity the amount of data that can be stored on media

CAPTCHA a computer program or system that can identify whether a user is a human or a computer

cell padding the space *inside* cells in a table that determines how close to the edge of the cell the content will be placed

cell reference the name of a cell in a spreadsheet, created by combining a cell's column letter and its row number

cell spacing the space *between* cells

characters the letters and symbols used to write text

chat room a place on the internet where users can write messages to other people and receive messages back from them immediately

cipher a code

client a computer connected to a server

clip art images, photographs or pictures that you can copy and use in your own computer documents

clock cycles per second used to measure processor speed; the number of times per second the processor can carry out one or more instructions

closed circuit television (CCTV) a system of cameras placed in public buildings or in the street that are used to help prevent crime

cloud a term used to describe the internet when it is used to provide software or space for storing information, rather than relying on local storage

cloud storage storage provided by servers that are connected to the internet

colour value 6 hexadecimal characters which represent the amount of red, green and blue in the colour

compatibility ability of one device, system or application to work with other devices, systems and applications

connections people or accounts to whom a user is connected

connectivity a device's ability to connect to networks and other devices

contact list a virtual address book that allows users to access quickly the contact details of friends and colleagues

contactless users can use a contactless debit card or credit card to pay for things by waving it over a machine, without using a PIN number

convergence when the designs of devices change so that they become similar to one another

cookies information that a website leaves on a user's computer so that the website will recognise that user when they use it again

copper cable a cable that carries data as electrical signals, which are conducted through copper wires

cropping using a photo editor to remove the edges of an image

crowd-mapped data provided by many individuals, forming a 'crowd', which is then superimposed on a map using geolocation information from users' data submissions

cursor a movable point on a screen that identifies the point that the user's input will affect

cyberbullying using the internet to send text, images or multimedia in order to upset or embarrass someone

data access speed how quickly data can be read from or written to media

data values that represent information

database object an object in a database, such as a table, query or report

decompress extract data from a compressed file so that it can be read

deselect remove something from a list of choices on a computer

design characteristics properties of a database object

dialogue box a box that appears on your computer screen when the program needs to ask the user a question before it can continue to do something; the user clicks on one part of the box to give their answer

diaphragm a thin round object, especially in a telephone or loudspeaker, that is moved by sound or that moves when it produces sound

digital assets content used in a digital product such as digital images

digital divide the gap between people who have access to digital devices and the internet and people who do not

digital footprint all the data that a user creates online, which is recorded and stored

digital humanitarian movement a group of people working together using ICT to promote human welfare

digital zoom enlarging part of an image to produce the effect of zooming in

discharge remove the electrical charge from something

domain the name used to identify a web server

domain name server a computer connected to the internet that translates domain names, such as pearson.com, into IP addresses

dragging moving with the mouse

Dynamic Host Configuration Protocol (DHCP) server a networked computer that automatically assigns an IP address to other computers when they join the network

dynaset a dynamic set of data used to store the results of a query

eavesdropper an unauthorised person or piece of software that intercepts data from a private communication

element an individual component of a web page (for example, '<a> ... ' is a hyperlink element)

embedded computing computing hardware that is fixed into position and carries out a specialist task

encrypt protect information by putting it into a special code that only some people can read

entity in a database, something that exists as a single and complete unit

entrance effect an animation effect that occurs when an object is brought onto a slide during a presentation

Ethernet a network connectivity standard that provides a way for computers to communicate

e-ticket an electronic version of a paper ticket

executable file a computer file that can be run as a program

exit effect an animation effect that occurs when an object leaves a slide during a presentation

experience points credits earned by a user for completing one part of a game

fibre optic cable a cable that sends data using light signals

file a collection of data that represents a document, image, database or similar

file extension the letters at the end of the name of a computer file, which show what type of file it is, such as .docx

file transfer protocol (FTP) a method of exchanging data and files across a network

filter a computer program that only allows certain types of information to pass through it

firewall a system (hardware or software) that protects a computer network from being used or looked at by people who do not have permission to do so

firmware a type of software that controls a hardware device

flash memory a form of storage that stores data as electrical charge held in tiny electrical cells

font the size and style of text

footer the area at the bottom of a page or slide that contains content to be repeated on every page or slide

foreign key in a database, a primary key from one table that is used in another table

formula (plural **formulae**) a mathematical operation or calculation

forum a web page, usually dedicated to a particular topic, where users can post comments and information and reply to other users' comments

free software software that can be modified or distributed by a user

freeware proprietary software provided free of charge to users

frequency the waveband at which a radio signal is transmitted

function in mathematics, an expression that represents a complex calculation or process

games console an electronic device that is used for playing computerised video games on a screen

Gbit/s the amount of data that can be transferred per second, measured in Gigabits (1 Gb = 1 000 000 000 bits)

geoblocking limiting access to internet content based on the user's geographical location (also known as geolocation rights management)

geolocation rights management the use of a user's location to block access to services online; for example, a television channel might only allow users in their own country to watch their shows online

geotag (verb) to add location data to a piece of content

GPS the Global Positioning System that uses radio signals from satellites to show your exact position on the Earth on a special piece of equipment

gradebook a virtual method of recording students' scores on assignments and tests

graphics card a device built into or added into a computer to allow it to display visual graphics

greyscale using a range of shades of grey

hacking accessing a system using malicious methods

HDMI (High-Definition Multimedia Interface) used to transmit video and audio data

header the area at the top of a page or slide that contains content to be repeated on every page or slide

hexadecimal a base-16 number system that uses the numbers 0–9 and the letters A–F

home page the introductory or main page of a website

hotspot a place in a public building where there is a computer system with an access point, which allows people in the building with a wireless computer or smartphone to connect to a network such as the internet

house style an organisation's preferred style of writing and layout of content

hover over use the mouse to position the cursor on an object on a computer screen without clicking on the object

hyperlink a link that can be clicked in order to go to another location (often a webpage on the internet)

Hypertext Markup Language (HTML) a computing language read by web browser software

Hypertext Transfer Protocol (HTTP) a set of standards that control how computer documents that are written in HTML connect to each other

Hypertext Transfer Protocol Secure (HTTPS) a secure form of HTTP

identifier a group of letters, numbers, or symbols that a computer has been programmed to recognise and uses to process information

indent marker a point at which text can be moved in from a margin, used to produce standardised spacing between text

information data that has been processed to be given context and meaning

information poor people with limited access to information provided by communication technologies

information rich people with good access to information provided by communication technologies

infra-red a type of electromagnetic radiation with a longer frequency than that of visible light

interactivity the ability of a computer to respond to input

Internet Service Provider (ISP) a company that provides customers with access to the internet

internet traffic data transferred between computers connected to the internet

intranet a computer network used for exchanging or seeing information within a company

in-video advertising advertisements that appear within online videos

IP address a unique address that networked devices use to send data to each other

keywords the words or search terms that a user types into a search engine in order to look for matching information

latency the amount of time it takes to send data between devices

layers individual sections of the image; each layer is positioned on a separate plane of the z axis and can contain one or more elements

layout the way in which text and/or images appear on a page

legitimate website a website that is operated legally

lock to make a device inactive, so that it requires a password or PIN to reactivate it

log a record of events

login user or account information, such as a user name or account name

macro a function that runs a set of instructions to perform a task

mainframe computer a large, powerful computer that can do a lot of complicated jobs quickly and can be used by a lot of people at the same time.

malware software that is created with the intention to do harm

mapped linked

markup the tags surrounding elements of text that indicate the purpose of the text and instruct the browser how to handle (display) them

masked hidden

Massive Open Online Course (MOOC) a course that is free of charge and available to large numbers of people over the internet

Massively Multiplayer Online Role-Playing Game (MMORPG) an online video game that allows large numbers of people to play together

Mbit/s the amount of data that can be transferred per second, measured in Megabits (1 Mb = 1 million bits)

media items such as memory cards and CD-ROMs that make information available to people and devices

media player an electronic device that can store and play digital music and videos and show digital photographs

member someone who is part of an online community

meme a photo, a piece of video, a joke or similar content that spreads quickly on the internet

Merge and Centre a spreadsheet tool that combines selected cells and centre-aligns the text in the new combined cell

microprocessor the device that controls what a computer does; it takes data as input, does something with it and provides output

minijack a plug and socket widely used for analogue audio signals in portable devices

moderate decide whether content is appropriate; moderators decide whether posts should be removed or posted at all

monochrome using a range of shades of one colour

multifunctional device a device that can perform a range of different functions

multimedia a combination of more than one medium, such as audio and video together

multiplayer games games that are played by more than one person, usually online

mute silence an audio track

navigate move around a piece of application software or a system, such as from one slide to the next

netbook a computer that does not have lots of storage or fast processors

network administrator a person who manages an organisation's network

normalisation increasing the average volume of a piece of audio to a defined maximum level

notification an alert that tells a user about a new interaction or new content within a social networking community

online community a group of people with a shared common interest who communicate online

open-source software software for which the source code is made available to users

operator a symbol that represents a mathematical action

optical zoom adjusting a lens in order to zoom in on an object

packet a unit of data packaged to travel across a network

page margin in a word processing application, where the white area of the ruler ends and the grey part begins

pair connect two devices (usually only with each other)

pathway the sequence of or route through a series of slides

payment processor a service that authorises financial transactions

payment server a computer that authorises financial transactions

peripheral device equipment that is connected to a computer and used with it

permissions authorisation settings that provide the ability for a user or users to access files, folders or drives

phishing the criminal activity of sending emails or having a website that is intended to trick someone into giving away personal information such as their bank account number or their computer password; this information is then used to get money or goods

physical computing interactive systems that can sense and respond to the world around them

pixel a small dot that helps to make up an image

pixelated an effect that creates an unclear image consisting of large individual pixels that are visible to the human eye

plain text text that is not specially formatted

platters circular plates

pointer an on-screen indicator used to select displayed objects

port a socket into which cables and devices can be plugged

post (noun) a message sent to an internet discussion group so that all members of the group can read it

post (verb) to put a message or computer document on the internet so that other people can see it

primary key field in a database, the field that holds the primary key, which is a unique attribute stored for each entity

print spooling the process of keeping pages queued so they are ready to be printed by a printer

privacy settings a method that allows users to control who can see information from their online profiles

processor cycle the process of fetching a program instruction from memory, decoding the actions required by the instruction and then executing those actions

processor one or more Central Processing Units (CPUs) that carry out software instructions

profile (noun) a collection of information about a user

profile (verb) to compile personal information about an individual

program a set of instructions that are carried out by the processor

proprietary software software that is marketed and distributed by its owner under a brand name

protocols rules that allow the exchange and transmission of data between devices

purpose what something (such as a digital product) is required to do

quarantine isolate a suspected virus in a protected area of storage where it cannot harm other files

query a request for data results

RAM Random Access Memory, which is the memory in a computer system that is used for running software

range a group of adjacent cells in a worksheet

reader a piece of electronic equipment that can read information that is stored or recorded somewhere, for example, on a card

read/write head the part of a disk drive that passes (or floats) across the platters on a very thin layer of gas above the platters

real time processing data within milliseconds of it being input and making the output available almost immediately

referential integrity the state of a database when any foreign key field in a table refers to a primary key in a related table

referring website the website previously visited, which linked the user to the website that they are currently visiting

required attribute an attribute that must be present for the element to work as intended

resize point a point at which an image can be dragged to be resized

resolution the number of pixels used by a screen to display an image

rotate point the pivot point around which an image can be rotated

seek time the time it takes for a read/write head to locate the area on the disk where the data to be read is stored

sensor an input peripheral that inputs data about the physical environment

service agreement contract

set-top box (STB) a device that sends video and audio received from a broadcaster to a television

shortcut a combination of key presses used to access common software commands quickly

single operator formula a formula that uses only one operator

single sign on (SSO) a system that allows users to log in once to access a number of related websites and systems

slide a single page of a presentation

slide master (or master slide) in a slide presentation, a template slide that specifies the layout and appearance of content slides

smart home a home equipped with devices that can connect to the internet and be controlled remotely by a computer or smartphone app

smart the ability of a device to use data from sensors to perform a programmed action independently; smart devices are often also able to connect to the internet

smartwatch a watch that provides data connectivity and often uses sensors to provide feedback to the device about its environment

snake case using underscores (underlines) to separate words in a filename

social bookmarking using tags to categorise web documents and URLs so that other people can find content by using the tags in a search

social network applications software that allows social interaction and the creation of links between users based on shared characteristics and interests

software licence a legal arrangement that gives a user the right to install and use software

solo only play one particular audio track

sort in a spreadsheet, arrange data in a particular order

sound card a device built into or added into a computer to allow it to playback and record audio

source code a collection of instructions that forms a piece of software

sprites on-screen graphics in a computer game, used to represent characters, vehicles and objects

spyware software that monitors and records data and user input

status update a message or post that a user adds to their own social media page to inform others of something that they consider important enough to share

stereo sound sound that comes from two sides (left and right)

stream play a file on your computer while it is being downloaded from the internet, rather than saving it as a file and then playing it

stylus a pen-shaped device

surround sound sound that comes from four or more sides, used so that sounds from a film or television programme come from all around the viewer as they would in real life

syntax the rules that describe how words and phrases are used in a language

tag a label that you can add to a post in order to categorise it

targeted marketing advertising that is matched to internet users based on their attributes, such as their age group or their gender, or their internet browsing history

telecommunications infrastructure the networks of hardware facilities, owned by private and public organisations, that are used to transfer data

tempo the speed at which a track is played

tethering connecting a host device that uses a mobile broadband connection with other devices so that they can use the host's broadband connection

third-party developer a software developer who is not the developer of the original application

third-party integration including features from an online service that do not belong to the original website and are not related to the user of the website

thread a series of messages concerning the same subject, written by members of an internet discussion group

toner a powder used by some printer types

toolbar a row of buttons or options at the top of a window that allow you to perform common actions within an application

track collect and analyse data

transaction the exchange of data

transactional data data that is sent between digital devices

transitions in a slide presentation, the effects that occur when a user moves from one slide to the next

transparency areas of an image that do not have a colour value, including white, are made see-through; if transparency is unavailable due to the file format, those areas are displayed as white

ungroup break apart the grouped components of an image

update (noun) a change or addition to a computer file so that it has the most recent information; (verb) to make a change or addition to a computer file so that it has the most recent information

upgrade to make a computer better and able to do more things

URL (uniform resource locator) a website address

usability the ease with which a device or application can be used

usage rights the way in which a piece of information is permitted to be used

USB (Universal Serial Bus) a standard for wired connectivity that can also supply electric power

user interface the system that allows a user to interact with a device

user the person who uses a computer system

user-generated created by the individuals who use the service

utility software (also known as utilities) system software that carries out configuration and maintenance tasks

validation the process of checking if something is acceptable when compared to given criteria

vector graphic a graphical image made up of points and lines

viral marketing a type of advertising used by internet companies in which computer users pass on advertising messages or images through email, sometimes without realising that they are doing so

viral something that is circulated widely by being shared through networks to large numbers of internet users

virtual keyboard a keyboard displayed on and used via a touchscreen

virtual learning environments (VLEs) websites that contain teaching and learning tools

virtual memory storage used by the processor once the space in RAM has run out

virus definitions sequences of code that are found in computer viruses

virus malware that uses networks to spread to connected devices

vlog (short for video log) a video blog

Voice over Internet Protocol (VoIP) the protocol that allows people to make telephone calls using the internet

voice recognition a system in which a computer obeys instructions spoken by a human voice

voucher code a code representing a specified amount of money that can be used on a shopping website to purchase goods or services

web server a computer that stores web pages and sends them to other devices that request them (often using web browser software)

webform a data entry form on a web page

white space the part of a page or digital product that is left blank

whitelist a list of acceptable URLs

Wi-Fi a way of connecting computers or other electronic machines to a network or the internet by using radio signals rather than wires

wiki a website or database that is developed by a number of collaborating users, all of whom can add and edit content

wipe to delete data from storage

WYSIWYG ('What You See Is What You Get') when something is presented on screen exactly as it will look when the finished product is output or published

z axis the dimension that provides an image's depth and allows elements to appear 'in front of' or 'behind' other elements

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